Adaptive Data Analysis with α^{-2} polylog Overhead, Interval Validation, and Adversarially Robust Turnstile F_2 Estimation in ε^{-2} polylog Bits

Abstract

We give a common equal-impulse estimator–corrector–learner framework across privacy-graded sample hierarchies. Our first result is an efficient mechanism for k adaptive statistical queries $q_1, \dots, q_k : X \to [0,1]$, for any distribution D over any measurable domain X. For $\alpha \in (0, 1/16)$ and $\beta \in (0, 1/2)$, it achieves simultaneous accuracy α with probability at least $1-\beta$, using

$$N_{\mathrm{cas}} = O\Bigg(\frac{\sqrt{k}\chi}{\alpha} + \frac{1}{\alpha^2}\Big[\ln^3(1/(\alpha\beta)) + \ln^{7/2}(1/\alpha) \ln\Big(\frac{k \ln(1/\alpha)}{\alpha\beta}\Big) \sqrt{\ln\Big(\frac{\ln(1/\alpha)}{\alpha\beta}\Big)}\Big]\Bigg)$$

samples. Here $\chi = \sqrt{\ln(1/(\alpha\beta))}$ when $\beta \geq \alpha^5$, and $\chi = \ln(1/(\alpha\beta))$ otherwise. This removes every $\log k$ factor from the $\sqrt{k}$ leading term and every quasipolynomial $2^{\Theta(\sqrt{\ln(1/\alpha)})}$ factor from the additive overhead.

Our second result is a finite-precision interval-validation mechanism over a finite domain. It handles M adaptive interval checks, at most J nonterminal rejections, and R numerical releases using $\widetilde{O}(\sqrt{J+R+1}/a_{\mathrm{req}} + a_{\mathrm{req}}^{-2})$ samples, with M entering only logarithmically. We deploy this mechanism inside a binary time tree of geometric learners. The resulting single-pass integer turnstile algorithm estimates F_2 against a computationally unbounded adaptive adversary using $O(\varepsilon^{-2}\Lambda^{15}) = \varepsilon^{-2}\,\mathrm{polylog}(n, m, U, 1/\varepsilon, 1/\beta)$ bits (certified at most $2^{512}\varepsilon^{-2}\Lambda^{15}$ bits in the accompanying Lean 4 artifact) on every execution, where Λ is logarithmic in these parameters. All statistical, runtime, and certified bit-space clauses of the three main theorems are mechanically verified in Lean 4.

1 Introduction

An estimator can succeed on every fixed input and still fail when its next input depends on its previous answers. This problem appears in both data analysis and streaming. In each setting, we want to reuse a small random object without letting an adaptive interaction exhaust what that object can safely reveal.

Adaptive data analysis. For statistical queries fixed independently of the sample, Hoeffding's inequality gives simultaneous error α from $O(\ln(k/\beta)/\alpha^2)$ observations. Adaptivity changes the problem. On a sample of size r, an analyst can invalidate naive empirical means after only $O(r)$ queries [16, 24, 34]. A useful mechanism must therefore control how its answers reveal the sample, rather than merely concentrate each answer in isolation [16, 17, 4, 13].

(a) Adaptive data analysis: sample complexity	
Private stability [4, 28]	Leading term $O(\sqrt{k \ln k}\, \ln(1/(\alpha\beta))/\alpha)$; a logarithmic loss remains on $\sqrt{k}$.
Bounded noise and cascades [13]	No leading $\log k$ loss; larger additive overhead, including $2^{\Theta(\sqrt{\ln(1/\alpha)})}$ factors in cascade tradeoffs.
Our cascade	$O(\sqrt{k}\chi/\alpha)$ leading term, with $\alpha^{-2}\,\mathrm{polylog}(1/\alpha, k, 1/\beta)$ additive overhead.
(b) Adversarially robust F_2: space and scope	
Oblivious sketches [2, 30]	Optimal oblivious F_2 space in polynomial-range streams; not an adaptive guarantee.
Robust wrappers [26, 36]	Polynomial dependence on λ; general turnstile streams can have $\lambda = \Theta(m)$.
Learner–corrector [23]	Insertion–deletion framework; persistence and finite precision require separate control for the bound below.
Our streaming algorithm	$\varepsilon^{-2}\,\mathrm{polylog}(n, m, U, 1/\varepsilon, 1/\beta)$ bits for integer turnstile streams, on every execution.

Table 1: The two comparisons. The ADA rows separate the leading $\sqrt{k}$ cost from the additive overhead. The streaming rows distinguish an oblivious guarantee, dependence on stream variation, and a bound on actual bits.

Differential privacy and stability give mechanisms whose leading sample term is of order $(\sqrt{k \ln k}/\alpha)\ln(1/(\alpha\beta))$ [16, 9, 4, 28]; retaining bounded memory between adaptive queries adds a separate structural constraint [15]. Bounded-noise privacy removes the $\sqrt{\ln k}$ loss [13], but obtaining a small additive term requires more care. Previous tradeoffs leave a larger k-independent overhead, or use cascades whose additive cost contains a factor $2^{\Theta(\sqrt{\ln(1/\alpha)})}$. The obstruction is accumulated drift. A proposal can pass many validators, moving slightly farther from the desired value at each stage. With $\Theta(\ln(1/\alpha))$ stages, charging each correction separately repeatedly charges the same drift.

We remove this charge by coupling adjacent learners. When a comparison detects a discrepancy, we update both endpoints with equal weighted impulse and opposite signs. Their unknown population terms cancel exactly. This allows logarithmic depth with only polynomial dependence on that depth.

Adversarially robust streaming. The same reuse problem arises when a stream depends on earlier estimates [7, 26, 36, 11, 22]. Classical linear sketches achieve optimal $O(\varepsilon^{-2}\log n)$-type bounds for oblivious, polynomially bounded streams at constant failure probability [2, 30]. Against an adaptive adversary, however, linear sketches and oblivious summaries can reveal enough information to expose their kernels [25, 29, 12, 21].

Sketch switching, privacy-based wrappers, and difference estimators protect sketches by limiting their use [7, 26, 36, 6]. Their costs typically depend polynomially on a flip number λ, which counts significant changes in the statistic. Depending on the regime, these methods pay ε^{-3} or ε^{-4}, or lose their advantage when $\lambda = \Theta(m)$. General turnstile streams permit exactly this behavior: a stream can repeatedly alternate between zero and a unit vector.

Learner–corrector methods offer a different route [23]. They spend information when a prediction fails, rather than whenever time advances. For full integer turnstile streams, two issues remain: ancestor corrections must not restart children in the middle of their lifetimes, and the implementation must bound every counter's precision. A word-space analysis alone does not settle either issue.

Our three theorems address different costs: answering every query, validating many intervals while releasing little, and maintaining a norm throughout an adaptive stream.

Theorem 1.1 (Adaptive Data Analysis Without $\log k$ Overhead)**.** *For every integer $k \geq 1$, accuracy $\alpha \in (0, 1/16)$, and failure probability $\beta \in (0, 1/2)$, the cascade mechanism $\mathcal{M}_{\mathrm{cas}}$ of Algorithm 1, instantiated with the locked parameters of Definition A.1, answers k adaptively chosen statistical queries $q_1, \ldots, q_k : X \to [0, 1]$ on any probability distribution D over any measurable domain X. Against any possibly randomized, computationally unbounded, jointly measurable adaptive analyst observing the full public transcript $(a_i, \tau_{\mathrm{aux},i})_{i=1}^k$, it satisfies*

$$\mathbb{P}\left[\max_{1 \leq i \leq k} |a_i - \mathbb{E}_D[q_i]| \leq \alpha\right] \geq 1 - \beta.$$

It uses $N_{\mathrm{cas}} = n_1 + \sum_{r=2}^{L-1} n_r + n_L$ i.i.d. samples from D, where

$$\begin{aligned} N_{\mathrm{cas}} &\leq \frac{32000\sqrt{k}\chi_1}{\alpha} + \frac{16000 L_1^3}{\alpha^2} + 1 \\ &\quad + \frac{10^8}{\alpha^2} L^{7/2} \ln\left(\frac{64 L k M_{2,\max}}{\delta_{\mathrm{val}}}\right) \sqrt{L_{\mathrm{val}}} \\ &= O\Bigg(\frac{\sqrt{k}\chi}{\alpha} + \frac{1}{\alpha^2}\Big[\ln^3\left(\frac{1}{\alpha\beta}\right) \\ &\qquad + \ln^{7/2}(1/\alpha) \ln\left(\frac{k \ln(1/\alpha)}{\alpha\beta}\right) \sqrt{\ln\left(\frac{\ln(1/\alpha)}{\alpha\beta}\right)}\Big]\Bigg). \end{aligned}$$

Here $\chi = \sqrt{\ln(1/(\alpha\beta))}$ if $\beta \geq \alpha^5$, and $\chi = \ln(1/(\alpha\beta))$ if $\beta < \alpha^5$. Starting from the preloaded public input $(k, \alpha, \beta, L, p_1, n_1, \ldots, n_L)$, a single uniform rational-literal real-RAM program—with unit-cost field arithmetic, comparisons, $\exp$, $\ln$, $\sqrt{\cdot}$, query-evaluation, and noise-table oracle calls—implements $\mathcal{M}_{\mathrm{cas}}$ in $O(kN_{\mathrm{cas}})$ operations ($O(N_{\mathrm{cas}})$ at initialization and $O(N_{\mathrm{cas}})$ per round) and $O(N_{\mathrm{cas}})$ working-space cells, streaming $(a_i, \tau_{\mathrm{aux},i})$ to a write-only output.

Corollary 4.7 absorbs the remaining additive $\ln k$ into $\sqrt{k}$, giving an alternative bound with k-independent additive overhead.

Theorem 1.2 (Finite-Precision Interval-Validation ADA)**.** *Let $\mathcal{U}$ be a finite domain with injective $b_{\mathcal{U}}$-bit record encodings, and let $\mathcal{D}$ be an unknown distribution on $\mathcal{U}$. Fix $a_{\mathrm{req}} \in (0, 1/10)$, $\xi_{\mathrm{req}} \in (0, 1/2)$, and nonnegative integer caps M, J, R on interval requests, nonterminal rejections, and numerical requests. The mechanism $\mathcal{M}_{\mathrm{val}}$ of Algorithm 2 uses the locked parameters of Definition A.3, including*

$$a = 2^{-\lceil \log_2(1/a_{\mathrm{req}}) \rceil}, \qquad \xi_A = 2^{-\lceil \log_2(1/\xi_{\mathrm{req}}) \rceil}, \qquad p = \Theta(\ln(1/a_{\mathrm{req}})), \qquad K = J + R + 1,$$

and χ as fixed there. It interacts with any adaptive controller submitting queries $f_t : \mathcal{U} \to [0, 1]$, accessed only on the sample, paired with either an interval $[l_t, u_t]$ with $l_t \leq u_t$ or a numerical request. Covered requests respect the caps M, R and stop after a possible terminal $(J+1)$-st interval rejection, including that rejection.

Given N independent samples from $\mathcal{D}$ and a single sufficiently long block of fresh independent fair bits read only by its bounded internal noise samplers, where

$$\begin{aligned} N = \sum_{j=0}^{p} n_j &\leq 2^{64} \left[\left(\frac{p^{7/2}}{a^2} + \frac{p^{3/2}\sqrt{K}}{a}\right)\chi^{3/2} + \frac{p^3\chi}{a^2}\right] \\ &= O\left[\left(\frac{\sqrt{J+R+1}}{a_{\mathrm{req}}} + \frac{1}{a_{\mathrm{req}}^2}\right) \ln^3(1/a_{\mathrm{req}}) \ln^2\left(\frac{M+J+R+2}{a_{\mathrm{req}}\xi_{\mathrm{req}}}\right)\right], \end{aligned}$$

with probability at least $1-\xi_{\mathrm{req}}$ *the mechanism never reaches an internal correction cap* $T_j^{\max}$ *or aborts on any covered request. Every covered interval response satisfies*

$$\textsc{Accept}: \quad \mathcal{D}f_t \in [l_t - a_{\mathrm{req}}, u_t + a_{\mathrm{req}}],$$
$$\textsc{High}: \quad \mathcal{D}f_t \geq u_t - a_{\mathrm{req}}, \qquad \textsc{Low}: \quad \mathcal{D}f_t \leq l_t + a_{\mathrm{req}}$$

(and strictly $\mathcal{D}f_t > u_t$, $\mathcal{D}f_t < l_t$ *whenever* $l_t, u_t \in [-1,2]$*). Every covered numerical response satisfies* $\hat{\mu}_t \in [0,1]$ *and* $|\hat{\mu}_t - \mathcal{D}f_t| \leq a_{\mathrm{req}}$.

A uniform implementation on the oracle bit-machine reproduces $\mathcal{M}_{\mathrm{val}}$ *within* $O(N(b_{\mathcal{U}}+\chi^2))$ *bits—with certified upper bound* $2^{30}N(b_{\mathcal{U}}+\chi^2)$ *bits (Section 7)—at every step, plus the internal working space of its evaluation oracle. Between requests it retains only an encoding of the sample and current in-block learner states, never historical queries.*

Theorem 1.3 (Adversarially Robust Turnstile F_2 Estimation)**.** *There is a single uniform finite-precision single-pass streaming algorithm that, for all integers* $n, m, U \geq 1$ *and reals* $\varepsilon, \beta \in (0, 1/10)$ *with*

$$\Lambda := \log_2 \left\lceil \frac{2^{100}(n+2)(m+2)(U+2)}{\varepsilon\beta} \right\rceil,$$

against any possibly randomized, computationally unbounded adaptive adversary in the integer turnstile model maintaining $\left\|x^{(t)}\right\|_\infty \leq U$ *for* $1 \leq t \leq m$*, outputs* $\widehat{F}_t \geq 0$ *after every update and satisfies*

$$\mathbb{P}\left[\forall\, t \in \{1,\ldots,m\}: \ \frac{F_2(x^{(t)})}{1+\varepsilon} \leq \widehat{F}_t \leq (1+\varepsilon)F_2(x^{(t)})\right] \geq 1-\beta.$$

It uses fresh independent fair bits and $O(\varepsilon^{-2}\Lambda^{15})$ *bits—with certified upper bound*

$$2^{512}\varepsilon^{-2}\Lambda^{15} = \varepsilon^{-2}\,\mathrm{polylog}(n, m, U, 1/\varepsilon, 1/\beta)$$

bits (Section 7)—on every execution, counting all samples, seeds, counters, learner states, scale tags, caches, noise samplers, public parameters, and step-level arithmetic and control workspace.

Why two ADA theorems? Theorem 1.1 does not give Theorem 1.2 as a black box. Publishing a working-sample proposal for each of M checks pays $\sqrt{M}/a$; in streaming, M can equal m. Interval validation instead pays for J rejections and R numerical releases, with M only inside logarithms. Conversely, setting $M = J = 0$ and $R = k$ in Theorem 1.2 gives $\widetilde{O}(\sqrt{k}/\alpha + \alpha^{-2})$, but its stated bound has a $\log^2 k$ factor on the leading term. Theorem 1.1 eliminates that factor completely. The two mechanisms share a cancellation argument, not an interchangeable output interface.

Credit. Both results were discovered by Gemini Colosseum. The update $u^+ = u + a_{\mathrm{step}}\, y$ (with $y = u + q$) of the geometric learners in Section 6 has the same form as the learner update of [23]. The identity $y^+ = (1 + a_{\mathrm{step}})\, y$, which lets descendant learners persist across ancestor corrections instead of being restarted, was discovered by Gemini Deep Think.

Formal verification. The specification proposition `Theorem_main_full` in `F2e2e/Spec.lean`, proved as `theorem_main_full` in `F2e2e/Headline.lean`, formalizes the statistical guarantees, the real-RAM runtime clause, and the exact 2^{30} and 2^{512} small-step bit-space clauses above. The accompanying Lean 4 package contains 654 modules and 84,399 kernel declarations, checked by Lean's kernel, `nanoda`, and `con-leche`. Section 7 records the formal interfaces, source hashes, and trust boundary.

2 Technical Overview

A cheap prediction is useful only if we can detect and repair its failures without discarding everything learned so far. Our mechanisms separate these jobs. An estimator proposes an answer, a protected comparison decides whether to trust it, and a learner changes state only when the comparison provides evidence of progress.

2.1 Equal Impulses Remove Pass Drift

A hierarchy can fail even when each validator is accurate in isolation. Suppose a proposal is scanned across validators sequentially, replacing the working proposal whenever a validator triggers, and only the final triggered validator r^* moves toward a proxy for the population mean μ_i. An adversary can make stage $r^* - 1$ pass (or match an earlier trigger), carrying drift up to $\tau_{(u,r^*-1)}$, while stage r^* triggers. A one-sided correction then inherits all the earlier slack. The usual repair spaces thresholds geometrically; optimizing the depth leaves a $2^{\Theta(\sqrt{\ln(1/\alpha)})}$ additive loss. With $L = \Theta(\ln(1/\alpha))$ stages, we instead couple adjacent learners so that the inherited drift cancels.

The right weights are inverse squared privacy capacities on the learned levels:

$$w_\ell = \varepsilon_\ell^{-2} \ (2 \le \ell < L) \text{ in the cascade}, \qquad \lambda_j = 4^j = (p\theta_j^2)^{-1} \ (0 \le j < p) \text{ in interval validation}.$$

When a protected adjacent comparison finds a discrepancy of sign $s \in \{-1, +1\}$, we move the two learners in opposite directions. Their weighted impulses agree:

$$w_{r^*-1}\eta_{r^*-1} = w_{r^*}\tilde{\eta}_{r^*} \quad (3 \le r^* < L), \qquad \lambda_j(+q) + \lambda_{j+1}(-q/4) = 0.$$

In the cascade, s has the algorithm's discrepancy-sign convention, so the score increments are $\delta_{r^*-1,i} = -s\eta_{r^*-1}$ and $\delta_{r^*,i} = +s\tilde{\eta}_{r^*}$ (whose negatives enter the regret sum below); in interval validation, q is the signed query increment to the learner's score vector.

Here is the cancellation in its simplest form. Let z_-, z_+ be adjacent predictions, let $w_-\eta_- = w_+\tilde{\eta}_+ = I$, and choose s toward $z_- - z_+$. The population-dependent part of the weighted regret sum is

$$\begin{aligned} w_- s\eta_-(z_- - \mu_i) - w_+ s\tilde{\eta}_+(z_+ - \mu_i) \\ = Is(z_- - z_+). \end{aligned}$$

The unknown μ_i disappears identically. We do not bound its contribution or charge it to another stage. The observed discrepancy now pays for the curvature term in the FTRL regret bound. What remains is a pure progress gain: $\Omega(\gamma^2)$ in the cascade and $\Omega(\tau^2)$ in interval validation, with the corresponding level weight. Here γ is the cascade's comparison margin, and τ is the interval hierarchy's margin.

This is a pathwise argument. It applies to the actual adaptive query sequence, without first proving that the learners' predictions are unbiased. Privacy enters separately, when we construct a low-cost comparator that tracks population expectations.

2.2 Two Ways to Protect the Hierarchy

Cancellation alone does not determine what the mechanism should reveal. Releasing a number on every round and checking an interval on every round have different privacy costs. We therefore build two interfaces around the same weighted progress argument; Figure 1 shows the distinction.

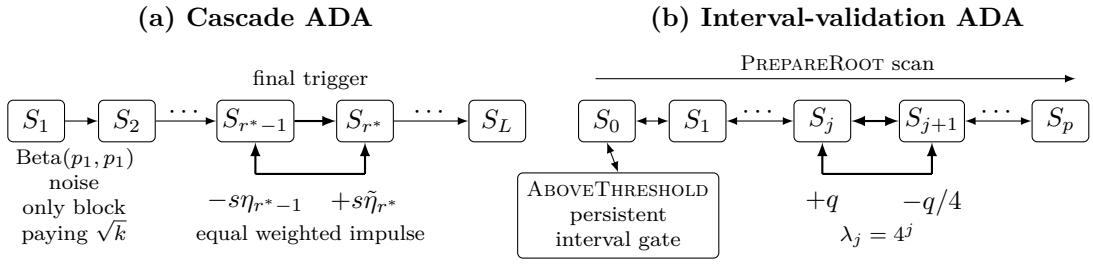


Figure 1: Two privacy architectures around the same cancellation. The cascade scans every stage, replacing the proposal at each trigger, and couples the final triggered stage to its predecessor. Interval validation repairs adjacent discrepancies while keeping its outer interval gate alive across accepted requests. Dotted portions omit intermediate levels.

The cascade: isolate the cost of answering every query. We partition the sample into $S_1, \ldots, S_L$. Only S_1 pays the $\sqrt{k}$ cost. It uses centered, scaled Beta(p_1, p_1) noise, whose density exponent is $q_B = p_1 - 1$. Bounded support avoids a union bound over k answer-noise tails. We analyze the cumulative privacy loss on this block without multiplying it by the hierarchy depth.

The remaining blocks form a forward trigger-and-replace chain, followed by an update at the final trigger $r^* = u_{\text{final}}$. On a final trigger at $r^* \geq 3$, the mechanism also uses a clipped predecessor release $c_i^{(r^*-1)}$ (drawn retroactively if $u_{\text{prev}}(r^*) < r^* - 1$). Clipping prevents a backward correction from importing uncontrolled pass drift into the predecessor's privacy accounting. The two learners receive opposite updates with equal weighted impulse; the mechanism does not restart either learner.

Entropic FTRL bounds the total progress expenditure. A tilted comparator then transfers this pathwise bound to population accuracy. This transfer accounts for the entire public transcript $\Pi = (a_i, \tau_{\text{aux},i})_{i=1}^k$, not merely the numerical answers. The separation matters: root privacy pays for k releases, while the validator analysis pays for a bounded amount of correction progress. Lemmas 4.2 to 4.5 make these four charges explicit. Their combination leaves only powers of L in the additive cost.

Interval validation: pay when information is released. Here a controller often needs only to know whether its proposed interval is plausible. Returning a fresh numerical estimate would reveal much more than the controller requested. We instead keep a persistent outer AboveThreshold phase at S_0. An acceptance does not end the phase and consumes no new root phase. Rejections and numerical releases account for the $K = J + R + 1$ phase budget.

Inside the hierarchy, PrepareRoot scans adjacent learners until the root is ready for the outer check. A correction can restart this scan, but it need not invalidate an edge whose endpoints have not changed. We cache such edges. Without this cache, repeated scans could charge M tests to every level even when no learner changes.

Quadratic FTRL makes the accounting explicit. For the learned levels $0 \leq j < p$, the privacy capacity is $\theta_j = 2^{-j}/\sqrt{p}$ with weight $\lambda_j = 4^j$, while the uniform anchor uses $\theta_p = h/(2\sqrt{2})$ outside the dynamic potential, and the weights lie in the $8/n_j$-capped simplex W_j. We construct the comparator by clipping inverse posterior likelihoods to $[e^{-\theta_j}, e^{\theta_j}]$ and normalizing. For a block of size r, Bhatia–Davis gives the deterministic bound

$$R_r(u_j) \leq \frac{1}{2} \sinh^2 \theta_j \leq \theta_j^2.$$

Thus the initial weighted budget is $\sum_{j<p} \lambda_j \theta_j^2 \leq 1$. A correction at edge j spends $\Omega(4^j \tau^2)$ of this

budget, giving $C_j = O(4^{-j}\tau^{-2})$, independently of M.

The choices $\tau = a/(8p)$, $h = \tau/64$, and $\kappa = \tau/4$ leave room for lattice rounding. Bounded samplers implement the noise using finitely many fair bits. Large M affects the failure allocation and logarithmic precision, not the correction budget. The mechanism retains the current learner states rather than a list of old queries; this is also what makes its bit-space guarantee useful inside a stream.

2.3 From a Fixed Sketch to a Turnstile Stream

A fixed sketch is easiest to protect when it will answer only a limited number of questions. The difficulty is to keep the same space bound when time can advance far more often than we can afford to release a private number.

For a warm-up, let $K = \lceil \varepsilon^{-2} \rceil$. Fix a sign matrix $A \in \{-1, +1\}^{N \times n}$ whose rows are independent draws from a $2s$-wise independent sign family. Reserve disjoint fresh AMS groups before the interaction. For each adaptive vector v_t, a previously unused group of $O(\log(K/\beta))$ rows supplies a constant-factor normalizer c_t. The remaining rows answer the bounded statistical query

$$b \longmapsto \min\left\{ \frac{\langle b, v_t \rangle^2}{L_{\mathrm{clip}} c_t}, 1 \right\}.$$

Moment bounds control the clipping bias. At bounded-query accuracy $\Theta(\varepsilon/L_{\mathrm{clip}})$, ADA uses $\widetilde{O}(\sqrt{K}/\varepsilon + \varepsilon^{-2}) = \widetilde{O}(\varepsilon^{-2})$ rows. The fresh normalizers also cost only $\widetilde{O}(K)$ rows. Four obstacles arise when we replace these K rounds by an m-step integer turnstile stream.

1. Normalize without spending a fresh sketch on every round. We cannot reserve a coarse AMS group for each accepted round. The scale-free regularized Norm Bank of Algorithm 4 instead starts from an unverified nonnegative candidate $S = P + \widehat{Q}$, where P is a residual energy cache and $\widehat{Q}$ is a child estimate. Append an unscaled coordinate $\sigma = \sqrt{\omega}$ to v, and set

$$F' = \|v\|_2^2 + \omega, \qquad c = S + \omega, \qquad r = F'/c.$$

The regularizer keeps the denominator positive, but does not assume that S is accurate.

For the dyadic local accuracy η, use $s = \log_2(1/\eta) + 22$ and $L_{\mathrm{clip}} = 16s$. If f is the clipped query on $(v, \sqrt{\omega})$ with denominator $L_{\mathrm{clip}} c$, a fourth-moment Cauchy–Schwarz argument gives

$$\mathbb{E}[f] \geq \frac{r}{L_{\mathrm{clip}} + 3r}.$$

This bound holds at every scale. It rules out $r \geq 2$ whenever the bank returns Accept or Low. Once $r \leq 2$, the $2s$-th moment bound controls the clipping tail at the requested local accuracy. We thus join a global coarse argument to a local accurate argument, rather than assuming a good normalizer before validating it. Theorem 6.2 formalizes this interface.

2. Correct an ancestor without restarting its children. Restarting a child after every ancestor correction would spend new samples and destroy the lifetime accounting. We instead place learners on a fixed binary time tree of depth $H = O(\log m)$; see Algorithm 5. At depth d, write

$$x_d = a_d + q_d = r_d + y_d, \qquad r_d = a_d - u_d, \qquad y_d = u_d + q_d = x_{d+1}.$$

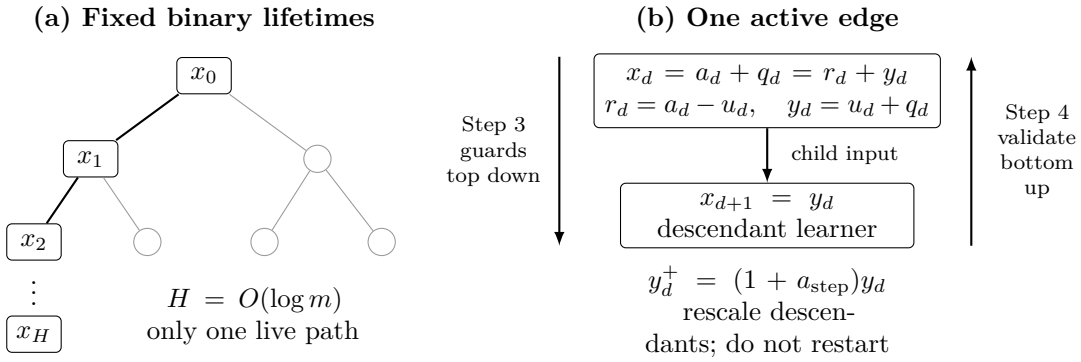


Figure 2: The live path in the time tree. Step 3 injects orthogonal guards from the root downward. Step 4 validates from the leaves upward. A correction changes a child's input by a scalar, so all descendant states persist.

The anchor is a_d, the subsequent input is q_d, and u_d is the learned correction. Set $E_d = \|r_d\|_2^2$ and $Q_d = \|y_d\|_2^2$. The identity $\|x_d\|_2^2 = E_d + Q_d + 2\langle r_d, y_d\rangle$ explains both the prediction and its repair: $P + \widehat{Q}$ estimates the first two terms, while a rejection detects a large signed cross term.

Specifically, a rejection certifies, for a direction $s_{\text{dir}} \in \{-1, +1\}$,

$$2s_{\text{dir}}\langle r_d, y_d\rangle \geq \frac{\eta}{2}(E_d + Q_d).$$

The locked step $u_d^+ = u_d + a_{\text{step}} y_d$ then satisfies

$$E_d^+ \leq (1 - \eta^2/64)E_d, \qquad y_d^+ = (1 + a_{\text{step}})y_d.$$

The learner transfers $a_{\text{step}} y_d$ from the residual to the child, preserving their sum. More than preserving the sum, it changes the child's entire input homogeneously. A tag $c = 1 + a_{\text{step}} \in [1/2, 3/2]$ rescales all descendants in place: linear counters scale by c, and quadratic caches by c^2. No child restarts in the middle of its lifetime (Lemma 6.3).

Only the active path occupies space. Contraction bounds the charged rejections within a lifetime; fresh banks are created at scheduled births, not at every failed prediction.

3. Establish scale floors before using contraction. A zero vector has no relative error budget, and a tiny child input can demand excessive precision. We address both before bottom-up validation. A homomorphic degree-n fingerprint modulo a prime q detects zero physical rounds using $O(\Lambda)$ bits (Algorithm 3 and Lemma 6.1). On its success event, every active round has $\left\|x_0^{(t)}\right\|_2 \geq 1$. Its analysis bounds the first erroneous zero decision, so it remains valid against adaptive updates.

Next, Step 3 injects top-down orthogonal guards $\delta_{g,d} e_{g(d,t)}$. These propagate $\|x_d\|_2 \geq \nu$ before Step 4 begins. The resulting two-step rejection floor $\min(E, Q) \geq E_{\min}$ is non-circular: it does not assume the contraction whose validity it must establish (Lemma 6.4 and Theorem 6.5).

We also need to control products of rescaling tags. Evaluate the guard argument at the first local correction t_* of an interval. If $\mathcal{U}_{d,k}$ is its stored energy scale and $C_d(t_*)$ is its ancestor tag, then

$$C_d(t_*)\sqrt{\mathcal{U}_{d,k}} \geq \|r_d(t_*)\|_2 \geq (\eta/16)\nu.$$

Expressing both sides in physical units cancels the ancestor tag from the scale comparison. Theorem 6.6 then bounds every forward interval product by $B = 16/\gamma_g = \Theta(\sqrt{T})$, where T is the locked event horizon. We control forward products from an interval's birth, not inverse products back to time zero.

4. Bound precision on certified prefixes and space on every execution. Repeated rescaling can amplify a tiny rounding error. On a prefix whose forward products and target norms satisfy the certified bounds, we compare the rounded execution with a shadow vector following that same sequence of decisions. Forward unnormalized counters avoid division by small accumulated tags. Their pathwise drift is at most

$$\xi = G(T+1)h_b, \qquad G = B^{H+2},$$

where h_b is the counter grid.

Residual caches need a different argument. Between local corrections, guard energy is nonnegative: $\delta_{g,d}^2 \geq 0$. If Π_{anc} is the intervening ancestor product, then $\Pi_{\mathrm{anc}}^2 E_{t_0} \leq E_t$. Thus an initial relative cache error remains a relative error after rescaling; it does not become an absolute error charged afresh at every update. A grid $h_P \propto \eta^3\nu^2$ suffices (Lemma 6.7).

Although G can be large, its logarithm is only polynomial in Λ. The locked grids keep every register within $O(\Lambda^3)$ bits. Combining this precision with the samples in the live banks gives $O(\varepsilon^{-2}\Lambda^{15})$ bits—with certified upper bound $2^{512}\varepsilon^{-2}\Lambda^{15}$ bits—across the active path (Lemma A.7). Explicit counter saturation, scalar range checks, hard correction/request caps, and bounded samplers hold even on executions where a statistical test fails. Only accuracy is a high-probability claim; the space bound is unconditional.

3 Preliminaries

Fixed-query concentration is not enough for an adaptive interaction. We use privacy to control the transcript, pathwise regret to charge corrections, and martingale bounds to analyze the remaining randomness. We record the versions and constants used later.

We write ln for the natural logarithm and $\log_2$ for the binary logarithm. For a distribution D, write $Dq = \mathbb{E}_D[q]$. All relevant maps are jointly measurable. In the standard tools below, r denotes a generic sample size; n, m, U remain the stream dimension, length, and coordinate bound. An integer turnstile stream starts at $x^{(0)} = 0$ and applies updates $x^{(t)} = x^{(t-1)} + \Delta_t e_{i_t}$, with $\Delta_t \in \mathbb{Z}$. The adversary chooses the next update after seeing the preceding outputs. We use $F_2(x) = \|x\|_2^2$.

Definition 3.1 (Single-Record Replacement Differential Privacy)**.** For $S, S' \in X^r$, write $S \sim_1 S'$ when they differ in at most one coordinate. A randomized mechanism $\mathcal{A}$ is (ε, δ)-differentially private if, for every such pair and every measurable output event E,

$$\mathbb{P}[\mathcal{A}(S) \in E] \leq e^{\varepsilon}\mathbb{P}[\mathcal{A}(S') \in E] + \delta.$$

For an interactive mechanism, the output is its full public transcript. Any measurable randomized post-processing independent of the private sample preserves this guarantee.

Randomized analysts and adversaries have an independent coin space. A bound proved uniformly for every fixed choice of their coins extends to randomized strategies by Tonelli–Fubini. We use this reduction for both privacy and failure probabilities.

Lemma 3.2 (Homogeneous Advanced Composition [19, 18])**.** *For $M \geq 1$, an M-fold adaptive composition of $(\varepsilon_0, 0)$-DP mechanisms is (ε, δ')-DP for every $\delta' \in (0,1)$, where*

$$\varepsilon = \varepsilon_0\sqrt{2M\ln(1/\delta')} + M\varepsilon_0(e^{\varepsilon_0} - 1).$$

In the small-failure regime $\delta' \leq 1/2$ used here, it is $(\tilde{\varepsilon}, \delta')$-DP whenever $\tilde{\varepsilon} \leq 1$ and $\varepsilon_0 \leq \tilde{\varepsilon}/(2\sqrt{2M\ln(1/\delta')})$.

More generally, along a chain of C hybrids from $Y(S)$ to $Y(S')$, if comparison r is $(\varepsilon_r, \delta_r)$-DP, then

$$\mathbb{P}[Y(S) \in E] \le e^{\sum_{r=1}^{C} \varepsilon_r} \mathbb{P}[Y(S') \in E] + \sum_{r=1}^{C} e^{\sum_{j<r} \varepsilon_j} \delta_r.$$

Lemma 3.3 (Sparse Vector AboveThreshold [18])**.** *One episode of* AboveThreshold *tests adaptive Δ-sensitive queries against $\tau + \rho$, where $\rho \sim \mathrm{Lap}(b)$ and the independent per-query noise is $\nu_i \sim \mathrm{Lap}(2b)$. Stopping at the first crossing is $(2\Delta/b, 0)$-DP. An optional Δ-sensitive scalar release with independent $\mathrm{Lap}(b)$ noise makes the episode $(3\Delta/b, 0)$-DP.*

On the lattice $h_A\mathbb{Z}$, let

$$\mathbb{P}[Z_{b,h_A} = k h_A] = \frac{1-\varrho}{1+\varrho}\varrho^{|k|}, \qquad \varrho = \frac{b}{b+h_A}.$$

For $z, s \in h_A\mathbb{Z}$ and $u \ge 0$, $\mathbb{P}[Z = z] \le e^{|s|/b}\mathbb{P}[Z = z + s]$ and $\mathbb{P}[|Z| \ge u] \le 2e^{-u/(b+h_A)}$. A two-sided directed AboveThreshold *phase with threshold scale $2b$, per-branch query-noise scale $4b$, and Δ-sensitive real statistics ($|v_i(S) - v_i(S')| \le \Delta$) is $(\Delta'/b, 0)$-DP for any rounded sensitivity bound $\Delta' \in h_A\mathbb{Z}$ with $\Delta' \ge \Delta$; neither $v_i(S)$ nor τ need lie in $h_A\mathbb{Z}$. Likewise, for $Z \sim Z_{b,h_A}$ and $\mathrm{round}_{h_A}(z) = h_A\lfloor z/h_A + 1/2\rfloor$, $\mathrm{round}_{h_A}(\mu + Z) = \mathrm{round}_{h_A}(\mu) + Z$ is $(\Delta'/b, 0)$-DP with $\Delta' = h_A\lceil \Delta/h_A \rceil$.*

To check the directed phase claim, order the two branches deterministically and fix a transcript whose crossing branch is $*$. The following map on the lattice noise values is injective:

$$\begin{aligned}
\rho' = \rho + \Delta', \quad \nu'_* = \nu_* + 2\Delta', \quad \nu'_i = \nu_i \quad (i \ne *);\\
v_i(S') + \nu'_i \le v_i(S) + \nu_i + \Delta' < \tau + \rho' \qquad (i < *);\\
v_*(S') + \nu'_* \ge v_*(S) + \nu_* + \Delta' \ge \tau + \rho';\\
\frac{\mathbb{P}[\rho, \nu]}{\mathbb{P}[\rho', \nu']} \le \exp\left(\frac{\Delta'}{2b} + \frac{2\Delta'}{4b}\right) = e^{\Delta'/b}.
\end{aligned}$$

For a transcript with no crossing, omit the crossing-noise shift.

Lemma 3.4 (Concurrent Pure-DP Composition [35])**.** *Concurrent adaptive interleaving of C independently randomized pure-DP interactive mechanisms with parameters $\varepsilon_1, \ldots, \varepsilon_C$ on the same dataset is $(\sum_{r=1}^{C} \varepsilon_r, 0)$-DP. Concurrent homogeneous pure-DP phases also obey homogeneous advanced composition. In particular, for $\theta \in (0, 1]$ and $\delta' \in (0, 1/2]$, C phases are (θ, δ')-DP whenever*

$$\varepsilon_0 \le \frac{\theta}{\sqrt{8C \ln(1/\delta')}}.$$

This applies to concurrent AboveThreshold *streams at the root.*

Lemma 3.5 (Martingale Tail Bounds [27, 3, 20])**.** *If $Y \in [a, b]$, then Hoeffding's lemma gives $\mathbb{E}[e^{\lambda(Y - \mathbb{E}Y)}] \le e^{\lambda^2(b-a)^2/8}$ for every $\lambda \in \mathbb{R}$.*

Let (D_i) be martingale differences with respect to $(\mathcal{F}_i)$. Azuma–Hoeffding gives

$$\mathbb{P}\left[\left|\sum_{i=1}^{r} D_i\right| \ge t\right] \le \begin{cases} 2\exp\left(-\dfrac{t^2}{2\sum_{i=1}^{r} c_i^2}\right), & |D_i| \le c_i, \\ 2\exp\left(-\dfrac{2t^2}{rw^2}\right), & \text{each conditional range has width at most } w. \end{cases}$$

For Freedman's inequality, suppose $D_i \le b$ almost surely and set $V_k = \sum_{i=1}^{k} \mathbb{E}[D_i^2 \mid \mathcal{F}_{i-1}]$. For every $L_1 > 0$ and $\sigma^2 > 0$,

$$\mathbb{P}\left[\sum_{i=1}^{k} D_i \ge \sqrt{2\sigma^2 L_1} + \frac{2}{3} b L_1 \text{ and } V_k \le \sigma^2\right] \le e^{-L_1}.$$

Lemma 3.6 (Bhatia–Davis Variance Inequality [8])**.** *For $v_1, \dots, v_r \in [m_-, m_+]$, with $\bar{v} = r^{-1} \sum_{i=1}^{r} v_i$,*

$$\frac{1}{r} \sum_{i=1}^{r} (v_i - \bar{v})^2 \le (m_+ - \bar{v})(\bar{v} - m_-).$$

The next bounds are deterministic. They apply to the realized sequence of adaptive queries and signed updates; no independence assumption is needed.

Lemma 3.7 (Entropic and Quadratic FTRL Regret [33])**.** *Let Δ_r be the probability simplex on r indexed records.*

(a) Entropic case. *For an indexed sample S, define $\mathcal{K}_2(S) = \{P \in \Delta_r : P(\ell) \le 2/r\}$. For $q_i(S) \in [0,1]^r$ and $\delta_i \in \mathbb{R}$, let*

$$P_{(i)} = \operatorname*{arg\,min}_{P \in \mathcal{K}_2(S)} \left\{ -\left\langle P, \sum_{j \le i} \delta_j q_j(S) \right\rangle + D_{\mathrm{KL}}(P \,\|\, \mathrm{Unif}(S)) \right\}.$$

For every $Q \in \mathcal{K}_2(S)$,

$$\sum_{i=1}^{k} (-\delta_i)(\left\langle P_{(i-1)}, q_i(S) \right\rangle - \langle Q, q_i(S) \rangle) \le D_{\mathrm{KL}}(Q \,\|\, \mathrm{Unif}(S)) + \frac{1}{8} \sum_{i=1}^{k} \delta_i^2.$$

The constant $1/8$ is the range-width-1 Hoeffding stability constant. Scaling the regularizer by $1/\eta$, for $\eta > 0$, changes the right side to $D_{\mathrm{KL}}(Q \,\|\, \mathrm{Unif}(S))/\eta + (\eta/8) \sum_i \delta_i^2$.

(b) Quadratic case. *Let $W_r = \{w \in \Delta_r : w_i \le 8/r\}$ and $R_r(w) = \frac{r}{2} \sum_{i=1}^{r} (w_i - 1/r)^2$. This regularizer is 1-strongly convex in $\|\cdot\|_1$. Let $w(G)$ maximize $\kappa \langle G, w \rangle - R_r(w)$ over W_r. For every $u \in W_r$ and $g \in \mathbb{R}^r$, the potential*

$$\begin{aligned} \Psi(G) &:= R_r(u) + \max_{w \in W_r} \{\kappa \langle G, w \rangle - R_r(w)\} - \kappa \langle G, u \rangle \\ &\ge \frac{1}{2} \|u - w(G)\|_1^2 \ge 0 \end{aligned}$$

satisfies

$$\Psi(G + g) - \Psi(G) \le -\kappa \langle u - w(G), g \rangle + \frac{\kappa^2}{2} \|g\|_\infty^2.$$

Lemma 3.8 (2s-Wise Independent Rademacher Families and Coarse AMS [10, 1, 5, 2])**.** *For $D_* \ge 1$ and integer $s \ge 2$, set $w = \lceil \log_2(D_* + 1) \rceil$. Evaluate a uniformly random polynomial of degree less than $2s$ over $\mathbb{F}_{2^w}$ at D_* distinct nonzero points, apply a nonzero $\mathbb{F}_2$-linear map, and identify its two outputs with signs. This uses $2sw$ seed bits and gives a $2s$-wise independent vector $b \in \{-1, +1\}^{D_*}$.*

For every $v \in \mathbb{R}^{D_*}$, the random variable $Z = \langle b, v \rangle$ satisfies

$$\begin{aligned} \mathbb{E}[Z^2] &= \|v\|_2^2, \qquad \mathbb{E}[Z^4] \le 3 \|v\|_2^4, \\ \mathbb{E}[Z^{2s}] &\le \frac{(2s)!}{2^s s!} \|v\|_2^{2s} \le (2s)^s \|v\|_2^{2s}. \end{aligned}$$

Moreover, the median of g independent empirical means, each formed from 32 independent rows' values of Z^2, belongs to $[\frac{9}{16} \|v\|_2^2, \frac{23}{16} \|v\|_2^2]$ except with probability at most $e^{-g/18}$.

Finally, a polynomial fingerprint needs only the root bound: a nonzero univariate polynomial of degree at most d over $\mathbb{F}_q$ vanishes at a uniform field element with probability at most d/q [32, 37]. In the adaptive zero test, we apply this bound to the first erroneous decision, rather than condition on a transcript that already contains an error.

4 Adaptive Data Analysis Without $\log k$ Factors on $\sqrt{k}$

Multi-level validation hierarchies face a tension between leading-term privacy loss and pass-drift accumulation. Perturbing a working sample with unbounded Gaussian or Laplace noise introduces a $\sqrt{\ln k}$ union-bound factor on $\sqrt{k}$, whereas testing proposals sequentially across $L = \Theta(\ln(1/\alpha))$ validators and updating only the triggered validator lets earlier validators pass with drift near their thresholds, forcing thresholds to double per level and paying $2^{\Theta(L)}$ samples. Our equal-impulse cascade $\mathcal{M}_{\mathrm{cas}}$ resolves both issues: Stage 1 adds bounded symmetric Beta noise [13] so its privacy is independent of L and $\ln k$, while any final trigger at stage $r^* \geq 3$ applies opposite entropic-FTRL updates of equal capacity-weighted impulse to levels $r^* - 1$ and r^*, canceling the unknown mean μ_i identically.

Fix an integer $k \geq 1$, $\alpha \in (0, 1/16)$, $\beta \in (0, 1/2)$, a measurable space $(X, \mathcal{F}_X)$, and a distribution D on X. Adaptive queries $q_1, \ldots, q_k : X \to [0,1]$ have population means $\mu_i := \mathbb{E}_{X\sim D}[q_i(X)] \in [0,1]$. We partition $N_{\mathrm{cas}} = n_1 + \sum_{r=2}^{L-1} n_r + n_L$ i.i.d. draws from D into blocks $S_r = (X_{r,1}, \ldots, X_{r,n_r}) \sim D^{n_r}$ $(1 \leq r \leq L)$ and maintain weights on the 2-capped simplex $\mathcal{K}_2(S_r) := \{P \in \mathbb{R}^{n_r} : 0 \leq P(j) \leq 2/n_r,\ \sum_{j=1}^{n_r} P(j) = 1\}$. By the Tonelli–Fubini reduction (Definition 3.1), we fix the analyst's independent coin tape so q_i and μ_i are deterministic measurable functions of the public output prefix $(a_j, \tau_{\mathrm{aux},j})_{j<i}$; averaging over those coins yields all claims for randomized analysts. Write $\Pi = (\Pi_1, \ldots, \Pi_k)$ with $\Pi_i = (a_i, \tau_{\mathrm{aux},i})$ for the analysis transcript; in the implementation, Π_i is streamed to a write-only output rather than retained in memory.

4.1 The Cascade Mechanism and Water-Filling Sensitivity

Parameters. The full locked schedule appears in Definition A.1 (Section A.1); we record all formulas needed below. Set depth $L := \max(4, \lceil \ln(1/\alpha) \rceil + 2) \geq 5$, margin $\gamma := \alpha/(8L)$, and noise unit $u_0 := \gamma/32 = \alpha/(256L)$. Stage 1 uses $\varepsilon_1 := 0.05$, $\delta_1 := \alpha\beta/10240$, $L_1 := \ln(2/\delta_1) = \ln(20480/(\alpha\beta)) > 13.39$, $R_1 := \alpha/32$, $\eta_1 := \alpha/16$, $T_{1,\max} := \lceil 600/\alpha^2 \rceil$, $(\chi_1, p_1) := (\sqrt{L_1}, 8)$ if $\beta \geq \alpha^5$ and $(L_1, \lceil L_1 \rceil + 1)$ if $\beta < \alpha^5$, and $n_1 := \lceil 32000\sqrt{k}\chi_1/\alpha + 16000 L_1^3/\alpha^2 \rceil$. Validators $(2 \leq r < L)$ and the anchor $(r = L)$ use

$$\varepsilon_r := 0.05\,\alpha^{(r-1)/(L-1)}\ (1 \leq r < L), \qquad \rho := \frac{\varepsilon_{r-1}}{\varepsilon_r} = \alpha^{-1/(L-1)} \in (e^{3/5}, e)\ (2 \leq r < L),$$

$$\varepsilon_L := \frac{u_0}{2}, \qquad r_{\mathrm{anc}} := 4\varepsilon_L = 2u_0, \qquad \delta_{\mathrm{val}} := \frac{u_0^3\beta}{10^8 L^3}, \qquad L_{\mathrm{val}} := \ln\left(\frac{4}{\delta_{\mathrm{val}}}\right) > 25.$$

Pairwise thresholds are $\tau_{(r,r)} := 0$, $\tau_{(1,r)} := \alpha/2 + 2(r-2)\gamma$ $(2 \leq r \leq L)$, and $\tau_{(u,r)} := 2(r-u)\gamma$ $(2 \leq u < r \leq L)$; thus $\tau_{(u,r^*)} - \tau_{(u,r^*-1)} = 2\gamma$ for $1 \leq u < r^* \leq L$ with $r^* \geq 3$, and $\tau_{(u,L)} + 4u_0 + r_{\mathrm{anc}} \leq \tau_{(1,L)} + 6u_0 < \frac{3}{4}\alpha + \frac{6\alpha}{256L} < \alpha$. Set $w_\ell := \varepsilon_\ell^{-2}$ and $\eta_\ell := \gamma/8$ $(2 \leq \ell < L)$, and $\tilde{\eta}_{r^*} := (w_{r^*-1}/w_{r^*})\eta_{r^*-1} = \rho^{-2}\gamma/8$ for $3 \leq r^* < L$ $(\tilde{\eta}_L := 0)$, giving $8w_{r^*-1}\eta_{r^*-1} = 8w_{r^*}\tilde{\eta}_{r^*} = w_{r^*-1}\gamma$ for $3 \leq r^* < L$. Finally, set $N_{r^*,\max} := \lceil 128 L \varepsilon_{r^*-1}^2/\gamma^2 \rceil$ $(3 \leq r^* \leq L)$, $M_{2,\max} := 1 + T_{1,\max} + \sum_{s=3}^{L} N_{s,\max}$, $M_{r,\max} := 1 + \sum_{s=r}^{L} N_{s,\max}$ $(3 \leq r \leq L)$, $b_{\mathrm{SVT}} := u_0/\ln(64LkM_{2,\max}/\delta_{\mathrm{val}})$, and

$$n_r := \left\lceil \frac{110\sqrt{M_{r,\max} L_{\mathrm{val}}}}{b_{\mathrm{SVT}}\,\varepsilon_r} \right\rceil \quad (2 \leq r < L), \qquad n_L := \left\lceil \frac{55\sqrt{M_{L,\max} L_{\mathrm{val}}}}{b_{\mathrm{SVT}}\,\varepsilon_L} + \frac{16\ln(6k/\delta_{\mathrm{val}})}{r_{\mathrm{anc}}\,\varepsilon_L} \right\rceil.$$

Algorithm 1 Equal-Impulse Coupled Entropic-FTRL Cascade $\mathcal{M}_{\mathrm{cas}}$

1: **Initialize:** Partition N_{cas} draws into $S_1, \dots, S_L$. Set $P^{(r)} \leftarrow \mathrm{Unif}(S_r)$ $(1 \le r < L)$, $P^{(L)} \equiv \mathrm{Unif}(S_L)$, $G_0^{(1)} \leftarrow 0$, $L^{(\ell)} \leftarrow 0$ $(2 \le \ell < L)$, $t_1 \leftarrow 0$, $N_{r^*} \leftarrow 0$ $(3 \le r^* \le L)$, $M_r \leftarrow 1$ and $\rho^{(r)} \sim \mathrm{Lap}(b_{\mathrm{SVT}})$ $(2 \le r \le L)$, and $\mathsf{HALT} \leftarrow \mathsf{false}$.

2: **Round** $i = 1, \dots, k$**:** If $\mathsf{HALT} = \mathsf{true}$, emit $a_i := 0$ and the inactive auxiliary record $\tau_{\mathrm{aux},i}$ ($\mathsf{active} = 0$, $\delta_{\ell,i} := 0$, all fields zero), and continue to round $i+1$ without accessing S. Otherwise evaluate $\theta_i^{(r)}(S_r) := \langle P^{(r)}, q_i(S_r) \rangle$ for $1 \le r \le L$.

3: **Step 1 (Proposal):** Draw $B_i \sim \mathrm{Beta}(p_1, p_1)$, $\xi_i := R_1(2B_i - 1)$, and $c_i^{(1)} := \mathrm{clip}(\theta_i^{(1)}(S_1) + \xi_i, [0,1])$.

4: **Step 2 (Trigger-and-replace chain):** Set $u \leftarrow 1$. For $r = 2, \dots, L$, draw $\nu_i^{(r)} \sim \mathrm{Lap}(2b_{\mathrm{SVT}})$ and test $\left| c_i^{(u)} - \theta_i^{(r)}(S_r) \right| + \nu_i^{(r)} \ge \tau_{(u,r)} + \rho^{(r)}$. If false (PASS), record $b_i^{(r)} := 0$. If true (TRIGGER), record $b_i^{(r)} := 1$ and $u_{\mathrm{prev}}(r) := u$, draw $\zeta_i^{(r)} \sim \mathrm{Lap}(b_{\mathrm{SVT}})$, release $v_i^{(r)} := \theta_i^{(r)}(S_r) + \zeta_i^{(r)}$ and $c_i^{(r)} := \mathrm{clip}(v_i^{(r)}, [0,1])$, and set $u \leftarrow r$. Set $u_{\mathrm{final}} := u$ and output $a_i := c_i^{(u_{\mathrm{final}})}$.

5: **Step 3 (Updates and resets):** Set $\delta_{\ell,i} \leftarrow 0$ for all $2 \le \ell < L$.

- *Case 0 ($u_{\mathrm{final}} = 1$):* Perform no updates or resets.
- *Case A ($u_{\mathrm{final}} = 2$):* Set $t_1 \leftarrow t_1 + 1$, $s_{t_1}^{(1)} := \mathrm{sign}(c_i^{(1)} - a_i)$, $G_{t_1}^{(1)} := G_{t_1-1}^{(1)} - s_{t_1}^{(1)} q_i(S_1)$, and $P^{(1)} \leftarrow P^*(G_{t_1}^{(1)})$ at η_1. Set $M_2 \leftarrow M_2 + 1$, redraw $\rho^{(2)} \sim \mathrm{Lap}(b_{\mathrm{SVT}})$, and mark halt if $t_1 = T_{1,\max}$.
- *Case B ($u_{\mathrm{final}} = r^* \in \{3, \dots, L\}$):* (B1) If $u_{\mathrm{prev}}(r^*) < r^* - 1$, draw $\zeta_i^{(r^*-1)} \sim \mathrm{Lap}(b_{\mathrm{SVT}})$ and release $v_i^{(r^*-1)} := \theta_i^{(r^*-1)}(S_{r^*-1}) + \zeta_i^{(r^*-1)}$ and $c_i^{(r^*-1)} := \mathrm{clip}(v_i^{(r^*-1)}, [0,1])$; else reuse its Step-2 release. (B2) Set $s_{i,r^*} := \mathrm{sign}(c_i^{(u_{\mathrm{prev}}(r^*))} - c_i^{(r^*)})$, $\delta_{r^*-1,i} := -\eta_{r^*-1} s_{i,r^*}$, and (if $r^* < L$) $\delta_{r^*,i} := +\tilde{\eta}_{r^*} s_{i,r^*}$; for $\ell \in \{r^* - 1, r^*\} \cap \{2, \dots, L-1\}$, set $L^{(\ell)} \leftarrow L^{(\ell)} + \delta_{\ell,i} q_i(S_\ell)$ and $P^{(\ell)} \leftarrow \arg\min_{P \in \mathcal{K}_2(S_\ell)} \{ \langle P, -L^{(\ell)} \rangle + D_{\mathrm{KL}}(P \,\|\, \mathrm{Unif}(S_\ell)) \}$. (B3) Set $N_{r^*} \leftarrow N_{r^*} + 1$; for $2 \le s \le r^*$, set $M_s \leftarrow M_s + 1$ and redraw $\rho^{(s)} \sim \mathrm{Lap}(b_{\mathrm{SVT}})$; mark halt if $N_{r^*} = N_{r^*,\max}$.

6: **Emit and check halt:** Stream $\Pi_i = (a_i, \tau_{\mathrm{aux},i})$—recording $\mathsf{active} = 1$, $c_i^{(1)}$, trigger bits $b_i^{(r)}$, predecessor indices $u_{\mathrm{prev}}(r)$, release flags and values $(v_i^{(r)}, c_i^{(r)})$, discrepancy sign, impulses $\delta_{\ell,i}$, and post-round halt flag (`ADAField`, from which counters are derived; never raw noises or $\theta_i^{(r)}$)—and if halt was marked, set $\mathsf{HALT} \leftarrow \mathsf{true}$.

Write $\mathrm{sign}(x) := +1$ for $x \ge 0$ and -1 for $x < 0$, and $\mathrm{clip}(x, [0,1]) := \min(1, \max(0, x))$.

Three invariants hold on every run of Algorithm 1: (i) each level receives at most one nonzero impulse per round; (ii) whenever validator $r \ge 2$ triggers ($r \le u_{\mathrm{final}}$) or updates ($r \in \{r^* - 1, r^*\}$ in Case B), stage r lies in $\{2, \dots, u_{\mathrm{final}}\}$ and resets its episode at round end, so $P^{(r)}$ is strictly constant within every active episode; and (iii) at most one counter among $t_1, N_3, \dots, N_L$ increments by 1 per round before the halt check, so $t_1 \le T_{1,\max}$, $N_{r^*} \le N_{r^*,\max}$, and $M_r \le M_{r,\max}$, with any halting round fully recorded in Π.

Conditioned on $\Pi_{<i}$, all past queries and scalar weights forming $G(S_r)$ are fixed. Replacing one sample point j_0 alters only $G(j_0)$, so water-filling normalization shifts all other coordinates monotonically in the same direction and keeps the ℓ_1 sensitivity at $4/n_r$ independently of the number of updates.

Lemma 4.1 (Water-Filling Sensitivity and Linear-Time Projection on $\mathcal{K}_2(S_r)$)**.** *For $n_r \ge 2$, $\eta_r > 0$, and $G \in \mathbb{R}^{n_r}$, the objective $P \mapsto \langle P, -G \rangle + \frac{1}{\eta_r} D_{\mathrm{KL}}(P \,\|\, \mathrm{Unif}(S_r))$ over $\mathcal{K}_2(S_r)$ (with $\eta_r G = L^{(r)}$ on validators) has the unique minimizer $P^*(G)(j) = \min(2/n_r, e^{\eta_r G(j) - \psi(G)})$, where $\psi(G) \in \mathbb{R}$ is the unique root of $\sum_{j=1}^{n_r} \min(2/n_r, e^{\eta_r G(j) - \psi}) = 1$. Starting from the preloaded public input $(k, \alpha, \beta, L, p_1, n_1, \dots, n_L)$, computing $P^*(G)$ takes $O(n_r)$ deterministic operations, giving $O(N_{\mathrm{cas}})$ operations at initialization and per round, $O(kN_{\mathrm{cas}})$ total time, and $O(N_{\mathrm{cas}})$ workspace cells in the*

uniform rational-literal real-RAM oracle model (storing only current score vectors and counters while streaming Π_i). Conditioned on the public prefix, for any neighbors $S_r \sim_1 S'_r$,

$$\left\|P^*(G(S_r)) - P^*(G(S'_r))\right\|_1 \le \frac{4}{n_r} \quad (r < L), \qquad \left|\theta^{(r)}(S_r) - \theta^{(r)}(S'_r)\right| \le \begin{cases} 4/n_r & (r < L), \\ 1/n_L & (r = L). \end{cases}$$

Proof. Compactness of $\mathcal{K}_2(S_r)$ ($\mathrm{Unif}(S_r) \in \mathcal{K}_2(S_r)$ for $n_r \ge 2$), strict convexity of $x \mapsto x \ln x$, and infinite slope at 0 yield a unique positive minimizer P^*. With Lagrange multiplier λ for $\sum_j P(j) = 1$ and KKT multipliers $\nu_j \ge 0$ for $P(j) \le 2/n_r$, stationarity gives $-G(j) + \frac{1}{\eta_r}(\ln(n_r P^*(j)) + 1) + \lambda + \nu_j = 0$ and $\nu_j(2/n_r - P^*(j)) = 0$. Setting $\psi := \eta_r\lambda + 1 + \ln n_r$ yields $P^*(j) = \min(2/n_r, e^{\eta_r G(j) - \psi})$. Since $\Phi(\psi) := \sum_j \min(2/n_r, e^{\eta_r G(j) - \psi})$ decreases strictly from 2 to 0 where $\Phi \in (0, 2)$, $\psi(G)$ is unique. Writing $a_j := e^{\eta_r G(j)}$ and $t := e^{-\psi}$, median-of-medians selection on the breakpoints $2/(n_r a_j)$ tests $\sum_j \min(2/n_r, t_{\mathrm{med}} a_j)$ against 1 and halves the active set each pass while maintaining capped counts and uncapped slopes, finding t in $O(n_r)$ operations. Thus from `AEff.adaInput`, each round uses $O(\sum_{r=1}^{L} n_r) = O(N_{\mathrm{cas}})$ operations and $O(N_{\mathrm{cas}})$ workspace cells with rational program literals and unit-cost query, noise-table, and $e^x, \ln x, \sqrt{x}$ calls, matching `Theorem_A_eff` in Section 7.

If $S_r \sim_1 S'_r$ ($r < L$) differ at j_0, fixed public weights give $G(S_r)(j) = G(S'_r)(j)$ for $j \ne j_0$. Monotonicity of $z \mapsto \min(2/n_r, e^{\eta_r G(j) - z})$ gives $P(j) - P'(j)$ a common sign across all $j \ne j_0$, so $\sum_{j \ne j_0} |P(j) - P'(j)| = \left|\sum_{j \ne j_0}(P(j) - P'(j))\right| = |P(j_0) - P'(j_0)| \le 2/n_r$ and $\|P - P'\|_1 \le 4/n_r$. For $q : X \to [0, 1]$, centering $\bar{q} := q - 1/2 \in [-1/2, 1/2]$ gives $|\langle P, q(S_r)\rangle - \langle P', q(S'_r)\rangle| \le \frac{1}{2}\sum_{j \ne j_0} |P(j) - P'(j)| + \frac{1}{2}(P(j_0) + P'(j_0)) \le 2/n_r \le 4/n_r$; uniform weights on S_L give $1/n_L$. □

4.2 Depth-Decoupled Martingale Privacy on S_1 and Validator Clean Event

Stage 1 perturbs $\theta_i^{(1)}(S_1)$ by $\xi_i = R_1(2B_i - 1)$ with $B_i \sim \mathrm{Beta}(p_1, p_1)$ [13], whose density vanishes to order $q_B = p_1 - 1$ at $\pm R_1$. Excising an interior collar of width $\kappa \propto dL_1$ bounds every step's truncated privacy loss by $O(1/L_1)$ with zero k-dependence, while two-sided Young cancellation between the $\sqrt{k}$ and L_1^3/α^2 terms of n_1 absorbs the k-round collar union bound for all $k \ge 1$.

Lemma 4.2 (Stage-1 Bounded-Noise Martingale Privacy)**.** *Under single-coordinate replacement $S_1 \sim_1 S'_1$, $S_1 \mapsto \Pi$ is $(\varepsilon_1 = 0.05, \delta_1 = \alpha\beta/10240)$-DP independently of L, η_1, and $T_{1,\max}$, and every active round satisfies $\left|c_i^{(1)} - \theta_i^{(1)}(S_1)\right| < R_1 = \alpha/32$.*

Proof. Since $\theta_i^{(1)}(S_1) \in [0, 1]$ and $\mathrm{clip}(\cdot, [0, 1])$ is 1-Lipschitz, $\left|c_i^{(1)} - \theta_i^{(1)}(S_1)\right| \le |\xi_i| < R_1$ on active rounds.

Set $q_B := p_1 - 1 \ge 7$. Then $V = 2B - 1 \in (-1, 1)$ has density $f_{q_B}(v) = Z_{q_B}^{-1}(1 - v^2)^{q_B}\mathbf{1}_{(-1,1)}(v)$. Integrating $[v(1 - v^2)^a]_{-1}^{1} = 0$ by parts gives $Z_{a-1}/Z_a = (2a + 1)/(2a)$ (so $Z_7 = 2\prod_{a=1}^{7}\frac{2a}{2a+1} = \frac{4096}{6435}$ and $8Z_7 > 5.09$), while $Z_{q_B} \ge 2\int_0^1 v(1 - v^2)^{q_B}\,dv = \frac{1}{q_B+1}$. Thus $\mathbb{E}[(1 - V^2)^{-1}] = \frac{2q_B+1}{2q_B}$, $\mathbb{E}[(1 - V^2)^{-2}] = \frac{4q_B^2-1}{4q_B(q_B-1)}$, and Fisher information is $\mathcal{I}_{q_B} := \mathbb{E}[(\frac{2q_B V}{1-V^2})^2] = 4q_B^2(\frac{4q_B^2-1}{4q_B(q_B-1)} - \frac{2q_B+1}{2q_B}) = \frac{q_B(2q_B+1)}{q_B-1}$ ($\mathcal{I}_7 = 17.5$). For $\kappa \in (0, 1)$, setting $v_0 := \sqrt{1 - \kappa}$ and using $1 \le v/v_0$ on $[v_0, 1)$ yields

$$\mathbb{P}[1 - V^2 < \kappa] = \frac{2}{Z_{q_B}}\int_{v_0}^{1}(1 - v^2)^{q_B}\,dv \le \frac{2}{Z_{q_B}v_0}\int_{v_0}^{1} v(1 - v^2)^{q_B}\,dv = \frac{\kappa^{q_B+1}}{(q_B+1)Z_{q_B}\sqrt{1-\kappa}} \le \frac{\kappa^{q_B+1}}{\sqrt{1-\kappa}}.$$

Fix $(S_2, \ldots, S_L)$ and validator/analyst coins, and let $S_1 \sim_1 S'_1$. Define $\tilde{a}^{(1)} = (\tilde{a}_1^{(1)}, \ldots, \tilde{a}_k^{(1)})$ by $\tilde{a}_i^{(1)} := \theta_i^{(1)}(S_1) + R_1 V_i$ on active rounds and $1/2$ after halting. By induction on i, Π is a deterministic

post-processing of $\tilde{a}^{(1)}$ (Definition 3.1), and given $\mathcal{F}_{i-1} := \sigma(\tilde{a}_1^{(1)}, \dots, \tilde{a}_{i-1}^{(1)})$, Lemma 4.1 bounds the predictable shift $u_i := (\theta_i^{(1)}(S_1) - \theta_i^{(1)}(S_1'))/R_1$ (0 after halting) by $|u_i| \le \frac{4/n_1}{\alpha/32} = \frac{128}{\alpha n_1} =: d$, with $V_i \sim f_{q_B}(\cdot + u_i)$ under S_1'.

Set $\kappa := 64 q_B L_1 d = \frac{8192 q_B L_1}{\alpha n_1}$ and $\mathcal{G}_k := \bigcap_{i=1}^k G_i$ with $G_i := \{1 - V_i^2 \ge \kappa\}$ ($V_i := 0$ after halting):

- *Regime $\beta \ge \alpha^5$ ($p_1 = 8$, $q_B = 7$, $\chi_1 = \sqrt{L_1}$):* Write $\alpha n_1 \ge A_1 + B_1$ with $A_1 := 32000\sqrt{kL_1}$ and $B_1 := 16000 L_1^3/\alpha$. Since $L_1 > 13.39$, $\kappa \le 57344 L_1/B_1 < 0.00125$, giving $(8 Z_7 \sqrt{1-\kappa})^{-1} < 0.20$ and $|V_i + u_i| \le \sqrt{1-\kappa} + d < 1$ on G_i. Two-sided $(4, 4/3)$ Young's inequality $A_1 + B_1 \ge (4A_1)^{1/4}(\frac{4}{3}B_1)^{3/4}$ gives $(\alpha n_1)^8 \ge \frac{4^8}{3^6} A_1^2 B_1^6 > A_1^2 B_1^6$; splitting $\kappa^8 = \kappa^2 \cdot \kappa^6$ cancels k identically across all $k \ge 1$:

$$\mathbb{P}_{S_1}[\mathcal{G}_k^c] < 0.20\, k \left(\frac{57344 L_1}{32000\sqrt{kL_1}}\right)^2 \left(\frac{57344 L_1}{16000 L_1^3/\alpha}\right)^6 < \frac{1362\,\alpha^6}{L_1^{11}} < 5.5 \times 10^{-10} \alpha^6 < \frac{\delta_1}{2}.$$

- *Regime $\beta < \alpha^5$ ($p_1 = \lceil L_1 \rceil + 1$, $q_B = \lceil L_1 \rceil$, $\chi_1 = L_1$):* Since $\alpha < 1/16$ and $\beta < \alpha^5$, $L_1 > \ln(20480 \cdot 16^6) > 26.5$, so $q_B \ge 27$ and $q_B < \frac{27}{26} L_1$. Writing $\alpha n_1 \ge A_1' + B_1$ with $A_1' := 32000\sqrt{k} L_1$, two-sided $(\frac{q_B+1}{2}, \frac{q_B+1}{q_B-1})$ Young splitting on $\kappa^{q_B+1} = \kappa^2 \cdot \kappa^{q_B-1}$ (using $d \le \frac{0.008\alpha}{L_1^3}$ and $d \le \frac{0.004}{\sqrt{k} L_1}$) gives $\kappa \le \frac{0.532\alpha}{L_1} < e^{-6}$ ($|V_i + u_i| < 1$ on G_i, $\sqrt{1-\kappa} > 0.998$) and $k\kappa^2 \le k(64 \cdot \frac{27}{26} L_1^2 \cdot \frac{0.004}{\sqrt{k} L_1})^2 < 0.071 L_1^2$, so $\mathbb{P}_{S_1}[\mathcal{G}_k^c] < \frac{0.071}{0.998} L_1^2 e^{-6(L_1-1)} < e^{-L_1} = \frac{\delta_1}{2}$.

On G_i, $\ell_i(V_i) := \ln \frac{f_{q_B}(V_i)}{f_{q_B}(V_i + u_i)} = -q_B \ln(1 - x_i)$ with $x_i := \frac{2u_i V_i + u_i^2}{1 - V_i^2}$. Since $d/\kappa = \frac{1}{64 q_B L_1}$ is free of k, $|x_i| \le \frac{2d + d^2}{\kappa} < \frac{1}{2990}$ on G_i, where $x \le -\ln(1-x) \le x + 0.5004 x^2$. Set $Y_i := \ell_i(V_i) \mathbf{1}_{G_i}$, $\mu_i^{\text{drift}} := \mathbb{E}[Y_i \mid \mathcal{F}_{i-1}]$, and $D_i := Y_i - \mu_i^{\text{drift}}$. Oddness on symmetric G_i kills $\mathbb{E}[\frac{V_i}{(1-V_i^2)^s} \mathbf{1}_{G_i} \mid \mathcal{F}_{i-1}]$ ($s \in \{1, 2\}$), so $\mathbb{E}[x_i \mathbf{1}_{G_i} \mid \mathcal{F}_{i-1}] = u_i^2 \mathbb{E}[(1 - V^2)^{-1} \mathbf{1}_{G_i}] \ge 0$ and

$$0 \le \mu_i^{\text{drift}} \le q_B u_i^2\, \mathbb{E}\left[\frac{1}{1 - V^2} + \frac{2.0016 V^2 + 0.5004 d^2}{(1 - V^2)^2}\right] \le 0.67\, \mathcal{I}_{q_B} d^2, \qquad \mathbb{E}[D_i^2 \mid \mathcal{F}_{i-1}] \le 1.03\, \mathcal{I}_{q_B} d^2,$$

and $D_i \le Y_i \le \frac{2.002 q_B d}{\kappa} < \frac{1}{31 L_1} =: b$ a.s. For $\beta \ge \alpha^5$, $\mathcal{I}_7 = 17.5$ and $kd^2 \le \frac{1.6 \times 10^{-5}}{L_1}$ give $k \mathcal{I}_{q_B} d^2 \le \frac{0.00028}{L_1}$; for $\beta < \alpha^5$, $\mathcal{I}_{q_B} < 2.4 L_1$ and $kd^2 \le \frac{1.6 \times 10^{-5}}{L_1^2}$ give $k \mathcal{I}_{q_B} d^2 < \frac{3.84 \times 10^{-5}}{L_1}$. Thus in both regimes $\sum_{i=1}^k \mu_i^{\text{drift}} < 0.001$ and $\sum_{i=1}^k \mathbb{E}[D_i^2 \mid \mathcal{F}_{i-1}] \le \frac{0.0002885}{L_1} =: \sigma^2$. By Lemma 3.5 [20], $\sum_{i=1}^k D_i < \sqrt{2\sigma^2 L_1} + \frac{2}{3} b L_1 < 0.0457$ except with probability at most $e^{-L_1} = \delta_1/2$. On $\mathcal{G}_k \cap \{\sum_{i=1}^k D_i < 0.0457\}$ (failing with probability $\le \delta_1$), the density ratio of $\tilde{a}^{(1)}$ against $\bigotimes_{i=1}^k (\text{Leb} + \delta_{1/2})$ satisfies $\ln \frac{p_{S_1}}{p_{S_1'}}(\tilde{a}^{(1)}) = \sum_{i=1}^k Y_i < 0.001 + 0.0457 < 0.05 = \varepsilon_1$, proving $(\varepsilon_1, \delta_1)$-DP after averaging over external coins. □

Validators keep fixed weights within each episode, so Sparse Vector [18] pays privacy per episode rather than per query. For analysis only, define $\theta_i^{(L)}(S_L) := n_L^{-1} \sum_{j=1}^{n_L} q_i(X_{L,j})$ for all $1 \le i \le k$ (never evaluated by the algorithm after halting). To certify uniform accuracy of S_L across all k adaptive queries without union-bounding over queries, we run an analysis-only *Ghost Monitor* after Π: any anchor error $> r_{\text{anc}}$ would let AboveThreshold privately select a biased query and contradict replace-one privacy.

Lemma 4.3 (Validator Privacy, Clean Event $\mathcal{E}_{\text{clean}}$, and Active-Round Accuracy)**.** *Under single-coordinate replacement, $S_r \mapsto \Pi$ is $(\varepsilon_r, \delta_{\text{val}}/2)$-DP for $2 \le r < L$, and $S_L \mapsto \Pi$ is $(\varepsilon_L/2, \delta_{\text{val}}/2)$-DP. Let $\mathcal{E}_{\text{clean}} := \mathcal{E}_{\text{anc}} \cap \mathcal{E}_{\text{lap}}$, where $\mathcal{E}_{\text{anc}} := \{\max_{i \le k} \left|\theta_i^{(L)}(S_L) - \mu_i\right| \le r_{\text{anc}}\}$ and $\mathcal{E}_{\text{lap}}$ requires*

$\left|\rho^{(r)}\right|, \left|\zeta_i^{(r)}\right| \le u_0$ *and* $\left|\nu_i^{(r)}\right| \le 2u_0$ *for all drawn validator Laplace variables. Then*

$$p_c := \mathbb{P}[\mathcal{E}_{\text{clean}}^c] \le \frac{100\,\delta_{\text{val}}}{\varepsilon_L} + \delta_{\text{val}} < 10^{-13}\alpha^2\beta < 10^{-5}\beta,$$

and on $\mathcal{E}_{\text{clean}}$*, every validator release satisfies* $\left|c_i^{(r)} - \theta_i^{(r)}(S_r)\right| \le u_0$ *and every active round satisfies* $|a_i - \mu_i| < \alpha$*.*

Proof. Fix $r \in \{2, \dots, L\}$ and S_{-r}. Intra-episode weight invariance and Lemma 4.1 give query sensitivity $\Delta_r = 4/n_r$ $(r < L)$ and $\Delta_L = 1/n_L$. One episode runs AboveThreshold ($2\Delta_r/b_{\text{SVT}}$ pure DP by Lemma 3.3 [18]) plus at most one trigger release and one retroactive release (each $+\text{Lap}(b_{\text{SVT}})$, costing Δ_r/b_{SVT}), totaling $\varepsilon_{r,\text{ep}} = 4\Delta_r/b_{\text{SVT}}$. For $2 \le r < L$, n_r gives $\varepsilon_{r,\text{ep}} \le \frac{16}{110}\varepsilon_r/\sqrt{M_{r,\max}L_{\text{val}}}$; since $L_{\text{val}} > 25$ and $\varepsilon_r \le 0.05$, composing $M_{r,\max}$ episodes via Lemmas 3.2 and 3.4 [19, 35] at $\delta' = \delta_{\text{val}}/2$ gives privacy $(\frac{16\sqrt{2}}{110} + \frac{1.01\cdot 16^2}{110^2}\frac{\varepsilon_r}{L_{\text{val}}})\varepsilon_r < 0.21\varepsilon_r < \varepsilon_r$. At $r = L$, $\Delta_L = 1/n_L$ and the first term of n_L similarly give $(\varepsilon_L/2, \delta_{\text{val}}/2)$-DP.

After Π finishes, let a conceptual Ghost Monitor run AboveThreshold on the $2k$ signed discrepancies $\sigma(\theta_i^{(L)}(S_L) - \mu_i)$ $((i,\sigma) \in [k] \times \{\pm 1\})$ at threshold $\frac{3}{4}r_{\text{anc}}$ and scale $b_{\text{anc}} := r_{\text{anc}}/(8\ln(6k/\delta_{\text{val}}))$. By Lemma 3.3, this costs $2/(n_L b_{\text{anc}}) = 16\ln(6k/\delta_{\text{val}})/(n_L r_{\text{anc}}) < 0.03\varepsilon_L < \varepsilon_L/2$ (since $r_{\text{anc}} = 2u_0$ and $55\sqrt{M_{L,\max}L_{\text{val}}} > 275$ in n_L), so the augmented output Π_{aug} is $(\varepsilon_L, \delta_{\text{val}}/2)$-DP. Let T be the monitor trigger event on (i^*, σ^*), and set $F(x) := \mathbf{1}_T\sigma^*(q_{i^*}(x) - \mu_{i^*}) \in [-1, 1]$. Over $1 + 2k \le 3k$ draws, $G_{\text{mon}} := \{|\rho_{\text{anc}}| \le r_{\text{anc}}/8\} \cap \bigcap\{|\nu_{\text{anc}}| \le r_{\text{anc}}/4\}$ fails with probability at most $3k(6k/\delta_{\text{val}})^{-1} = \delta_{\text{val}}/2$. On $T \cap G_{\text{mon}}$, $\frac{1}{n_L}\sum_j F(X_{L,j}) \ge (\frac{3}{4} - \frac{1}{8} - \frac{1}{4})r_{\text{anc}} = \frac{3}{2}\varepsilon_L$, while $F \ge -1$ on $T \cap G_{\text{mon}}^c$ and $F = 0$ on T^c. For fresh $X' \sim D$, $\mathbb{E}[F(X') \mid \Pi_{\text{aug}}] = 0$ and $\mathbb{E}[F_\pm(X')] \le \frac{1}{2}\mathbb{P}[T]$, so replace-one DP on $F_\pm \in [0,1]$ gives

$$\frac{3}{2}\varepsilon_L\mathbb{P}[T] - \left(1 + \frac{3}{2}\varepsilon_L\right)\frac{\delta_{\text{val}}}{2} \le \mathbb{E}\left[\frac{1}{n_L}\sum_{j=1}^{n_L} F(X_{L,j})\right] \le \sinh(\varepsilon_L)\mathbb{P}[T] + \delta_{\text{val}}.$$

Since $\sinh(\varepsilon_L) < 1.001\varepsilon_L$, $\mathbb{P}[T] < 4\delta_{\text{val}}/\varepsilon_L$. Conditioning on (S, Π) inside $\mathcal{E}_{\text{anc}}^c$ fixes the first (i_0, σ_0) with $\sigma_0(\theta_{i_0}^{(L)}(S_L) - \mu_{i_0}) > r_{\text{anc}}$ before exposing the fresh monitor coins; T holds if an earlier branch crosses or if $\rho_{\text{anc}} \le 0$ and $\nu_{\text{anc},(i_0,\sigma_0)} \ge 0$ (probability 1/4), so $\mathbb{P}[\mathcal{E}_{\text{anc}}^c] \le 4\mathbb{P}[T] < 100\delta_{\text{val}}/\varepsilon_L$.

Across k rounds and $L-1$ stages, Algorithm 1 draws at most $(L-1)(3k+1) \le 4Lk \le 32LkM_{2,\max}$ Laplace variables, each exceeding u_0 (or $2u_0$) with probability $\delta_{\text{val}}/(64LkM_{2,\max})$, so $\mathbb{P}[\mathcal{E}_{\text{lap}}^c] \le \delta_{\text{val}}/2$ and $p_c \le (200 + u_0)u_0^2\beta/(10^8L^3) < 10^{-13}\alpha^2\beta < 10^{-5}\beta$. On $\mathcal{E}_{\text{clean}}$, 1-Lipschitz clipping gives $\left|c_i^{(r)} - \theta_i^{(r)}(S_r)\right| \le \left|\zeta_i^{(r)}\right| \le u_0$, and Stage L either passes on a_i ($\left|a_i - \theta_i^{(L)}(S_L)\right| \le \tau_{(u_{\text{final}},L)} + 3u_0$) or triggers ($\left|a_i - \theta_i^{(L)}(S_L)\right| \le u_0$), giving $|a_i - \mu_i| \le \tau_{(u_{\text{final}},L)} + 4u_0 + r_{\text{anc}} < \alpha$. □

4.3 Equal-Impulse Cancellation, Tilted Comparator Transfer, and Proof of Theorem 1.1

When Case B has final trigger $r' \in \{3, \dots, L-1\}$ from u_{prev}, stage $r'-1$ either passed on $c_i^{(u_{\text{prev}})}$ or is u_{prev} itself, separating $c_i^{(r'-1)}$ from $c_i^{(r')}$ by $2\gamma - 8u_0$. Because $8w_{r'-1}\eta_{r'-1} = 8w_{r'}\tilde{\eta}_{r'}$, summing our dimensionless public progress functional cancels μ_i identically and forces some validator to reach progress ≥ 128 whenever any cap $N_{r^*,\max}$ fills.

Lemma 4.4 (Equal-Impulse μ_i-Cancellation and Halting Progress)**.** *For* $2 \le \ell < L$*, define the* $\sigma(\Pi)$*-measurable progress functional*

$$\widetilde{\text{Prog}}_\ell(\Pi) := 8w_\ell \sum_{i \in [k]:\delta_{\ell,i} \ne 0} \left[(-\delta_{\ell,i})(c_i^{(\ell)} - \mu_i) - \frac{1}{8}\delta_{\ell,i}^2 - |\delta_{\ell,i}|\,u_0\right],$$

and set $\mathcal{E}_{\mathrm{halt}}(\Pi) := \bigcup_{r^*=3}^{L}\{N_{r^*} = N_{r^*,\max}\}$ *and* $A_\ell(\Pi) := \mathcal{E}_{\mathrm{halt}}(\Pi) \cap \{\widetilde{\mathrm{Prog}}_\ell(\Pi) \geq 128\}$. *On* $\mathcal{E}_{\mathrm{halt}}(\Pi) \cap \mathcal{E}_{\mathrm{clean}}$, $\sum_{\ell=2}^{L-1} \widetilde{\mathrm{Prog}}_\ell(\Pi) \geq 192L > 128(L-2)$, *so* $\mathcal{E}_{\mathrm{halt}}(\Pi) \cap \mathcal{E}_{\mathrm{clean}} \subseteq \bigcup_{\ell=2}^{L-1}(A_\ell(\Pi) \cap \mathcal{E}_{\mathrm{clean}})$.

Proof. Whenever $\delta_{\ell,i} \neq 0$, round i is an active Case-B round (including any halting round) recording $c_i^{(\ell)}$ and $\delta_{\ell,i}$ in Π_i, so $\widetilde{\mathrm{Prog}}_\ell(\Pi)$ and $A_\ell(\Pi)$ are well-defined and $\sigma(\Pi)$-measurable.

In an active Case-B round i on $\mathcal{E}_{\mathrm{clean}}$ with $u_{\mathrm{final}} = r' \in \{3, \dots, L\}$, $u := u_{\mathrm{prev}}(r')$, and $s := s_{i,r'} = \mathrm{sign}(c_i^{(u)} - c_i^{(r')})$, stage r''s trigger gives $\left|c_i^{(u)} - c_i^{(r')}\right| \geq \tau_{(u,r')} - 4u_0$, while stage $r'-1$'s pass (if $u < r'-1$) or reuse (if $u = r'-1$) gives $\left|c_i^{(u)} - c_i^{(r'-1)}\right| \leq \tau_{(u,r'-1)} + 4u_0$. For $3 \leq r' < L$, subtracting gives $s(c_i^{(r'-1)} - c_i^{(r')}) \geq 2\gamma - 8u_0 = \frac{28}{16}\gamma$. Since $8w_{r'-1}(-\delta_{r'-1,i}) = w_{r'-1}\gamma s = -8w_{r'}(-\delta_{r',i})$, μ_i cancels identically between levels $r'-1$ and r'; using $\tilde{\eta}_{r'} < \eta_{r'-1} = \gamma/8$ and $u_0 = \gamma/32$, each level's quadratic and noise deduction is at most $\frac{3}{64}w_{r'-1}\gamma^2$, giving round-$i$ sum

$$\widetilde{\mathrm{Prog}}_{r'-1,i}(\Pi) + \widetilde{\mathrm{Prog}}_{r',i}(\Pi) \geq w_{r'-1}\gamma\, s(c_i^{(r'-1)} - c_i^{(r')}) - \frac{6}{64}w_{r'-1}\gamma^2 \geq \frac{106}{64}w_{r'-1}\gamma^2 > 1.5\, w_{r'-1}\gamma^2.$$

For $r' = L$, $\left|c_i^{(L)} - \mu_i\right| \leq u_0 + r_{\mathrm{anc}} = 3u_0$ gives $s(c_i^{(L-1)} - \mu_i) \geq 2\gamma - 11u_0 = \frac{53}{32}\gamma$ and $\widetilde{\mathrm{Prog}}_{L-1,i}(\Pi) \geq (\frac{53}{32} - \frac{3}{64})w_{L-1}\gamma^2 = \frac{103}{64}w_{L-1}\gamma^2 > 1.5w_{L-1}\gamma^2$. Summing over $i \in [k]$ on $\mathcal{E}_{\mathrm{halt}}(\Pi) \cap \mathcal{E}_{\mathrm{clean}}$ where $N_{r^*} = N_{r^*,\max}$ gives $\sum_{\ell=2}^{L-1}\widetilde{\mathrm{Prog}}_\ell(\Pi) \geq 1.5w_{r^*-1}\gamma^2 N_{r^*,\max} \geq 192L > 128(L-2)$, proving the Pigeonhole inclusion. □

We close all trigger caps by comparing entropic FTRL (Lemma 3.7(a)) [33] against an a posteriori two-level tilted witness $Q^{*(r)} \in \mathcal{K}_2(S_r)$: FTRL holds pathwise for every comparator, $D_{\mathrm{KL}} \leq \varepsilon_r^2/2$ cancels $w_\ell = \varepsilon_\ell^{-2}$ ($8w_\ell D_{\mathrm{KL}} \leq 4$), and we never condition DP on $\mathcal{E}_{\mathrm{clean}}$, charging $\mathcal{E}_{\mathrm{clean}}^c$ separately via bounded payoff.

Lemma 4.5 (Two-Level Tilted Comparator and Cap-Non-Exhaustion)**.** *Let* $S_r \sim D^{n_r}$ *($n_r \geq 2$) satisfy $(\varepsilon_r, \delta_r)$-DP for $S_r \mapsto \Pi$ with $0 \leq \varepsilon_r \leq 0.05$ and $\delta_r \geq 0$. For any $\sigma(\Pi) \otimes \mathcal{F}_X$-measurable $F(\Pi, x) \in [-1,1]$ with $\mathbb{E}_{X' \sim D}[F(\Pi, X') \mid \Pi] = 0$, set $w_j^{(r)} := e^{-\varepsilon_r}$ if $F(\Pi, X_{r,j}) \geq 0$ and e^{ε_r} otherwise, $Z_r(S_r) := \frac{1}{n_r}\sum_{j=1}^{n_r} w_j^{(r)}$, and $Q^{*(r)}(j) := \frac{w_j^{(r)}}{n_r Z_r(S_r)}$. Then $Z_r(S_r) \in [e^{-\varepsilon_r}, e^{\varepsilon_r}]$, $Q^{*(r)} \in \mathcal{K}_2(S_r)$,*

$$D_{\mathrm{KL}}(Q^{*(r)} \,\|\, \mathrm{Unif}(S_r)) \leq \frac{1}{2}\varepsilon_r^2, \qquad \textit{and} \qquad \mathbb{E}\Big[Z_r(S_r)\left\langle Q^{*(r)}, F(\Pi, S_r)\right\rangle\Big] \leq (1 + e^{-\varepsilon_r})\delta_r \leq 2\delta_r.$$

Consequently, in Algorithm 1, $\mathbb{P}[\{t_1 = T_{1,\max}\} \cap \mathcal{E}_{\mathrm{clean}}] < 0.002\beta$ *and* $\mathbb{P}[\mathcal{E}_{\mathrm{halt}}(\Pi) \cap \mathcal{E}_{\mathrm{clean}}] < 10^{-3}\beta$.

Proof. Since $e^{2\varepsilon_r} \leq e^{0.10} < 2$, $Z_r(S_r) \in [e^{-\varepsilon_r}, e^{\varepsilon_r}]$ and $Q^{*(r)}(j) \leq e^{2\varepsilon_r}/n_r < 2/n_r$. With $p_+ := \frac{1}{n_r}\sum_j \mathbf{1}_{\{F(\Pi, X_{r,j}) \geq 0\}}$ and $\varphi(t) := \ln(p_+ e^{-t} + (1-p_+)e^t)$, $\varphi''(t) = 1 - (\varphi'(t))^2 \in [0,1]$ gives $D_{\mathrm{KL}}(Q^{*(r)} \,\|\, \mathrm{Unif}(S_r)) = \varepsilon_r\varphi'(\varepsilon_r) - \varphi(\varepsilon_r) = \int_0^{\varepsilon_r} t\varphi''(t)\, dt \leq \frac{1}{2}\varepsilon_r^2$. Writing $F = f_+ - f_-$ ($f_\pm \in [0,1]$) gives $w_j^{(r)}F(\Pi, X_{r,j}) = e^{-\varepsilon_r}f_+(\Pi, X_{r,j}) - e^{\varepsilon_r}f_-(\Pi, X_{r,j})$; integrating replace-one DP tail bounds against fresh $X' \sim D$ gives $e^{-\varepsilon_r}\mathbb{E}[f_+(\Pi, X_{r,j})] \leq \mathbb{E}[f_+(\Pi, X')] + e^{-\varepsilon_r}\delta_r$ and $-e^{\varepsilon_r}\mathbb{E}[f_-(\Pi, X_{r,j})] \leq -\mathbb{E}[f_-(\Pi, X')] + \delta_r$, which sum via $\mathbb{E}[F(\Pi, X') \mid \Pi] = 0$ to $(1 + e^{-\varepsilon_r})\delta_r \leq 2\delta_r$. Thus if $F = 0$ off a $\sigma(\Pi)$-measurable event A and $\left\langle Q^{*(r)}, F(\Pi, S_r)\right\rangle \geq c > 0$ on $A \cap \mathcal{E}_{\mathrm{clean}}$, then $Y := Z_r(S_r)\left\langle Q^{*(r)}, F(\Pi, S_r)\right\rangle \mathbf{1}_{\mathcal{E}_{\mathrm{clean}}} \geq 0$ satisfies $\mathbb{E}[Y] \leq 2\delta_r + e^{\varepsilon_r}p_c$ and Markov's inequality gives

$$\mathbb{P}[A \cap \mathcal{E}_{\mathrm{clean}}] \leq \frac{e^{\varepsilon_r}(2\delta_r + e^{\varepsilon_r}p_c)}{c}. \tag{$\star$}$$

For Stage 1, abbreviate $T := T_{1,\max}$ and set $F^{(1)}(\Pi, x) := T^{-1}\mathbf{1}_{\{t_1=T\}} \sum_{t=1}^{T} s_t^{(1)}(q_{i_t}(x) - \mu_{i_t})$ over the Case-A rounds $i_1 < \cdots < i_{t_1}$. On $\mathcal{E}_{\text{clean}}$, each Case-A round has $\left|c_{i_t}^{(1)} - a_{i_t}\right| \geq \frac{\alpha}{2} - 4u_0$, $\left|c_{i_t}^{(1)} - \theta_{i_t}^{(1)}(S_1)\right| < \frac{\alpha}{32}$, and Stage-$L$ pass error $|a_{i_t} - \mu_{i_t}| \leq 2(L-2)\gamma + 3u_0 + r_{\text{anc}} < \frac{\alpha}{4} + 5u_0$, giving $s_t^{(1)}(\theta_{i_t}^{(1)}(S_1) - \mu_{i_t}) > \frac{7}{32}\alpha - 9u_0 > \frac{3}{16}\alpha$. By Lemma 3.7(a) ($\delta_{1,i_t} = -\eta_1 s_t^{(1)}$), on $\{t_1 = T\} \cap \mathcal{E}_{\text{clean}}$ we have $\left\langle Q^{*(1)}, F^{(1)}(\Pi, S_1)\right\rangle \geq \frac{3}{16}\alpha - \frac{\eta_1}{8} - \frac{\varepsilon_1^2}{2\eta_1 T} > \frac{\alpha}{8} > \frac{\alpha}{10}$, so $(\star)$ with $c = \alpha/8$ and Lemmas 4.2 and 4.3 gives $\mathbb{P}[\{t_1 = T\} \cap \mathcal{E}_{\text{clean}}] < 0.002\beta$.

For validator $\ell \in \{2, \ldots, L-1\}$, $8w_\ell \sum_{i=1}^{k} |\delta_{\ell,i}| \leq \widetilde{W}_{\max,\ell} := \gamma(w_\ell N_{\ell+1,\max} + w_{\ell-1} N_{\ell,\max}\mathbf{1}_{\{\ell\geq 3\}})$. Since $N_{r^*,\max} < \frac{128L\varepsilon_{r^*-1}^2}{\gamma^2} + 1$ and $\gamma^2 w_\ell < (\frac{\alpha}{8L})^2 \frac{400}{\alpha^2} < \frac{1}{4}$, $\widetilde{W}_{\max,\ell} < \frac{256L}{\gamma} + \gamma(w_\ell + w_{\ell-1}\mathbf{1}_{\{\ell\geq 3\}}) < \frac{257L}{\gamma} < \frac{9L}{u_0}$. Setting $F^{(\ell)}(\Pi, x) := \widetilde{W}_{\max,\ell}^{-1}\mathbf{1}_{A_\ell(\Pi)} 8w_\ell \sum_{i=1}^{k}(-\delta_{\ell,i})(q_i(x) - \mu_i) \in [-1, 1]$, on $A_\ell(\Pi) \cap \mathcal{E}_{\text{clean}}$ we have $\left|c_i^{(\ell)} - \theta_i^{(\ell)}(S_\ell)\right| \leq u_0$ whenever $\delta_{\ell,i} \neq 0$ and $8w_\ell D_{\text{KL}}(Q^{*(\ell)} \,\|\, \text{Unif}(S_\ell)) \leq 4$, so Lemma 3.7(a) gives $\widetilde{W}_{\max,\ell}\left\langle Q^{*(\ell)}, F^{(\ell)}(\Pi, S_\ell)\right\rangle \geq \widetilde{\text{Prog}}_\ell(\Pi) - 4 \geq 124$. Applying $(\star)$ with $c = 124/\widetilde{W}_{\max,\ell} > \frac{124u_0}{9L}$ and $\delta_\ell = \delta_{\text{val}}/2$ and summing over ℓ via Lemma 4.4 yields

$$\mathbb{P}[\mathcal{E}_{\text{halt}}(\Pi) \cap \mathcal{E}_{\text{clean}}] \leq \sum_{\ell=2}^{L-1} \mathbb{P}[A_\ell(\Pi) \cap \mathcal{E}_{\text{clean}}] \leq (L-2)\frac{9L\, e^{0.05}}{124\, u_0}(\delta_{\text{val}} + e^{0.05} p_c) < \frac{20L^2\delta_{\text{val}}}{u_0^2} < 10^{-3}\beta. \quad \square$$

Proof of Theorem 1.1. On $\mathcal{E}_{\text{clean}} \cap \{t_1 < T_{1,\max}\} \cap \mathcal{E}_{\text{halt}}(\Pi)^c$, HALT stays false through round k, so every round is active and satisfies $|a_i - \mu_i| < \alpha$ by Lemma 4.3. By Lemmas 4.2 to 4.5,

$$\mathbb{P}\left[\max_{1\leq i\leq k} |a_i - \mu_i| > \alpha\right] \leq \mathbb{P}[\mathcal{E}_{\text{clean}}^c] + \mathbb{P}[\{t_1 = T_{1,\max}\} \cap \mathcal{E}_{\text{clean}}] + \mathbb{P}[\mathcal{E}_{\text{halt}}(\Pi) \cap \mathcal{E}_{\text{clean}}] < 0.00301\beta < \beta,$$

and averaging over the analyst's coins covers randomized analysts. Lemma A.2 (Section A.1) proves the explicit and asymptotic sample bounds on N_{cas}, and Lemma 4.1 proves the $O(kN_{\text{cas}})$ time and $O(N_{\text{cas}})$ space bounds. $\square$

Since $\ln k$ enters N_{cas} only once inside b_{SVT}^{-1}, one scalar inequality absorbs it into $\sqrt{k}$.

Lemma 4.6 (Single-Shot $\ln k$ Absorption)**.** *For all $k \geq 1$ and $A, B > 0$, we have $A\sqrt{k} + B\ln k \leq 2A\sqrt{k} + 2B\ln(1 + 2B/A)$.*

Proof. For $x = \sqrt{k} \geq 1$, the concave function $2B\ln x - Ax$ on $(0, \infty)$ is maximized at $x^* = 2B/A$ with value $2B(\ln(2B/A) - 1) \leq 2B\ln(1 + 2B/A)$; adding $2A\sqrt{k}$ yields the bound. $\square$

Corollary 4.7 (Absorbed k-Independent Additive Overhead)**.** *With $\chi = \sqrt{\ln(1/(\alpha\beta))}$ for $\beta \geq \alpha^5$ and $\chi = \ln(1/(\alpha\beta))$ for $\beta < \alpha^5$, $\mathcal{M}_{\text{cas}}$ satisfies*

$$N_{\text{cas}} = O\left(\frac{\sqrt{k}\,\chi}{\alpha} + \frac{1}{\alpha^2}\left[\ln^3\left(\frac{1}{\alpha\beta}\right) + \ln^{7/2}(1/\alpha)\,\ln^{3/2}\left(\frac{\ln(1/\alpha)}{\alpha\beta}\right)\right]\right).$$

Proof. Set $H_0 := \ln(1/(\alpha\beta))$ and $J := \ln(L/(\alpha\beta)) = \Theta(\ln(\ln(1/\alpha)/(\alpha\beta)))$. By Theorem 1.1 (Lemma A.2), $N_{\text{cas}} = O(\frac{\sqrt{k}\chi}{\alpha} + \frac{H_0^3}{\alpha^2} + \frac{L^{7/2}\sqrt{J}}{\alpha^2}(J + \ln k))$. Applying Lemma 4.6 with $A = \chi/\alpha$ and $B = L^{7/2}\sqrt{J}/\alpha^2$ (where $\chi \geq 1$ gives $\ln(1+2B/A) = O(J)$) bounds the $\ln k$ term by $O(\frac{\sqrt{k}\chi}{\alpha} + \frac{L^{7/2}J^{3/2}}{\alpha^2})$, proving the claim since $L = \Theta(\ln(1/\alpha))$. $\square$

5 Finite-Precision Interval-Validation ADA

Streaming norm banks usually need to check whether an existing estimate remains plausible, rather than produce a new estimate after every update. Interval validation charges for rejections and numerical releases, not successful checks. With at most M interval requests, $J+1$ rejections, and R numerical requests, the sample complexity is

$$N = \widetilde{O}\left(\frac{\sqrt{J+R+1}}{a_{\mathrm{req}}} + \frac{1}{a_{\mathrm{req}}^2}\right);$$

M appears only inside logarithms.

Finite precision also affects the choice of regularizer. We use an $8/n_j$-capped simplex W_j and a quadratic regularizer R_j. Their KKT projection is exact rational water-filling: recomputing weights from accumulated scores avoids exponentially growing denominators and requires no transcendental evaluations. The analysis uses a clipped inverse-posterior comparator whose quadratic cost is bounded by Bhatia–Davis. This comparator is only an analysis witness; the mechanism never computes posterior probabilities.

5.1 Clipped Sequential Inverse-Posterior Comparator

For a positive integer r, write

$$W_r = \left\{ w \in \mathbb{R}^r_{\geq 0} : \sum_i w_i = 1,\ w_i \leq \frac{8}{r} \right\}, \qquad R_r(w) = \frac{r}{2} \sum_i \left(w_i - \frac{1}{r} \right)^2,$$

and write $\mathcal{D}f = \mathbb{E}_{x\sim\mathcal{D}} f(x)$.

Theorem 5.1 (Clipped Sequential Inverse-Posterior Comparator)**.** *Let $\mathcal{U}$ be finite, let $\mathcal{D}$ be a probability distribution on $\mathcal{U}$, and let $X = (X_1, \ldots, X_r) \sim \mathcal{D}^r$. Suppose a countable randomized transcript $T = T(X; C_T) \in \mathcal{T}$, with independent mechanism randomness C_T, is (θ, δ)-DP in the sense of Definition 3.1, where $0 < \theta \leq 1$, and determines queries $f_1, \ldots, f_k : \mathcal{U} \to [0,1]$. Put $f_0 \equiv 1$. For $z \in \mathcal{U}^{i-1}$, $x \in \mathcal{U}$, and $t \in \mathcal{T}$, write $p_{t,z}(x) = \mathbb{P}_{C_T,\, X_{>i}\sim\mathcal{D}^{r-i}}[T(z, x, X_{>i}; C_T) = t]$ and $q(t,z) = \sum_{y\in\mathcal{U}} \mathcal{D}(y) p_{t,z}(y)$. Set $a_i(x) = 1$ for all x when $q(t,z) = 0$ (such histories have zero probability under the joint law), and otherwise define (with $\mathrm{clip}_{[e^{-\theta},e^{\theta}]}(1/0) := e^{\theta}$)*

$$a_i(x) = \frac{p_{t,z}(x)}{q(t,z)}\Big|_{t=T,\, z=X_{<i}}, \qquad v_i = \mathrm{clip}_{[e^{-\theta},e^{\theta}]}\left(\frac{1}{a_i(X_i)}\right),$$
$$\bar{v} = \frac{1}{r}\sum_i v_i \in [e^{-\theta}, e^{\theta}], \qquad u_i = \frac{v_i}{r\bar{v}}.$$

If $0 < h \leq 1$, $0 < \xi < 1$, and

$$r \geq 72 e^{2\theta} h^{-2} \ln \frac{k+2}{\xi}, \qquad \delta \leq \frac{h\xi}{24},$$

then deterministically

$$r u_i \in [e^{-2\theta}, e^{2\theta}] \subset [0, 8], \qquad u \in W_r, \qquad R_r(u) \leq \frac{1}{2}\sinh^2\theta \leq 0.691\,\theta^2 \leq \theta^2.$$

Moreover, with probability at least $1-\xi$,

$$\max_{1\leq t\leq k} \left| \sum_i u_i f_t(X_i) - \mathcal{D}f_t \right| \leq \frac{2}{3}h < h.$$

Proof. *1. Prefix augmentation.* Fix a prefix $z = x_{<i}$. Average the replacement-DP inequality for $T(z, x, X_{>i}; C_T)$ and $T(z, x', X_{>i}; C_T)$ over the *prior* suffix $X_{>i} \sim \mathcal{D}^{r-i}$ (with C_T already averaged inside replacement DP). For every $x, x' \in \mathcal{U}$ and every subset $E \subseteq \mathcal{T}$, this gives $\sum_{t\in E} p_{t,z}(x) \le e^{\theta} \sum_{t\in E} p_{t,z}(x') + \delta$. For fixed z, x, averaging these inequalities over $x' \sim \mathcal{D}$ compares the laws $p_{\cdot,z}(x)$ and $q(\cdot, z)$ in both directions, without assuming full support of $\mathcal{D}$ or conditional independence given T.

2. Uniform drift control. At a fixed history (t, z), the conditional drift for any $f : \mathcal{U} \to [0, 1]$ is

$$\mathbb{E}[v_i f(X_i) \mid T = t, X_{<i} = z] - \mathcal{D}f = \sum_x \mathcal{D}(x) f(x) \left[a_i(x) \operatorname{clip}_{[e^{-\theta}, e^{\theta}]}(1/a_i(x)) - 1\right].$$

Use the convention $\operatorname{clip}_{[e^{-\theta}, e^{\theta}]}(1/0) = e^{\theta}$. The identity

$$s \operatorname{clip}_{[e^{-\theta}, e^{\theta}]}(1/s) - 1 = (e^{-\theta} s - 1)_+ - (1 - e^{\theta} s)_+$$

bounds the absolute drift, uniformly over all f, by

$$\Delta_i(t, z) = \sum_x \mathcal{D}(x) \left[(e^{-\theta} a_i(x) - 1)_+ + (1 - e^{\theta} a_i(x))_+\right].$$

Apply the two DP inequalities to the positive-part events

$$E_{+,x} = \{t : e^{-\theta} p_{t,z}(x) > q(t, z)\}, \qquad E_{-,x} = \{t : q(t, z) > e^{\theta} p_{t,z}(x)\}.$$

Their contributions, after multiplying by $q(t, z)$ and summing over t, are at most $e^{-\theta}\delta$ and δ, respectively. Consequently,

$$\sum_t q(t, z) \Delta_i(t, z) \le (1 + e^{-\theta})\delta, \qquad \mathbb{E}\left[\frac{1}{r} \sum_i \Delta_i(T, X_{<i})\right] \le 2\delta.$$

Markov's inequality gives $r^{-1} \sum_i \Delta_i \le h/6$, except with probability at most $12\delta/h \le \xi/2$. There is no union bound over queries or transcripts in this step.

3. Martingale concentration and normalization. Condition on $T = t$. For $0 \le m \le k$, let

$$D_{m,i} = v_i f_m^{(t)}(X_i) - \mathbb{E}[v_i f_m^{(t)}(X_i) \mid T = t, X_{<i}].$$

These are martingale differences for $\mathcal{F}_i = \sigma(T, X_{\le i})$, with conditional range width at most e^{θ}. The range-width Azuma–Hoeffding bound in Lemma 3.5, followed by a union bound, bounds the probability that any $|r^{-1} \sum_i D_{m,i}| > h/6$ by

$$2(k + 1) \exp\left(-\frac{rh^2}{18e^{2\theta}}\right) \le 2(k + 1) \left(\frac{\xi}{k + 2}\right)^4 < \frac{\xi}{2}.$$

This bound is uniform in t. On the two good events, writing $V_m = r^{-1} \sum_i v_i f_m(X_i)$, we have $|V_m - \mathcal{D}f_m| \le h/3$, including $|\bar{v} - 1| \le h/3$ for $m = 0$. For $A_m = \langle u, f_m(X)\rangle \in [0, 1]$, the division-free identity

$$A_m - \mathcal{D}f_m = V_m - \mathcal{D}f_m + (1 - \bar{v})A_m$$

therefore gives $|A_m - \mathcal{D}f_m| \le 2h/3$.

4. Quadratic cost. The box bound follows directly from the bounds on $v_i, \bar{v}$. Applying Lemma 3.6 to the uniform empirical distribution on $v_1, \ldots, v_r$ gives

$$\begin{aligned} R_r(u) &= \frac{1}{2r\bar{v}^2} \sum_i (v_i - \bar{v})^2 \le \frac{(e^{\theta} - \bar{v})(\bar{v} - e^{-\theta})}{2\bar{v}^2} \\ &= \frac{1}{2}\left[\sinh^2\theta - (\bar{v}^{-1} - \cosh\theta)^2\right] \le \frac{1}{2}\sinh^2\theta. \end{aligned}$$

Finally, $\sinh\theta \le \theta \sinh 1$ on $[0, 1]$, and $\frac{1}{2}\sinh^2 1 < 0.691$. □

5.2 Algorithm, Sampler, and Exact Water-Filling

The protocol permits M interval requests and R numerical requests. It stops after returning the $(J+1)$-st interval rejection. A request is *covered* if it respects its request cap and occurs before this stop, including the terminal rejection itself. An ACCEPT response certifies $\mathcal{D}f \in [\ell - a_{\mathrm{req}}, u + a_{\mathrm{req}}]$; HIGH and LOW certify $\mathcal{D}f > u \geq u - a_{\mathrm{req}}$ and $\mathcal{D}f < \ell \leq \ell + a_{\mathrm{req}}$, respectively. Numerical answers have additive error a_{req}.

We use Definition A.3 in Section A.2. Here are the relations needed in this section:

$$a = 2^{-\lceil \log_2(1/a_{\mathrm{req}}) \rceil}, \qquad \xi_A = 2^{-\lceil \log_2(1/\xi_{\mathrm{req}}) \rceil}, \qquad \tau = \frac{a}{8p}, \quad h = \frac{\tau}{64}, \quad \kappa = \frac{\tau}{4},$$

$$T_j^{\max} = \left\lceil \frac{6}{4^j \tau^2} \right\rceil, \qquad \lambda_j = 4^j, \quad \theta_j = (p4^j)^{-1/2}, \quad d_j = \lceil \sqrt{p4^j} \rceil \qquad (0 \leq j < p),$$

$$\theta_p = \frac{h}{2\sqrt{2}}, \qquad d_p = \lceil 4/h \rceil, \qquad \vartheta_j = \frac{1}{d_j} \leq \theta_j \qquad (0 \leq j \leq p).$$

Here $p \geq 2$ is the least integer with $4^{p-1}(a/(8p))^2 \geq 1$. The irrational θ_j's are analysis bounds; the implementation uses the integers d_j. Set $T_r^{\max} = 0$ outside $0 \leq r < p$, and use phase budgets

$$C_0 = T_0^{\max} + T_1^{\max} + J + R + 2, \qquad C_p = 1 + T_{p-2}^{\max} + T_{p-1}^{\max},$$
$$C_j = 2 + T_{j-2}^{\max} + 2T_{j-1}^{\max} + 2T_j^{\max} + T_{j+1}^{\max} \qquad (1 \leq j < p).$$

The integer draw bound Q exceeds $2p(M+R) + 2M + R + p + J + 1 + 9\sum_{j<p} T_j^{\max}$. The remaining choices satisfy

$$\xi_j = \frac{\xi_A}{8(p+1)}, \qquad \delta = \frac{h\xi_j}{24}, \qquad \ln \frac{1000Q}{\delta} \leq \chi = O\left(\log \frac{M+J+R+2}{a\xi_A} \right),$$

$$b_{\mathrm{int}} = \frac{h}{128\chi}, \qquad b_{\mathrm{out}} = \frac{a}{512\chi}, \qquad b_{\mathrm{num}} = \frac{a}{64\chi},$$

$$n_j \geq 72e^2 h^{-2} \ln \frac{M+R+2}{\xi_j}, \qquad \frac{9}{n_j b_{\mathrm{int}}} \leq \frac{\vartheta_j}{8\sqrt{C_j \ln(4/\delta)}},$$

$$h_A \in 2^{-\mathbb{N}}, \qquad h_A \leq \min\{b_{\mathrm{int}}, 1/n_{\max}\}, \qquad \log n_{\max} + \log h_A^{-1} = O(\chi), \quad n_{\max} = \max_j n_j.$$

Up to logarithmic factors, $n_j = \widetilde{O}(h^{-2} + d_j\sqrt{C_j}/h)$. The exact integer choices and their arithmetic verification are deferred to Lemma A.4. Split $N = \sum_{j=0}^{p} n_j$ independent samples into blocks S_j; write $W_j = W_{n_j}$ and $R_j = R_{n_j}$.

Lemma 5.2 (Bounded Lattice-Noise Sampler)**.** *For rational $b > 0$, write $\varrho = b/(b + h_A) = A/D_\varrho$ in lowest terms, and put $\zeta = \delta/(1000Q)$. Set*

$$B_0 = \left\lceil \log_2 \frac{16}{\zeta} \right\rceil, \qquad K_{\mathrm{s}} = \left\lceil \frac{b + h_A}{h_A} B_0 \right\rceil, \qquad r_{\mathrm{s}} = \left\lceil \log_2 \frac{32(K_{\mathrm{s}} + 1)}{\zeta} \right\rceil.$$

Two geometric continuation counts, each capped at K_{s}, can be generated using at most r_{s} rejection attempts per Bernoulli trial. Returning $h_A(G_1 - G_2)$ terminates on every fair-bit tape, has TV distance $< \zeta$ from the ideal lattice law Z_{b,h_A} of Lemma 3.3, and uses

$$O(\log D_\varrho + \log(K_{\mathrm{s}} + 1) + \log(1/\zeta))$$

workspace bits.

Proof. A Bernoulli trial draws $\lceil \log_2 D_\varrho \rceil$ bits, rejects integers at least D_ϱ, and continues exactly when the accepted integer is below A. After r_{s} failed attempts it stops the count. Without either cap, the two geometric counts have difference law Z_{b,h_A}. Each rejection attempt succeeds with probability at least $1/2$. Coupling to uncapped counts bounds the TV error by

$$2\varrho^{K_{\mathrm{s}}} + 2(K_{\mathrm{s}}+1)2^{-r_{\mathrm{s}}} \le 2e^{-B_0} + \frac{\zeta}{16} \le \frac{3\zeta}{16} < \frac{\zeta}{2}.$$

Here $1-\varrho = h_A/(b+h_A)$. The count, candidate integer, and loop counters require the stated space; all loops have deterministic bounds. □

Write $\mathrm{round}_h(z) = h\lfloor z/h + 1/2 \rfloor$ (ties toward $+\infty$). On the oracle bit-machine (Section 7), a request is accessed through a binary-truncation oracle (`OracleFor`): at precision index $q \in \mathbb{N}$, it returns $\lfloor 2^q f(x) \rfloor$ on encoded sample records and $\lfloor 2^q \,\mathrm{clip}_{[-1,2]}(\ell) \rfloor, \lfloor 2^q \,\mathrm{clip}_{[-1,2]}(u) \rfloor$ on endpoints, so querying at $q_A + 1$ ($h_A = 2^{-q_A}$) computes $\mathrm{round}_{h_A}(z) = h_A \lfloor (\lfloor 2^{q_A+1} z \rfloor + 1)/2 \rfloor$ exactly in integer arithmetic without touching off-sample inputs. Every noise draw in Algorithm 2 uses Lemma 5.2 reading fair bits MSB-first per rejection trial, G_1 before G_2, initial and reopened thresholds in increasing edge order, and both directed branch noises before testing the priority branch. Between requests, only the sample codes, in-block scores G_j ($j < p$), thresholds, counters, and flags persist (`alg2RetainedOf`); weights w_j, query buffers, and edge-acceptance caches are recomputed temporarily during a request.

Lemma 5.3 (Water-Filling Sensitivity and Non-Accumulating Denominators on W_j)**.** *Fix a comparison-history prefix and a rounded query f. Replacing one point of S_j, for $j < p$, changes the weights by at most $16/n_j$ in ℓ_1, and changes $\mu_j(f) = \langle w_j, f(S_j) \rangle$ by at most $8/n_j$. The anchor sensitivity is $1/n_p$. All weights are computable exactly with $O(\chi)$ bits per coordinate, without denominator accumulation across updates.*

Proof. The KKT equations give, for a common water level ν_{lag},

$$w_{j,i} = \mathrm{clip}_{[0,8/n_j]} \left(\frac{1 + \kappa G_{j,i} - \nu_{\mathrm{lag}}}{n_j} \right).$$

At a fixed history, replacing point i_0 changes only G_{j,i_0}. Increasing this score can only increase the water level, so all off-i_0 weights decrease; decreasing it reverses these signs. Mass conservation therefore gives, with $\Delta = w'_{j,i_0} - w_{j,i_0}$,

$$\left\| w'_j - w_j \right\|_1 = 2|\Delta| \le \frac{16}{n_j}.$$

Let $a_0, a'_0 \in [0,1]$ be the two query evaluations at i_0. The off-coordinate changes have one sign and sum to $-\Delta$; their query contribution is thus $-\Delta \bar{f}$ for some $\bar{f} \in [0,1]$. If $\Delta = 0$, take any such $\bar{f}$. Hence

$$\begin{aligned} |\mu'_j - \mu_j| &= \left| w'_{j,i_0}(a'_0 - \bar{f}) + w_{j,i_0}(\bar{f} - a_0) \right| \\ &\le \max\{w_{j,i_0}, w'_{j,i_0}\} \le \frac{8}{n_j}. \end{aligned}$$

The uniform anchor has sensitivity $1/n_p$.

For exact arithmetic, every score lies in $(h_A/4)\mathbb{Z}$. Put $D_G = \mathrm{denom}(\kappa h_A/4)$ and $s_i = 1 + \kappa G_{j,i} \in D_G^{-1}\mathbb{Z}$. Sort the breakpoints $s_i, s_i - 8$. If I is the active set, B the set at the upper cap, and $k_{\mathrm{act}} = |I| > 0$, the mass equation is

$$\nu_{\mathrm{lag}} = \frac{\sum_{i \in I} s_i + 8|B| - n_j}{k_{\mathrm{act}}} \in (k_{\mathrm{act}} D_G)^{-1}\mathbb{Z}, \qquad w_{j,i} \in (k_{\mathrm{act}} n_j D_G)^{-1}\mathbb{Z}.$$

Algorithm 2 Capped-Simplex Validation ADA Mechanism $\mathcal{M}_{\mathrm{val}}$

1: **Retained state and total-response rules.** Store the blocks S_j. For $0 \le j < p$, retain $G_j \in \mathbb{R}^{n_j}$, initially zero; during a request define the exact weights

$$w_j = \arg\max_{w \in W_j} \{\kappa \langle G_j, w\rangle - R_j(w)\}; \qquad w_p = \mathbf{1}/n_p.$$

Retain counters T_j, interval/numerical request counters, a rejection counter, and sticky `aborted`/`halted` flags, initially zero or false. Draw persistent thresholds $\hat{\vartheta}_{\mathrm{int},j}$ $(j = 0, \dots, p-1)$ at scale $2b_{\mathrm{int}}$ and $\hat{\vartheta}_{\mathrm{out}}$ at scale $2b_{\mathrm{out}}$ (if initialization exhausts the fair-bit tape, set both flags). Every call returns a total response, with dummy value ACCEPT for interval requests and 0 for numerical requests: if `halted` or `aborted` is set, or the called request-type cap (M or R) is already reached, return the dummy response without mutating state or tape (exceeding one request-type cap refuses only that type).

2: PREPAREROOT(f)**.** Evaluate $F_j = \mathrm{round}_{h_A}(f(S_j))$, compute $\mu_j = \langle w_j, F_j\rangle$, and clear all edge acceptance-cache bits. Scan $j = 0, \dots, p-1$, skipping cached edges. For an uncached edge draw fresh ν_+, ν_-, each at scale $4b_{\mathrm{int}}$, and test in this order:

$$\mu_{j+1} - \mu_j + \nu_+ \ge \tau + \hat{\vartheta}_{\mathrm{int},j} \quad (q = +1), \qquad \mu_j - \mu_{j+1} + \nu_- \ge \tau + \hat{\vartheta}_{\mathrm{int},j} \quad (q = -1).$$

If neither crosses, cache the edge as accepted and advance. On a crossing, if $T_j + 1 > T_j^{\max}$ (or if any sampler draw in PREPAREROOT exhausts the tape), restore the pre-request learner/threshold state (keeping the incremented request counter), set `aborted` and `halted`, and abort to dummy return. Otherwise update

$$G_j \leftarrow G_j + qF_j, \qquad G_{j+1} \leftarrow G_{j+1} - (q/4)F_{j+1} \quad \text{if } j+1 < p, \qquad T_j \leftarrow T_j + 1.$$

Recompute the modified weights and means. Unconditionally close and redraw the internal thresholds, and clear the cache bits, on every edge incident to a modified block in increasing order: $\{j-1, j, j+1\} \cap \{0, \dots, p-1\}$ (just $\{p-2, p-1\}$ at the anchor). Restart the scan at $j = 0$. Return μ_0 when the scan finishes.

3: Interval($f, [\ell, u]$)**.** Enforce the interval cap M and flags, increment the interval counter, clip endpoints to $[-1, 2]$ and round to $h_A\mathbb{Z}$, and run PREPAREROOT(f). Draw fresh $\nu_{\mathrm{out},+}, \nu_{\mathrm{out},-}$ at scale $4b_{\mathrm{out}}$ (if the tape runs out, keep post-PREPAREROOT state, set both flags, and return ACCEPT), and test HIGH first:

$$\mu_0 - u + \nu_{\mathrm{out},+} \ge a/2 + \hat{\vartheta}_{\mathrm{out}} \quad (\textsc{High}), \qquad \ell - \mu_0 + \nu_{\mathrm{out},-} \ge a/2 + \hat{\vartheta}_{\mathrm{out}} \quad (\textsc{Low}).$$

If neither crosses, return ACCEPT and retain $\hat{\vartheta}_{\mathrm{out}}$. Otherwise increment the rejection counter and close the outer phase: on the $(J+1)$-st rejection, set `halted` and return the rejection without redrawing $\hat{\vartheta}_{\mathrm{out}}$; else redraw $\hat{\vartheta}_{\mathrm{out}}$ (setting both flags if that redraw exhausts the tape) and return the rejection.

4: Numerical(f)**.** Enforce the numerical cap R and flags, increment the numerical counter, run PREPAREROOT(f), draw Z at scale b_{num} (if the tape runs out, keep post-PREPAREROOT state, set both flags, and return 0), and return $\mathrm{clip}_{[0,1]}(\mathrm{round}_{h_A}(\mu_0 + Z))$ without closing the outer phase.

Thus each denominator divides $k_{\mathrm{act}} n_j D_G \le n_j^2 D_G$. If $k_{\mathrm{act}} = 0$, every weight is 0 or $8/n_j$. The capped update counts and the parameter bounds give $O(\chi)$ bits for all these numerators and denominators. Each projection starts from the score lattice, not from the previous weights. □

5.3 Coupled Potential and Correctness

Lemma 5.4 (Coupled 4^j Bregman Potential Contraction)**.** *Suppose fixed comparators $u_j \in W_j$, $0 \le j \le p$, satisfy $R_j(u_j) \le \theta_j^2$ and $|\langle u_j, f(S_j)\rangle - \mathcal{D}f| \le h$ for every rounded submitted query f. Suppose also that every internal comparison has total absolute threshold/query noise at most $2h$.*

For $j < p$, define

$$\Psi_j(G) = R_j(u_j) + \max_{w \in W_j} \{\kappa \langle G, w\rangle - R_j(w)\} - \kappa \langle G, u_j\rangle, \qquad P = \sum_{j<p} 4^j \Psi_j(G_j).$$

Then the following hold throughout the run.

(i) $P \geq 0$, *and* $P_0 = \sum_{j<p} 4^j R_j(u_j) \leq 1$.

(ii) An accepted edge satisfies $|\mu_{j+1} - \mu_j| \leq \tau + 2h$. *A rejected edge with sign* q *satisfies* $q(\mu_{j+1} - \mu_j) \geq \tau - 2h$.

(iii) An interior correction decreases P *by at least* $(25/128)4^j\tau^2$.

(iv) An anchor correction decreases P *by at least* $(103/512)4^{p-1}\tau^2 > (25/128)4^{p-1}\tau^2$.

(v) $\sum_{j<p} 4^j T_j \leq (128/25)\tau^{-2} = 5.12\,\tau^{-2} < 6\tau^{-2}$, *and* $T_j < T_j^{\max}$ *for every* $j < p$.

Proof. Part (i) follows from the defining maximum and $R_j(\mathbf{1}/n_j) = 0$; initially, $\sum_{j<p} 4^j\theta_j^2 = 1$. Part (ii) follows from the tests. A cached acceptance remains valid because an incident weight change clears its cache bit.

The regularizer R_j is 1-strongly convex in ℓ_1. By the quadratic case of Lemma 3.7, a score increment g satisfies

$$\Psi_j(G_j + g) - \Psi_j(G_j) \leq \kappa \langle g, w_j - u_j\rangle + \frac{\kappa^2}{2} \|g\|_\infty^2.$$

For an interior correction, use $g_j = qf(S_j)$ and $g_{j+1} = -(q/4)f(S_{j+1})$. The population terms cancel because $4^j q + 4^{j+1}(-q/4) = 0$. Part (ii) and the two comparator errors give

$$\begin{aligned}\Delta P &= 4^j(\Delta\Psi_j + 4\Delta\Psi_{j+1}) \\ &\leq 4^j\left[-\kappa(\tau - 4h) + \frac{5}{8}\kappa^2\right] = -\frac{25}{128}4^j\tau^2.\end{aligned}$$

This proves (iii).

For the anchor, Cauchy–Schwarz gives

$$|\mu_p - \langle u_p, f(S_p)\rangle| \leq \|u_p - \mathbf{1}/n_p\|_1 \leq \sqrt{2R_p(u_p)} \leq \sqrt{2}\,\theta_p = \frac{h}{2}.$$

Thus $|\mu_p - \mathcal{D}f| \leq 1.5h$. Only block $p-1$ is updated, so

$$\begin{aligned}\Delta P &\leq 4^{p-1}\left[-\kappa(\tau - 4.5h) + \frac{1}{2}\kappa^2\right] \\ &= -\frac{103}{512}4^{p-1}\tau^2 = -\frac{51.5}{256}4^{p-1}\tau^2 < -\frac{25}{128}4^{p-1}\tau^2.\end{aligned}$$

This proves (iv). Summing the decreases and using $0 \leq P \leq P_0$ gives (v), including the strict bound

$$T_j \leq \frac{128}{25\,4^j\tau^2} < \frac{6}{4^j\tau^2} \leq \left\lceil \frac{6}{4^j\tau^2} \right\rceil = T_j^{\max}.$$

□

Proof of Theorem 1.2. Phase and draw counts. A new request clears caches, not thresholds. Only a correction on an incident block closes an internal phase. Thus edge r has at most $1+T_{r-1}^{\max}+T_r^{\max}+T_{r+1}^{\max}$ phases. There are at most $J+1$ outer phases and R numerical releases, giving exactly the block budgets C_j above. Minimality of p gives $1 \le 4^{p-1}\tau^2 < 4$, from which

$$C_j \le 190\, 4^{-j}\tau^{-2} \quad (1 \le j < p), \qquad C_p \le 31.$$

For each request, at most p initially uncached edges are tested; each correction reopens at most three edges. Counting two directed noises per test and all threshold redraws gives fewer than Q scalar draws on every capped run.

Block privacy. Fix the controller's coins, and augment the public transcript by the internal comparison outcomes, but not by weights or noise values. Because each response is a discrete outcome or lattice rational in $h_A\mathbb{Z}$, this finite sequence of discrete outcomes is countable and determines the sequence of submitted queries and endpoints without serializing arbitrary real functions. Fixing the other blocks, Lemma 5.3 gives effective lattice sensitivity

$$\Delta_j' = h_A \left\lceil \frac{8/n_j}{h_A} \right\rceil \le \frac{9}{n_j}$$

(and likewise for the anchor), which suffices in Lemma 3.3 even though the rational means μ_j need not lie in $h_A\mathbb{Z}$. Under ideal lattice noise, Lemma 3.3 makes an internal or outer phase $(\Delta_j'/b, 0)$-DP, and a numerical release $\mathrm{round}_{h_A}(\mu_0+Z) = \mathrm{round}_{h_A}(\mu_0)+Z$ is $(\Delta_0'/b_{\mathrm{num}}, 0)$-DP. Since b_{int} is the smallest scale, each participating phase has privacy parameter at most

$$\frac{9}{n_j b_{\mathrm{int}}} \le \frac{\vartheta_j}{8\sqrt{C_j \ln(4/\delta)}}.$$

Concurrent composition and advanced composition (Lemmas 3.2 and 3.4) therefore make the ideal transcript $(\vartheta_j, \delta/2)$-DP for block S_j.

The stronger estimate in the proof of Lemma 5.2 couples the full bounded and ideal runs at TV cost $< Q\zeta/2 < Q\zeta = \delta/1000$. By Definition 3.1, restoring the bounded sampler adds at most $(1+e^{\vartheta_j})Q\zeta/2 < 2Q\zeta$ to the DP failure term. Thus the actual transcript is (ϑ_j, δ)-DP for every block, with $\vartheta_j = 1/d_j \le \theta_j$.

Accuracy and cap non-exhaustion. Pad the at most $M+R$ rounded submitted queries by zero queries. Apply Theorem 5.1 to each block at failure probability ξ_j. With failure probability at most $\sum_{j=0}^{p}\xi_j = \xi_A/8$, this supplies a posteriori pathwise comparators satisfying the hypotheses of Lemma 5.4. The geometric tail of Z_{b,h_A}, together with $h_A \le b_{\mathrm{int}}$, bounds the probability that any of

$$|\hat{\vartheta}_{\mathrm{int},j}| + |\nu_{\mathrm{int}}| \le 2h, \qquad |\hat{\vartheta}_{\mathrm{out}}| + |\nu_{\mathrm{out}}| \le a/8, \qquad |Z_{\mathrm{num}}| \le a/4$$

fails by at most $2Qe^{-6\chi} + Q\zeta < \xi_A/8$ (and every draw fits within the deterministic tape cap). The joint failure probability is therefore $< \xi_A \le \xi_{\mathrm{req}}$.

On this joint event, Lemma 5.4(v) rules out both $T_j \ge T_j^{\max}$ and any internal or tape abort. After PrepareRoot(f), all cached edge certificates remain valid. Telescoping to the anchor, and accounting for query rounding, gives

$$|\mu_0(f) - \mathcal{D}f| \le p(\tau+2h) + 1.5h + h_A < 0.14a < a/4.$$

Endpoint clipping to $[-1,2]$ is harmless because $\mathcal{D}f \in [0,1]$, and endpoint rounding fits inside the unused $a/4 - 0.14a$ margin: an acceptance satisfies $\mathcal{D}f \in [\ell - a, u + a]$; on High, $a/2 - a/8 - a/4 > 0$

gives $\mathcal{D}f > \mathrm{clip}_{[-1,2]}(u)$, forcing $u < 1$ and hence $\mathrm{clip}_{[-1,2]}(u) = \max(-1, u) \geq u$, so $\mathcal{D}f > u > u - a_{\mathrm{req}}$ (and symmetrically $\mathcal{D}f < \ell < \ell + a_{\mathrm{req}}$ on Low). Numerical answers have error at most $a/4 + a/4 + h_A < a$; clipping to $[0, 1]$ cannot increase this error. Since $a \leq a_{\mathrm{req}}$, these guarantees hold for every covered request, including the terminal $(J + 1)$-st rejection.

Finally, Lemma A.4 in Section A.2 verifies the explicit sample bound with prefactor 2^{64} and the $O(N(b_{\mathcal{U}} + \chi^2))$-bit (certified $2^{30}N(b_{\mathcal{U}} + \chi^2)$-bit) space bound. The hard update/request caps, Lemma 5.2, and the exact arithmetic of Lemma 5.3 give termination and this space bound on every fair-bit tape, not only on the accuracy event. □

6 Adversarially Robust Turnstile F_2 Estimation

6.1 Model, Parameters, and the Zero Fingerprint

Integer coordinates let us handle the zero vector without a statistical query. We choose all scales and precision grids from public parameters before the stream begins; none depends on a realized norm or on the outcome of a validation.

The stream is

$$x^{(t)} = x^{(t-1)} + \Delta_t e_{i_t} \in \mathbb{Z}^n, \qquad x^{(0)} = 0, \qquad \left\|x^{(t)}\right\|_\infty \leq U \quad (1 \leq t \leq m).$$

Here $n, m \geq 1$, and $\varepsilon, \beta \in (0, 1/10)$. We may take $U \geq 1$ integral, since otherwise the stream is identically zero. Thus $|\Delta_t| \leq 2U$ and $\left\|x^{(t)}\right\|_2 < nU + 1$. The adversary chooses each update from its coins and the previous reports $\widehat{F}_1, \ldots, \widehat{F}_{t-1}$. Our algorithm processes every physical update once, without replay. Banks are born only at fixed binary-tree times and are never restarted during their lifetimes.

Write

$$\Lambda = \log_2\left(\frac{2^{100}(n+2)(m+2)(U+2)}{\varepsilon\beta}\right), \qquad \varepsilon_0 = 2^{-\lceil \log_2(1/\varepsilon)\rceil}, \qquad \beta_0 = 2^{-\lceil \log_2(1/\beta)\rceil}.$$

We use the sequential dyadic choices in Definitions A.5 and A.6 of Section A.3. The structural parameters needed here are

$$\begin{gathered}
H = 1 + \lceil \log_2 \max\{1, m\} \rceil, \qquad T = 2^{12}(m+1)(H+1)^3, \qquad D_* = n + T + 1, \qquad W = nU + 1, \\
\eta = 2^{-\lceil \log_2(2^{20}(H+1)^2/\varepsilon_0)\rceil}, \qquad \gamma_g = 2^{-\lceil \log_2(8\sqrt{T})\rceil}, \qquad B = 16/\gamma_g, \\
V_0 = W, \qquad V_{d+1} = (1 + 2B^d)V_d \quad (0 \leq d \leq H), \\
\mathcal{U}_{d,0} = 2^{\lceil \log_2(2V_d^2)\rceil}, \qquad \mathcal{U}_{d,k} = 2^{-k}\mathcal{U}_{d,0} \quad (0 \leq d < H), \\
\mathcal{U}_* = \mathcal{U}_{H-1,0}, \qquad V = 8B^{H+1}V_H + 2, \qquad G = B^{H+2}, \\
\nu = \gamma_g \eta\, 2^{-H-20}, \qquad E_{\min} = 2^{-8}\eta^2\nu^2, \\
K_0 = \lceil 128/\eta^2 \rceil, \qquad K_{\mathrm{epoch}} = \left\lceil \log_2\left(\frac{2^{20}\mathcal{U}_* B^{2H}}{\eta^2\nu^2}\right)\right\rceil, \qquad J_* = K_0(K_{\mathrm{epoch}} + 1), \\
s = \log_2(1/\eta) + 22, \qquad L_{\mathrm{clip}} = 16s, \\
a_{\mathrm{num}} = \frac{\eta}{2^{14}L_{\mathrm{clip}}}, \qquad a_{\mathrm{val}} = \frac{\eta}{16L_{\mathrm{clip}}}, \qquad \beta_B = \frac{\beta_0}{64m(H+1)}.
\end{gathered} \tag{1}$$

Algorithm 3 Homomorphic Zero-Fingerprint Round Dispatch

1: At initialization, set

$$M_q = \left\lceil \frac{10nm(U+1)}{\beta_0} \right\rceil^3$$

and let q be the least prime at least M_q. Sample $r_{\mathrm{fp}} \in \mathbb{F}_q$ by rejection from $\{0, \dots, 2^{\lceil \log_2 q \rceil} - 1\}$, using at most $\lceil \log_2(1000/\beta_0) \rceil$ trials and falling back to 0. Initialize $F_{\mathrm{zero}} = 0$.

2: On update (i_t, Δ_t), execute Steps 1–2 of Algorithm 5, and set

$$F_{\mathrm{zero}} \leftarrow (F_{\mathrm{zero}} + \Delta_t r_{\mathrm{fp}}^{i_t}) \bmod q.$$

3: If $F_{\mathrm{zero}} = 0$, report $\widehat{F}_t = 0$. Otherwise execute Steps 3–4 and report the root's answer. Scheduled births and physical updates are processed in both cases.

In particular, $\gamma_g \in (1/(16\sqrt{T}), 1/(8\sqrt{T})]$ and $T\gamma_g^2 \le 1/64$. Each bank has caps $M = m$, $J_{\max} = 2J_*$, and $R_{\max} = 4J_* + 2$.

For clarity, the precision bounds satisfied by the committed dyadic grids of Definition A.6 are collected here; rounding is to the nearest grid point (ties toward $+\infty$):

$$\begin{gathered} h_g = \frac{\gamma_g \eta \nu B^{-2H}}{2^{20}} \le \frac{\gamma_g \eta \nu}{2^{20}}, \qquad e_C = \frac{h_g}{2^{10}\gamma_g V} \le \frac{h_g}{2^{10}(1+\sqrt{\mathcal{U}_*})}, \qquad h_C \le \frac{e_C}{GT}, \\ 0 < \omega = \sigma^2 \le \min\left\{1, \frac{\eta E_{\min}}{2^{40}G^2}\right\}, \qquad h_P \le \frac{\eta E_{\min}}{2^{32}TG^2D_*}, \\ h_b \le \frac{\eta\omega}{2^{40}D_*(V+1)G(T+1)}, \qquad h_A \le \frac{\eta}{2^{32}L_{\mathrm{clip}}}, \qquad h_{\mathrm{step}} = \frac{\eta^2}{2^{20}} \le \frac{\eta^2}{2^{12}}, \qquad h_{\mathrm{leaf}} = h_g^2 h_{\mathrm{step}}^{2H}. \end{gathered} \tag{2}$$

Square roots only select dyadic grid points and can be implemented by exact squared comparisons. The complete parameter order and bit exponents are deferred to Definitions A.5 and A.6 and Table 3.

Lemma 6.1 (Homomorphic Zero-Fingerprint Coupling)**.** *The fingerprint is always zero when $x^{(t)} = 0$. With probability greater than $1 - \beta_0/8$, its dispatch agrees on every round with the ideal oracle $\mathbf{1}[x^{(t)} = 0]$. Consequently, every round entering Steps 3–4 satisfies $\left\|x_0^{(t)}\right\|_2 = \left\|x^{(t)}\right\|_2 \ge 1 > \nu$. The fingerprint uses $O(\Lambda)$ bits.*

Proof. Homomorphism gives $F_{\mathrm{zero}}^{(t)} = \sum_i x_i^{(t)} r_{\mathrm{fp}}^i \bmod q$, including for negative updates. Bertrand's postulate gives $q \le 2M_q$, so the modulus, seed, and counter have $O(\Lambda)$ bits. The capped sampler differs from a uniform field element with probability at most $\beta_0/1000$.

Fix the adversary's coins and all algorithmic coins other than the fingerprint. Run the ideal zero dispatcher. This trajectory is independent of r_{fp}. At every nonzero round, $p_t(z) = \sum_i x_i^{(t)} z^i$ is a nonzero polynomial over $\mathbb{F}_q$: its degree is at most n, and $U < q$ prevents a nonzero coefficient from vanishing modulo q. Schwartz–Zippel [32, 37] and a union bound give failure at most mn/q. Until the first incorrect zero test, the actual and ideal executions coincide, so this bound also controls the first disagreement of the adaptive executions. The total failure is

$$mn/q + \beta_0/1000 < \beta_0/8.$$

Finally, a nonzero integer vector has Euclidean norm at least one. □

Algorithm 4 Two-Regime Regularized Norm Bank

1: At the bank's scheduled birth, allocate the row sample and ADA state of Theorem 1.2, with accuracy a_{num}, failure $\beta_B/2$, and caps $(M, J, R) = (m, J_{\max}, R_{\max})$. Also allocate $R_{\max}$ independent coarse groups, each containing $32g$ rows, where $g = 32\lceil \log_2(4R_{\max}/\beta_B)\rceil$. All these counters are maintained from birth; "fresh group" means previously unread, not newly initialized.

2: In `PRIMARY` mode use $\hat{z}(b) = X(b) + b_{D_*}\sigma$; in `DIFF` mode use $\hat{z}(b) = (X(b) - Y(b)) + b_{D_*}\sigma$. For a scale $c > 0$, supply ADA with the bounded row function

$$\hat{f}(b) = \text{round}_{h_A}\left(\min\left\{\frac{\hat{z}(b)^2}{L_{\text{clip}}c}, 1\right\}\right).$$

3: ValidateInterval(v, S, θ) checks that the bank's ADA state is neither halted nor aborted and has remaining interval budget, sets $c = S + \omega$, and invokes the interval test of Algorithm 2 on $[(1-\theta)/L_{\text{clip}}, (1+\theta)/L_{\text{clip}}]$. If the check fails or the call sets `aborted` or `halted` (including the terminal $(J_{\max}+1)$-st rejection), signal bank failure; otherwise return Accept, High, or Low.

4: ReleaseNumerical(v) checks that an unread coarse group remains and ADA is neither halted nor aborted; otherwise signal bank failure. Consume the next coarse group, let c_{raw} be the lower median of its g block averages (each averaging 32 values $\hat{z}(b)^2$), and put $c = \max\{c_{\text{raw}}, \omega/2\}$. Obtain an ADA numerical answer $\hat{\mu}$ for the row function above; if ADA sets `aborted`, signal bank failure, and otherwise return

$$N(v) = \max\{0, L_{\text{clip}}c\hat{\mu} - \omega\}.$$

6.2 Two-Regime Regularized Norm Bank

A bank validates a proposed scale, and obtains a new numerical scale only when needed. Its artificial coordinate is added at read time: storing it in a counter would rescale it under homogeneous updates, while adding it to both difference counters would cancel it.

All algorithmic vectors are padded with zeros to dimension D_*; the last coordinate is reserved for regularization. A row $b \in \{-1, 1\}^{D_*}$ has $2s$-wise independent Rademacher entries, and different rows are independent. Each row has primary and secondary counters, denoted $X(b)$ and $Y(b)$. For $g \geq 1$ real numbers, define the lower median as the $\lceil g/2\rceil$-th smallest element ($g/2$-th for even g).

Theorem 6.2 (Regularized Norm Bank Certificates)**.** *For each bank there is a statistical event of probability at least $1 - \beta_B$, defined without assuming counter accuracy. On this event, suppose the augmented target norm is at most V, primary counter errors are at most ξ, difference counter errors are at most 2ξ, and*

$$2\sqrt{D_*}V\xi + \xi^2 \leq \eta\omega/2^{30}.$$

For $S \geq 0$, $0 \leq \theta \leq 2^{-16}$, and $r = (\|v\|_2^2 + \omega)/(S + \omega)$, the certificates are

$$\begin{aligned} &\text{Accept}: && |r - 1| \leq \theta + \eta/4, \\ &\text{High}: && r - 1 \geq \theta - \eta/4, \qquad |N(v) - \|v\|_2^2| \leq 2^{-13}\eta\,\|v\|_2^2 + \omega. \\ &\text{Low}: && 1 - r \geq \theta - \eta/4. \end{aligned}$$

These assertions hold simultaneously up to the bank's caps.

Proof. Put $F' = \|v\|_2^2 + \omega$, $Z = \langle b, (v, \sigma)\rangle$, and $f_{v,c}(b) = \min\{Z^2/(L_{\text{clip}}c), 1\}$, where (v, σ) means replacing the reserved zero coordinate by σ. For $r = F'/c$ and $W_Z = Z^2/(L_{\text{clip}}c)$, Lemma 3.8 gives $\mathbb{E}W_Z = r/L_{\text{clip}}$ and $\mathbb{E}W_Z^2 \leq 3r^2/L_{\text{clip}}^2$. Since $(\mathbb{E}W_Z)^2 \leq \mathbb{E}[W_Z/(1+W_Z)]\,\mathbb{E}[W_Z(1+W_Z)]$ by Cauchy–Schwarz and $W_Z/(1+W_Z) \leq \min(W_Z, 1)$, we obtain

$$\frac{r}{L_{\text{clip}} + 3r} \leq \mathbb{E}\left[\frac{W_Z}{1+W_Z}\right] \leq \mathbb{E}f_{v,c} \leq \frac{r}{L_{\text{clip}}}. \tag{3}$$

For $r \le 2$, the higher-moment bound in Lemma 3.8 gives

$$0 \le \frac{r}{L_{\mathrm{clip}}} - \mathbb{E} f_{v,c} \le \mathbb{E} W_Z^s \le \frac{r}{L_{\mathrm{clip}}} 2s\, 4^{-(s-1)} \le 2^{-16}\eta \frac{r}{L_{\mathrm{clip}}}. \tag{4}$$

Next, $|Z| \le \sqrt{D_*}V$. A primary perturbation is at most ξ, and a difference perturbation is at most 2ξ. The assumed inequality, $\omega \le 1$, and $V \ge 1$ give

$$|\hat{z}^2 - Z^2| \le 3(2\sqrt{D_*}V\xi + \xi^2).$$

Clipping is nonexpansive. Including rounding to h_A, even for $c \ge \omega/2$ we obtain

$$|\hat{f} - f_{v,c}| \le \frac{\eta}{2^{27} L_{\mathrm{clip}}} = 2^{-13} a_{\mathrm{num}}. \tag{5}$$

For an exact coarse group, a 32-row block has variance $\le (F')^2/16$ by Lemma 3.8 and lies outside $[3F'/5, 7F'/5]$ with probability at most $(1/16)/(2/5)^2 = 25/64$. By Hoeffding's inequality, the lower median of g blocks lies outside $[3F'/5, 7F'/5]$ with probability at most $\exp(-2g(1/2 - 25/64)^2) = e^{-49g/2048} < \beta_B/(2R_{\max})$, and adding the counter perturbation puts c_{raw} in $[F'/2, 3F'/2]$. A union bound covers all coarse releases.

Apply Theorem 1.2 to the actual bounded functions $\hat{f}$. After (5), each validation certificate has population-mean error bounded by $(1 + 2^{-13})a_{\mathrm{num}} < a_{\mathrm{val}} = \eta/(16L_{\mathrm{clip}})$. If $r \ge 2$, (3) gives

$$\mathbb{E} f_{v,c} \ge \frac{2}{L_{\mathrm{clip}} + 6} > \frac{1+\theta}{L_{\mathrm{clip}}} + a_{\mathrm{val}}.$$

Thus ACCEPT and LOW force $r < 2$. In that regime, (4) and $L_{\mathrm{clip}} a_{\mathrm{val}} + 2 \cdot 2^{-16}\eta < \eta/4$ convert the mean certificates into the three asserted bounds on r. For HIGH with $r \ge 2$, its asserted bound is immediate.

For a numerical release, the coarse event gives $c \in [F'/2, 3F'/2]$, so $r \le 2$. Equations (4)–(5) and ADA numerical accuracy imply

$$|L_{\mathrm{clip}} c\hat{\mu} - F'| \le 2^{-16}\eta F' + \frac{3}{2} L_{\mathrm{clip}} F'(1 + 2^{-13}) a_{\mathrm{num}} < 2^{-13}\eta F'.$$

Subtracting ω and truncating at zero cannot increase error relative to $\|v\|_2^2$. The claimed numerical bound follows. The ADA and coarse events cost at most $\beta_B/2$ each. □

6.3 Binary Time Tree, Homogeneous Persistence, and Orthogonal Guards

The time tree fixes when a learner may start; a rejection changes a homogeneous scale rather than starting another learner. Fresh orthogonal coordinates record these scale changes, while an epoch envelope limits how much energy those coordinates can accumulate.

Place round t at leaf $2t - 1$ of a complete depth-H binary tree; even leaves and leaves past the stream are unused. A nonleaf at depth $d < H$ has lifetime 2^{H-1-d} physical rounds, and an active depth-H leaf has lifetime one. Write $\ell = H - d$ for height. A learner at depth $d - 1$ owns a bank $\mathcal{B}_{d-1}$ for its depth-d clients; the root uses an initial bank $\mathcal{B}_{-1}$. The bank's primary array X_d persists across its at most two sequential clients. Its secondary array Y_d resets at client birth, but its ADA state and unread coarse groups do not restart. The depth-H client instead maintains an exact squared-norm accumulator.

Algorithm 5 Within-Round Binary Time-Tree Controller (Steps 1–4)

1: **Scheduled births, top down.** Let $d_0 \le H$ be the shallowest depth whose lifetime starts at round t ($2^{H-1-d} \mid t-1$). Retire depths $\ge d_0$ before allocating new learners top down for $d = d_0, \dots, H-1$: set

$$a_d = x_d, \quad q_d = u_d = 0, \quad r_d = a_d, \quad y_d = u_d + q_d = x_{d+1} = 0, \quad C_d = \widehat{C}_d = 1, \quad k_d = 0.$$

Keep the inherited X_d and clear Y_d. If $d = 0$ or $d > d_0$ (so $a_d = 0$ deterministically), set $P_d = 0$; at $d = d_0 > 0$, initialize $P_d = \mathrm{round}_{h_P}(\mathcal{B}_{d-1}.\textsc{ReleaseNumerical}(\texttt{PRIMARY}))$ (matching `DIFF` since $Y_d = 0$). Allocate the fresh child bank $\mathcal{B}_d$ with zero counters ($d < H-1$), and reset the exact leaf accumulator to zero. Epoch and lifetime correction counts start at zero.

2: **Physical update.** Pass $\Delta_t e_{i_t}$ once down the active path, adding it to every q_d and hence to every input x_d and suffix y_d. Update all corresponding row counters, rounding to h_b. The leaf accumulator becomes Δ_t^2. The zero dispatcher now decides whether to execute Steps 3–4.

3: **Guards, top down.** For $d = 0, \dots, H-1$, set

$$g(d,t) = n + H(t-1) + d + 1, \qquad \tilde{\delta}_{g,d} = \frac{\gamma_g}{4}\sqrt{\mathcal{U}_{d,k_d}}.$$

If $\widehat{C}_d \tilde{\delta}_{g,d} < 2h_g$, set $\delta_{g,d} = 0$; otherwise set $\delta_{g,d} = \mathrm{round}_{h_g}(\widehat{C}_d \tilde{\delta}_{g,d})$. Add $\delta_{g,d} e_{g(d,t)}$ to u_d and to the descendant suffix (rounding row updates to $h_b\mathbb{Z}$). Thus r_d loses that vector and y_d gains it. Set $P_d \leftarrow \mathrm{round}_{h_P}(P_d + \delta_{g,d}^2)$; add $\delta_{g,d}^2$ to the leaf accumulator.

4: **Validation and correction, bottom up.** The leaf returns its exact norm in $h_{\mathrm{leaf}}\mathbb{Z}$. At depth $d < H$, let $\widehat{Q}_d = \widehat{F}_{d+1}$, $S_d = P_d + \widehat{Q}_d$, and $\theta_\ell = (8\ell+4)\eta$. Validate x_d in `PRIMARY` mode against (S_d, θ_ℓ). On Accept, return $\widehat{F}_d = S_d$. Otherwise set $\varsigma = 1$ for High and $\varsigma = -1$ for Low, check $J_{\mathrm{used},d} < J_*$, $P_d > 0$, and $\widehat{Q}_d > 0$ (entering the absorbing failure state if any fails), obtain the `PRIMARY` release $N_1(x_d)$, and compute

$$a_{\mathrm{step}} = \mathrm{clip}_{[-1/2,1/2]}\left(\mathrm{round}_{h_{\mathrm{step}}}\left(\varsigma \frac{\eta}{16}\sqrt{\frac{P_d}{\widehat{Q}_d}}\right)\right), \qquad c = 1 + a_{\mathrm{step}}.$$

Perform $u_d \leftarrow u_d + a_{\mathrm{step}} y_d$: multiply Y_d and all descendant row counters by c (rounded to $h_b\mathbb{Z}$), descendant tags $\widehat{C}_{d'}$ by c (rounded to $h_C\mathbb{Z}$), descendant caches $P_{d'}$ by c^2 (rounded to $h_P\mathbb{Z}$), and the leaf accumulator by c^2 (exact on $h_{\mathrm{leaf}}\mathbb{Z}$), without resetting any ADA state. Increment $J_{\mathrm{used},d}$; after each K_0 corrections in an epoch, increment k_d, reset the intra-epoch counter, and check $k_d \le K_{\mathrm{epoch}}$. Refresh $P_d = \mathrm{round}_{h_P}(N_0(r_d))$ by a `DIFF` release, and return $\widehat{F}_d = N_1(x_d)$ (which leaves x_d invariant since local updates preserve X_d).

Total-behavior and failure transitions. All vectors below are analysis variables; the implementation performs only homomorphic counter and scalar operations under Definition A.5. Every row counter update rounds to $h_b\mathbb{Z}$ and silently saturates to $[-Z_{\max}, Z_{\max}]$ without altering control flow or exposing unread rows. Stored tags $\widehat{C}_d \in h_C\mathbb{Z}$ are range-checked in $[0, C_{\max}]$, and residual caches $P_d \in h_P\mathbb{Z}$, candidates S_d, estimates $\widehat{F}_d$, and leaf accumulators in $h_{\mathrm{leaf}}\mathbb{Z}$ are range-checked in $[0, A_{\max}]$. If any range check, bank operation (Algorithm 4), correction/epoch cap ($J_{\mathrm{used},d} < J_*$ before rejection, $k_d \le K_{\mathrm{epoch}}$ after epoch increment), or Step-4 positivity check ($P_d > 0$ and $\widehat{Q}_d > 0$ on rejection) fails, the controller immediately enters a global absorbing failure state, returning 0 in that round and on every subsequent physical update without further bank requests or allocations. The proof shows that on the statistical event no clamp, saturation, or failure transition is ever triggered.

Lemma 6.3 (Homogeneous Child Persistence and Target Invariance)**.** *At every depth,*

$$x_d = a_d + q_d = r_d + y_d, \qquad r_d = a_d - u_d, \qquad y_d = u_d + q_d = x_{d+1}.$$

A local correction multiplies the existing child input by $1 + a_{\text{step}}$*; it does not replace the child. Suffix additions and strict-ancestor rescalings leave* r_d/C_d *unchanged.*

Proof. The identities hold at birth and are preserved by suffix additions. For the homogeneous update [23],

$$u_d^+ = u_d + a_{\text{step}} y_d, \qquad r_d^+ = r_d - a_{\text{step}} y_d, \qquad y_d^+ = u_d^+ + q_d = (1 + a_{\text{step}}) y_d.$$

A strict-ancestor rescaling by c multiplies $(a_d, q_d, u_d, r_d, y_d, C_d)$ by c. It therefore preserves both the decomposition and r_d/C_d. A suffix addition changes q_d but neither r_d nor C_d. □

Lemma 6.4 (Orthogonal Guards and Step-3 Top-Down Norm Floor)**.** *Assume the fingerprint is correct, the scale error is at most* e_C*, and previously completed corrections satisfy the rejection floor and epoch envelope proved below. Whenever* $\widehat{C}_d \tilde{\delta}_{g,d} \geq 2h_g$*,*

$$\frac{\delta_{g,d}}{C_d \sqrt{\mathcal{U}_{d,k_d}}} \in [\gamma_g/15, \gamma_g] \subseteq [\gamma_g/16, \gamma_g].$$

At the end of Step 3, every active input satisfies $\|x_d\|_2 > \nu$*.*

Proof. The coordinates $g(d,t) \leq n + T$ are pairwise distinct and have zero coefficient in every vector before their injection. The error in $\widehat{C}_d \tilde{\delta}_{g,d}$ is at most $e_C \gamma_g \sqrt{\mathcal{U}_*}/4 \leq h_g/2^{12}$, and nearest-grid rounding adds at most $h_g/2$. Dividing by the true amplitude, which is at least $2h_g - h_g/2^{12}$, proves the displayed interval. An inactive guard is zero, so the upper bound holds for every guard.

At the root, $C_0 = \widehat{C}_0 = 1$. If $k_0 = 0$, then $\mathcal{U}_{0,0} \geq 2W^2 \geq 8$. Otherwise, the last rejection of the preceding completed root epoch had, by its already proved Step-4 certificate,

$$E_0 \geq \frac{\eta^2}{32} \|x_0\|_2^2 \geq \frac{\eta^2}{32}, \qquad E_0 < \mathcal{U}_{0,k_0-1}.$$

Hence $\mathcal{U}_{0,k_0} \geq 2^{-6}\eta^2$ in both cases, and

$$\tilde{\delta}_{g,0} \geq 2^{-5}\gamma_g \eta = 2^{H+15}\nu, \qquad \delta_{g,0} \geq \frac{4}{5}\tilde{\delta}_{g,0} > 2^{H+14}\nu.$$

The grid condition ensures that this guard is active. All births precede Step 3. Thus every deeper anchor a_d and current u_d has zero coefficient on the new coordinate $g(0,t)$ when that coordinate arrives. Later guards use different coordinates, and no correction occurs until Step 4. Consequently $(x_d)_{g(0,t)} = \delta_{g,0}$ for every $1 \leq d \leq H$. The root itself has norm at least one by Lemma 6.1. Only earlier root corrections were used in this argument. □

Theorem 6.5 (Step-4 Height-ℓ Accuracy and Non-Circular Two-Step Contraction)**.** *Suppose the Step-3 norm floor, the applicable bank certificates, and the pre-query cache bound of Lemma 6.7 hold. A height-*ℓ *call returns*

$$|\widehat{F}_d - \|x_d\|_2^2| \leq 8(\ell + 1)\eta \|x_d\|_2^2.$$

Every rejection has positive residual and child energies, uses $1 + a_{\text{step}} \in [1/2, 3/2]$*, and contracts the residual energy by a factor at most* $1 - \eta^2/64$*.*

Proof. The height-zero accumulator is exact. During its one-round lifetime it receives one physical update and distinct orthogonal guards, followed by at most H dyadic rescalings. Its squared norm therefore lies on the grid h_{leaf}.

For the inductive step, abbreviate

$$x = x_d, \quad r = r_d, \quad y = y_d, \quad F = \|x\|_2^2, \quad E = \|r\|_2^2, \quad Q = \|y\|_2^2, \quad S = P + \widehat{Q}.$$

The child guarantee is $|\widehat{Q} - Q| \le 8\ell\eta Q$. On acceptance, Theorem 6.2 gives $|F - S| \le (\theta_\ell + \eta/4)(S + \omega)$. Since $F > \nu^2$, (1)–(2) yield $|F - S| \le 8(\ell + 1)\eta F$. On rejection, the numerical handoff will give the same accuracy once the correction is justified.

Step A: obtain energy floors before division. Let $A = 2\omega G^2 + TG^2 D_* h_P$. The cache bound and (2) give

$$|P - E| \le \frac{\eta}{2^{12}} E + A, \qquad A \le (2^{-32} + 2^{-39})\eta E_{\min} < 2^{-31}\eta E_{\min} < 2^{-39}\eta\nu^2.$$

Put $D = E + Q \ge F/2 > \nu^2/2$, so this additive error is absorbed without assuming a lower bound on either E or Q. With $|S - D| \le 8\ell\eta D + A$ and $b := \theta_\ell - \eta/4 = (8\ell + 15/4)\eta$, the rejection certificate $\varsigma(F - S) \ge b(S + \omega)$ gives $2\varsigma \langle r, y\rangle = \varsigma(F - D) \ge bD - (1 + b)|S - D|$. Since $8\ell(8\ell + 15/4) < (8\ell + 4)^2$ and $(8H + 4)^2\eta < 2^{-14}$, we obtain

$$\begin{aligned} 2\varsigma \langle r, y\rangle &\ge \left(\frac{15}{4} - 2^{-11} - (8\ell + 4)^2\eta\right)\eta(E + Q) \\ &\ge 2.4\eta(E + Q) \ge \frac{\eta}{2}(E + Q) \ge \eta\sqrt{EQ}. \end{aligned} \tag{6}$$

Cauchy–Schwarz first shows $E, Q > 0$. It then gives

$$\sqrt{E/Q} \in [\eta/4, 4/\eta], \qquad \min(E, Q) \ge \frac{EQ}{E + Q} \ge \frac{\eta^2}{16}(E + Q) \ge \frac{\eta^2}{32} F > E_{\min}. \tag{7}$$

Step B: now use relative estimates. Dividing only after (7), we get

$$|P/E - 1| \le \eta/2^{11}, \qquad |\widehat{Q}/Q - 1| \le 8\ell\eta < 2^{-15}.$$

Rounding the proposed step changes its normalized magnitude by at most $2h_{\mathrm{step}}/\eta$. Thus its sign remains ς, and

$$a_{\mathrm{step}} = \varsigma t_{\mathrm{eff}} \sqrt{E/Q}, \qquad t_{\mathrm{eff}} \in [\eta/32, \eta/8].$$

Together with (7), this gives $|a_{\mathrm{step}}| \le 1/2$. By (6),

$$E^+ = E - 2a_{\mathrm{step}} \langle r, y\rangle + a_{\mathrm{step}}^2 Q \le (1 - \eta t_{\mathrm{eff}} + t_{\mathrm{eff}}^2)E \le (1 - \eta^2/64)E. \tag{8}$$

No relative estimate of E or Q was used to obtain their floors.

The local update leaves x_d unchanged. Once its counter bounds have been extended through this update, the fresh handoff $N_1(x_d)$ has error at most $2^{-13}\eta F + \omega \le \eta F$ by Theorem 6.2. This completes the height induction. At the root, $8(H + 1)\eta \le \varepsilon/2$. □

Theorem 6.6 (Epoch Envelope, First-Correction Guard Bound B, and Lifetime Cap J_*)**.** *On a prefix satisfying the scale precision and the correction certificates above, each epoch starts with $\|r_d/C_d\|_2^2 \le \mathcal{U}_{d,k}/2$ and maintains $\|r_d/C_d\|_2^2 < \mathcal{U}_{d,k}$. The product of local child multipliers over any interval within one learner's lifetime is less than B. Consequently,*

$$C_d \le B^d, \qquad \|x_d\|_2 \le V_d,$$

all relevant augmented norms are at most V, all forward products needed for precision are at most G, and each learner makes at most J_* corrections.

Proof. At birth, $C_d = 1$ and $\|r_d\|_2^2 \le V_d^2 \le \mathcal{U}_{d,0}/2$. By Lemma 6.3, suffix additions and ancestor rescalings do not change the normalized residual. Each guard adds at most $\gamma_g^2 \mathcal{U}_{d,k}$ to its squared norm, and each local correction contracts it by (8). There are fewer than T guard injections. Throughout epoch k,

$$\|r_d/C_d\|_2^2 \le (1/2 + T\gamma_g^2)\mathcal{U}_{d,k} \le \frac{33}{64}\mathcal{U}_{d,k} < \mathcal{U}_{d,k}.$$

After K_0 corrections, the stronger bound is

$$\|r_d/C_d\|_2^2 \le \left(\frac{1}{2}e^{-2} + \frac{1}{64}\right)\mathcal{U}_{d,k} < \frac{1}{4}\mathcal{U}_{d,k} = \frac{1}{2}\mathcal{U}_{d,k+1}.$$

This proves the buffered two-level envelope.

Let $P_{\mathrm{int}}[t_1, t_2]$ be the product of this learner's local multipliers over an interval in its lifetime. If there is no correction, the product is $1 < B$. Otherwise let t_* be its first correction in that interval. At that correction, (7) and the envelope give, in physical units,

$$C_d(t_*)\sqrt{\mathcal{U}_{d,k(t_*)}} \ge \|r_d(t_*)\|_2 \ge \sqrt{E_{\min}} = \frac{\eta\nu}{16}.$$

Therefore

$$\widehat{C}_d(t_*)\tilde{\delta}_{g,d}(t_*) \ge \frac{\gamma_g}{4}\left(\frac{\eta\nu}{16} - e_C\sqrt{\mathcal{U}_*}\right) > \frac{\gamma_g\eta\nu}{128} > 2h_g.$$

This proves activation without any lower bound on $C_d(t_*)$. The guard lemma now gives $\delta_{g,d}(t_*)/C_d(t_*) \ge (\gamma_g/16)\sqrt{\mathcal{U}_{d,k(t_*)}}$.

On the coordinate $g(d, t_*)$, both a_d and q_d remain zero. Each later local correction multiplies that residual coordinate by its child multiplier; ancestor factors cancel against C_d. No later additive update touches the coordinate. Orthogonality and the nonincreasing epoch envelope thus give

$$\left(\frac{\gamma_g}{16}\sqrt{\mathcal{U}_{d,k(t_*)}}\, P_{\mathrm{int}}[t_*, t_2]\right)^2 \le \left\|r_d(t_2^+)/C_d(t_2^+)\right\|_2^2 < \mathcal{U}_{d,k(t_*)}.$$

There were no earlier corrections in the interval, so $P_{\mathrm{int}}[t_1, t_2] < 16/\gamma_g = B$.

The time intervals are nested. Hence C_d is a product of at most d such ancestor products and is at most B^d. The same reasoning bounds every forward counter or scale product by $B^{H+2} = G$; a reset only shortens the product. Since $\mathcal{U}_{d,0} < 4V_d^2$, inductively

$$\|x_{d+1}\|_2 \le \|x_d\|_2 + \|r_d\|_2 \le V_d + 2B^d V_d = V_{d+1}.$$

The bounds on residuals, differences, and $\sigma \le 1$ then give augmented norm at most $V = 8B^{H+1}V_H + 2$.

Finally, immediately before the last correction of a completed epoch k, (7) gives

$$2^{-k}\mathcal{U}_* \ge \mathcal{U}_{d,k} > \|r_d/C_d\|_2^2 \ge E_{\min}/B^{2H}.$$

Thus the epoch index cannot reach K_{epoch} with another completed epoch; its definition includes twelve bits of slack. There are at most $K_{\mathrm{epoch}} + 1$ epochs containing corrections, each with at most K_0 corrections, proving the lifetime cap J_*. □

6.4 Shadow Precision, Covariant Caches, and the Main Guarantee

Counter errors are propagated along the same homogeneous updates as the exact vectors. A cached norm needs a different argument: its relative error follows the current residual energy, rather than paying the largest possible squared scale factor. We establish these pathwise facts before assembling the statistical and geometric inductions.

Lemma 6.7 (Pathwise Shadow-Vector Drift and Covariant Cache Invariant)**.** *On a prefix with forward products bounded by G and augmented norms bounded by V,*

$$|\widehat{C}_d - C_d| \le e_C, \qquad \xi := G(T+1)h_b \le \frac{\eta\omega}{2^{40}D_*(V+1)}$$

bounds each primary counter error, and 2ξ bounds each difference counter error. After a valid numerical refresh, the residual cache maintains

$$|P_t - E_t| \le \frac{\eta}{2^{12}}E_t + 2\omega G^2 + TG^2D_*h_P. \tag{9}$$

These bounds also certify that silent counter saturation is not reached.

Proof. Between resets, a scalar rounding error satisfies $e_{\tau+1} = a_\tau e_\tau + \zeta_\tau$. There are fewer than T affine operations on any counter trajectory. Unrolling, with every forward product at most G, gives

$$|\widehat{C}_d - C_d| \le GTh_C \le e_C, \qquad |e_{\text{primary}}| \le GTh_b < \xi.$$

The inherited primary array is not reset at a new client, but this argument starts at its bank's birth and already covers both clients. Subtracting two counters gives error at most 2ξ. The grid bound implies

$$2\sqrt{D_*}V\xi + \xi^2 < \eta\omega/2^{38} < \eta\omega/2^{30}.$$

Exact row projections have magnitude at most $\sqrt{D_*}V$. The saturation level $Z_{\max} \ge 2\sqrt{D_*}V + 2$ from Definition A.5 is therefore never reached; equivalently, a first saturation would contradict the same unrolled bound.

For the cache, start at its last refresh t_0. Until the next local correction, its exact energy obeys

$$E_{\tau+1} = c_\tau^2 E_\tau + \delta_{g,\tau}^2.$$

The additive terms are nonnegative, so, writing $A_t = \prod_{\tau=t_0}^{t-1} c_\tau^2$, we have $A_t E_{t_0} \le E_t$. The initial numerical error $2^{-13}\eta E_{t_0} + \omega$ therefore contributes at most $2^{-13}\eta E_t + \omega G^2$, not $2^{-13}\eta G^2 E_{t_0}$. Unrolling the scalar rounding errors adds at most $TG^2D_*h_P$. Allowing slack for refresh rounding proves (9). By (2), its additive part is less than $2^{-31}\eta E_{\min}$, so when $E_t \ge E_{\min}$ its relative contribution is below $2^{-31}\eta$, well within the slack used in Step B. □

Proof of Theorem 1.3. Fix an adaptive adversary. First expose the statistical events, without assuming geometric correctness or accurate counters. For each scheduled bank, every ADA query is a bounded function of its row and the previously exposed transcript, even on tapes with inaccurate rounding or clamping. Unread coarse groups never influence the controller. Conditional on the history before such a group is read, its exact shadow target is independent of its rows. Thus Theorem 1.2 and Lemma 3.8 give the unconditional bank event in Theorem 6.2. There are at most $2m(H+1)$ scheduled bank births. Together with Lemma 6.1, a union bound gives

$$\mathbb{P}[\text{some statistical event fails}] < \frac{\beta_0}{8} + 2m(H+1)\beta_B = \frac{5\beta_0}{32} < \beta.$$

Condition on their intersection. We induct in exact execution order: by round t, and within Step 4 by increasing height ℓ. At scheduled births, child banks start with zero inputs, while an inherited primary array keeps its already certified shadow history. Steps 1–2 therefore require neither replay nor a new accuracy assumption. Numerical initialization is covered by the bank certificate, which also applies to zero or arbitrarily small targets.

On a nonzero round, Lemma 6.1 gives the root's integer norm floor. Lemma 6.4 uses only earlier completed root corrections to establish the Step-3 floor for every depth. Guard injection extends the epoch envelope and the shadow bounds without introducing a new homogeneous multiplier.

At a Step-4 visit, the child answer is already certified. The current shadow bounds give Theorem 6.2, and the current cache bound gives Step A of Theorem 6.5. Only after that step establishes $E, Q \geq E_{\min}$ do we use relative estimates and perform the correction. Equation (8) first extends the epoch envelope. The first-correction argument in Theorem 6.6 then bounds the newly extended scale products. Lemma 6.7 certifies the resulting counter state, after which the two post-correction numerical releases are valid and the height induction returns its certified answer. This order never uses post-correction counter accuracy to justify the correction itself.

Each client has at most J_* corrections and uses at most one initialization release and two releases per correction. Hence a bank's at most two clients satisfy

$$J \leq 2J_* = J_{\max}, \qquad R \leq 4J_* + 2 = R_{\max},$$

and there are at most m validation calls. The epoch slack excludes a first cap violation; the norm and precision bounds exclude any saturation, clamping, or absorbing failure transition. Thus the induction covers the entire execution.

The zero rounds are exact. On every other round, the root guarantee is $(1-\varepsilon/2)F_2(x^{(t)}) \leq \widehat{F}_t \leq (1+\varepsilon/2)F_2(x^{(t)})$. Since $\varepsilon < 1$, this gives simultaneously

$$\frac{F_2(x^{(t)})}{1+\varepsilon} \leq \widehat{F}_t \leq (1+\varepsilon)F_2(x^{(t)}) \qquad (1 \leq t \leq m).$$

Finally, Lemma A.7 and Table 3 in Section A.3 bound the total memory on every execution—including capped and absorbing-failure runs—by $O(\varepsilon^{-2}\Lambda^{15})$ bits (certified $2^{512}\varepsilon^{-2}\Lambda^{15}$ bits; Section 7). The controller processes each physical update once, and its bank births and retirements follow the fixed time tree; no rejection or epoch transition restarts a bank. □

7 Formal Verification in Lean 4

While Sections 3 to 6 cite standard concentration, differential privacy, online convex optimization, and $2s$-wise hashing lemmas [20, 27, 3, 8, 18, 19, 35, 33, 10, 2], the accompanying Lean 4 [14] archive `f2e2e_simple_lean.zip` (`F2e2e`, built on Mathlib [31] under `leanprover/lean4:v4.33.1`) proves Theorems 1.1 to 1.3—including the exact $2^{30}N(b_{\mathcal{U}} + \chi^2)$ and $2^{512}\varepsilon^{-2}\Lambda^{15}$ small-step bit-space bounds—end to end from Lean's three logical axioms (`propext`, `Classical.choice`, `Quot.sound`), with zero external paper assumptions and zero `sorry`s.

7.1 Formal Specification and Computational Models

All theorem statements, operational semantics, and cost models reside in the self-contained specification `F2e2e/Spec.lean`, and `F2e2e/Headline.lean` exports the proved six-part conjunction:

```
def Theorem_main_full : Prop :=
  Theorem_A_statistical ∧ Theorem_A_eff ∧
  Theorem_B_statistical ∧ Theorem_B_space ∧ Theorem_B_joint ∧ Theorem_C

theorem theorem_main_full : Theorem_main_full :=
  ⟨theorem_A_statistical, theorem_A_eff, theorem_B_statistical,
   theorem_B_space, theorem_B_joint, theorem_C⟩
```

The verbatim specification definitions in `F2e2e/Spec.lean` corresponding to Theorems 1.1 to 1.3 are:

```
def Theorem_A_statistical : Prop :=
  ∃ C : ℝ, 0 < C ∧
    ∀ (k : ℕ) (α β : ℝ), 1 ≤ k → 0 < α → α < 1 / 16 → 0 < β → β < 1 / 2 →
      IsProbabilityMeasure (adaCoinLaw k α β) ∧
        (adaCascade k α β).JointlyMeasurable ∧
        (adaSampleSize k α β : ℝ) ≤ adaExplicitBound k α β ∧
        (adaSampleSize k α β : ℝ) ≤ C * adaAsymp k α β ∧
        ADAPublicGuarantee (adaSampleSize k α β) k α β
          (adaCoinLaw k α β) (adaCascade k α β)

def Theorem_A_eff : Prop :=
  ∃ C : ℕ, ∃ P : AEff.RamProgram, P.RatLits ∧
    ∀ (k : ℕ) (α β : ℝ), 1 ≤ k → 0 < α → α < 1 / 16 → 0 < β → β < 1 / 2 →
      ∀ (qv : ℕ → Fin (adaSampleSize k α β) → ℝ)
        (ω : ADANoiseIndex k (adaL α) → ℝ),
        P.RunsWithin (AEff.adaInput k α β) k (AEff.adaOracle qv ω)
          (fun i out => AEff.ADAOutputStreamed (adaL α) out
            (adaCascade k α β i (fun j : Fin (i + 1) => qv j) ω))
          (C * adaSampleSize k α β) (C * adaSampleSize k α β)

def Theorem_B_statistical : Prop :=
  ∃ C : ℝ, 0 < C ∧
    ∀ (M J R : ℕ) (aReq ξReq : ℝ),
      0 < aReq → aReq < 1 / 10 → 0 < ξReq → ξReq < 1 / 2 →
      ((alg2Params M J R aReq ξReq).N : ℝ) ≤ valNExplicit M J R aReq ξReq ∧
        ((alg2Params M J R aReq ξReq).N : ℝ) ≤ C * valNAsymp M J R aReq ξReq ∧
        ∃ c : ℕ, VGuaranteeWithInternalCaps M J R aReq ξReq
          (alg2Mech M J R aReq ξReq c) (alg2Aborted M J R aReq ξReq c)
          (alg2PeakCorrections M J R aReq ξReq c)

def Theorem_B_space : Prop := BSpace.Theorem_B_space_with 30
def Theorem_B_joint : Prop := BSpace.Theorem_B_joint_with 30
```

```
def RunsWithinBits (prog : List Expr) (input : Val) (tape : List Bool)
    (out : Val) (S : ℝ) : Prop :=
  ∃ (n : ℕ) (rest : List Bool),
    stepN prog n (initConf input tape) = some ⟨Ctrl.ret out, [], rest⟩ ∧
    ∀ k < n, ∀ c c' : Conf,
      stepN prog k (initConf input tape) = some c → step prog c = some c' →
      ((confBits prog c + confBits prog c' +
        stepWork prog c + ctrlWork prog c c' : ℕ) : ℝ) ≤ S

def StreamAccurate {n : ℕ} (m U : ℕ) (ε : ℝ) (adv : StreamAdversary n)
    (outs : ℕ → ℚ) : Prop :=
  ∀ t < m, BoxUpTo U adv outs t →
    F2 (streamVec adv outs t) / (1 + ε) ≤ (outs t : ℝ) ∧
      (outs t : ℝ) ≤ (1 + ε) * F2 (streamVec adv outs t)

def Theorem_C : Prop :=
  ∃ alg : StreamAlgorithm,
    ∀ (n m U : ℕ) (ε β : ℝ),
      1 ≤ n → 1 ≤ m → 1 ≤ U → 0 < ε → ε < 1 / 10 → 0 < β → β < 1 / 10 →
      (∀ (adv : StreamAdversary (mkStreamParams n m U ε β).n)
          (tapes : Fin ((mkStreamParams n m U ε β).m + 1) →
            Fin (alg.tapeLen (mkStreamParams n m U ε β)) → Bool),
```

```
    ∃ (states : ℕ → Val) (outs : ℕ → ℚ),
      StreamExec alg (mkStreamParams n m U ε β)
        (spaceBoundC n m U ε β) adv tapes states outs) ∧
  (∀ (Ω : Type) [MeasurableSpace Ω] (P : Measure Ω)
      [IsProbabilityMeasure P]
      (adv : Ω → StreamAdversary (mkStreamParams n m U ε β).n),
    (∀ h : List ℚ, Measurable (fun ω => adv ω h)) →
      ∑ tapes : Fin ((mkStreamParams n m U ε β).m + 1) →
          Fin (alg.tapeLen (mkStreamParams n m U ε β)) → Bool,
        P {ω | ¬ ∃ (states : ℕ → Val) (outs : ℕ → ℚ),
            StreamExec alg (mkStreamParams n m U ε β)
              (spaceBoundC n m U ε β) (adv ω) tapes states outs ∧
            StreamAccurate m U ε (adv ω) outs} ≤
        ENNReal.ofReal β * 2 ^ (((mkStreamParams n m U ε β).m + 1) *
          alg.tapeLen (mkStreamParams n m U ε β)))
```

Four modeling invariants in `F2e2e/Spec.lean` foreclose vacuity or hidden relaxations:

- *Pinned mechanisms and joint tape witness.* In `Theorem_A_statistical`, the mechanism is literally `adaCascade` (Algorithm 1) under Definition A.1, with `ADAPublicGuarantee` exposing the full auxiliary record `ADAField` ($\tau_{\mathrm{aux},i}$) to the analyst. In `Theorem_B_joint` (defined as `BSpace.Theorem_B_joint_with 30`), a single threshold $c_0 \in \mathbb{N}$ witnesses both `VGuaranteeWithInternalCaps` (jointly bounding inaccuracy, $T_j^{\max}$ violations, and internal/tape aborts) and `ReproducesFor 30` on `alg2Mech M J R aReq ξReq c` for *every* fair-bit tape length $c \geq c_0$.
- *Uniform rational-literal real-RAM* (`Theorem_A_eff`). One program P fixed prior to (k, α, β) with rational constants (`P.RatLits`, ruling out real advice), starting from the preloaded public input `AEff.adaInput k` α β and querying external query/noise-table oracles `AEff.adaOracle`, streams $(a_i, \tau_{\mathrm{aux},i})$ within $C \cdot N_{\mathrm{cas}}$ operations per round and $C \cdot N_{\mathrm{cas}}$ workspace cells.
- *Small-step CEK bit machine* (`RunsWithinBits`, `RunsWithinBits0`). Space is charged in literal bits at every transition $c \to c'$: `confBits prog c + confBits prog c'` (where `bitEncodeVal` encodes integer z in $2\,\mathrm{bitLen}(z) + 3$ bits and a list of m items in $2 + m$ overhead bits plus element lengths), plus `stepWork` (live register-file trace `opWork` of bit-level integer arithmetic, satisfying `opWork` $\leq 8(|v_a| + |v_b|) + 64$) and `ctrlWork` (four code-position cursors plus 64 control bits); unread fair bits are uncharged and reading past the tape gets stuck. In `Theorem_B_space`, `OracleFor` answers $\lfloor 2^q f(s_\ell) \rfloor$ on encoded sample records and $\lfloor 2^q \,\mathrm{clip}_{[-1,2]}(\cdot) \rfloor$ on endpoints (giving $\mathrm{round}_{h_A}(z)$ via $(\lfloor 2^{q_A+1} z \rfloor + 1)/2$) while leaving off-sample codes unconstrained so no off-sample query can occur, and `ReproducesFor` requires every between-request state to equal a fixed sample-dependent encoding of `alg2RetainedOf` (retaining only in-block scores $G_{j,i}$ for $j < p, i < n_j$ and persistent counters/thresholds).
- *Scope alignment on Theorem 1.3.* In `Theorem_C`, `StreamExec` runs `alg.initProg` and each round's `alg.updProg` within `spaceBoundC` ($2^{512}\varepsilon^{-2}\Lambda^{15}$ bits) up to the first ℓ_∞-box violation (`BoxUpTo`), and `StreamAccurate` certifies simultaneous multiplicative accuracy $F_2(x^{(t)})/(1+\varepsilon) \leq \widehat{F}_t \leq (1+\varepsilon)F_2(x^{(t)})$ with probability $\geq 1 - \beta$ against every measurable randomized adaptive adversary (`F2e2e/C/Alg/StrictModel.lean` additionally proves single-pass no-replay and fixed-birth tree invariants on the constructed algorithm).

7.2 Trust Boundary and Independent Kernel Audits

Trusting `theorem_main_full` reduces to four items: (1) the Lean 4 kernel (`v4.33.1`), (2) the Mathlib definitions referenced in `F2e2e/Spec.lean` (`MeasureTheory.Measure`, `IsProbabilityMeasure`,

Measurable, Real.log, Real.sqrt, Fintype), (3) the specification F2e2e/Spec.lean (literal SHA-256 0bda75467c74b470966a37c00c6f3fa0c0c31437b67078e9c83ca5ac6def35f0, comment-stripped SHA-256 79c6cc1cbd8304adb52dd654db1facafa60aa1890e9af7b9bd1eaeebcde2263d via scripts/strip_check.py), and (4) Lean's standard axioms (propext, Classical.choice, Quot.sound). The 654 proof modules (84,399 exported kernel declarations; Table 2) in f2e2e_simple_lean.zip pass three reproducible audit gates: (1) *fresh replay* via lake build F2e2e.Headline and leanchecker --fresh F2e2e.Headline; (2) *static and axiom gates* via bash scripts/gate.sh --worktree and F2e2e.GateCheck (#print axioms), confirming zero occurrences of sorry, sorryAx, axiom, native_decide, implemented_by, extern, or unsafe and import-DAG isolation of F2e2e/Spec.lean; and (3) *independent external kernels* via sandboxed comparator in Lean's kernel, the Rust kernel nanoda, and con-leche --verified (84,399/84,399 declarations accepted).

Table 2: Main theorems and proof-source declarations in F2e2e (654 files, 84,399 declarations).

Paper Result	**Lean Module**	**Proof Declaration & Certified Guarantee**
Theorem 1.1 (stat.)	F2e2e/A/Final.lean	A.theorem_A_statistical_proof → theorem_A_statistical (adaCascade).
Theorem 1.1 (RAM)	F2e2e/AEff/Final.lean	AEff.theorem_A_eff_proof → theorem_A_eff (RatLits RAM, $O(N_{\text{cas}})$).
Theorem 1.2 (stat.)	F2e2e/B/Final.lean	B.theorem_B_statistical_proof → theorem_B_statistical (alg2Mech).
Theorem 1.2 (bits)	F2e2e/BSpace/O2AssembleFinal.lean	BSpace.theorem_B_{space,joint}_proof ($2^{30}N(b_{\mathcal{U}}+\chi^2)$ bits, alg2RetainedOf).
Theorem 1.3 (stat.)	F2e2e/C/Final.lean	C.Closure.theLawBound: adaptive accuracy of the tree/bank sketch.
Theorem 1.3 (bits)	F2e2e/C/Impl/Assemble.lean	C.Impl.Asm.theorem_C_of_lawBound → theorem_C ($2^{512}\varepsilon^{-2}\Lambda^{15}$ bits).
Full conjunction	F2e2e/Headline.lean	theorem_main_full : Theorem_main_full, assembling all six propositions.

Acknowledgments

The results of Theorems 1.1 to 1.3 were discovered by Gemini Colosseum. The update $u^+ = u + a_{\text{step}}\, y$ (with $y = u + q$) of the geometric learners in Section 6 has the same form as the learner update of [23]. The identity $y^+ = (1 + a_{\text{step}})\, y$, which lets descendant learners persist across ancestor corrections instead of being restarted, was discovered by Gemini Deep Think.

A Locked Parameter Schedules, Precision Grids, and Lean Dictionary

This appendix records the exact locked parameter schedules and dyadic precision grids certified in F2e2e/Spec.lean and F2e2e/C/ParamsDefs.lean and proves the explicit sample-size and bit-length bounds invoked in Sections 4 to 6.

A.1 Cascade ADA Parameters and Explicit Sample Bound

All logarithms in Section A.1 are natural logarithms (ln).

Definition A.1 (Locked Cascade Parameters)**.** For integers $k \geq 1$ and reals $\alpha \in (0, 1/16)$, $\beta \in (0, 1/2)$, Algorithm 1 (`adaCascade` in `F2e2e/Spec.lean` §1.1, §1.3) fixes:

(P1) *Depth and margins:* $L := \max(4, \lceil \ln(1/\alpha) \rceil + 2)$ (`adaL`), $\gamma := \alpha/(8L)$ (`adaGamma`), and $u_0 := \gamma/32 = \alpha/(256L)$ (`adaU0`).

(P2) *Stage-1 learner and Beta noise:* $\varepsilon_1 := 1/20$, $\delta_1 := \alpha\beta/10240$, $L_1 := \ln(2/\delta_1) = \ln(20480/(\alpha\beta))$ (`adaL1`), $\chi_1 := \sqrt{L_1}$ if $\beta \geq \alpha^5$ and $\chi_1 := L_1$ otherwise (`adaChi1`), $R_1 := \alpha/32$ (`adaR1`), $\eta_1 := \alpha/16$ (`adaEta1`), Beta(p_1, p_1) shape $p_1 := 8$ if $\beta \geq \alpha^5$ and $p_1 := \lceil L_1 \rceil + 1$ otherwise (`adaP1`, density exponent $q_B := p_1 - 1$), cap $T_{1,\max} := \lceil 600/\alpha^2 \rceil$ (`adaT1max`), and block size $n_1 := \lceil 32000\sqrt{k}\,\chi_1/\alpha + 16000L_1^3/\alpha^2 \rceil$ (`adaN1`).

(P3) *Validator and anchor privacy:* $\varepsilon_r := \frac{1}{20}\alpha^{(r-1)/(L-1)}$ for $2 \leq r \leq L-1$ (`adaEps`), $\varepsilon_L := u_0/2 = \alpha/(512L)$ (`adaEpsL`), $r_{\text{anc}} := 4\varepsilon_L = 2u_0$ (`adaRanc`), $\delta_{\text{val}} := u_0^3\beta/(10^8 L^3)$ (`adaDeltaVal`), and $L_{\text{val}} := \ln(4/\delta_{\text{val}})$ (`adaLval`).

(P4) *Threshold matrix (`adaTau`):* $\tau_{(r,r)} := 0$, $\tau_{(1,r)} := \alpha/2 + 2(r-2)\gamma$ $(2 \leq r \leq L)$, and $\tau_{(u,r)} := 2(r-u)\gamma$ $(2 \leq u < r \leq L)$.

(P5) *Coupled steps, caps, and block sizes:* $w_\ell := \varepsilon_\ell^{-2}$ (`adaW`) and $\eta_\ell := \gamma/8$ (`adaEta`) for $2 \leq \ell \leq L-1$; $\tilde{\eta}_{r^*} := (w_{r^*-1}/w_{r^*})\eta_{r^*-1}$ for $3 \leq r^* \leq L-1$ and $\tilde{\eta}_L := 0$ (`adaEtaTilde`); $N_{r^*,\max} := \lceil 128L\varepsilon_{r^*-1}^2/\gamma^2 \rceil$ $(3 \leq r^* \leq L$, `adaNmax`); $M_{2,\max} := 1 + T_{1,\max} + \sum_{s=3}^{L} N_{s,\max}$ and $M_{r,\max} := 1 + \sum_{s=r}^{L} N_{s,\max}$ $(3 \leq r \leq L$, `adaMmax`); $b_{\text{SVT}} := u_0/\ln(64LkM_{2,\max}/\delta_{\text{val}})$ (`adaBsvt`); validator sizes $n_r := \lceil 110\sqrt{M_{r,\max}L_{\text{val}}}/(b_{\text{SVT}}\varepsilon_r) \rceil$ $(2 \leq r \leq L-1$, `adaNr`); anchor size

$$n_L := \left\lceil \frac{55\sqrt{M_{L,\max}L_{\text{val}}}}{b_{\text{SVT}}\varepsilon_L} + \frac{16\ln(6k/\delta_{\text{val}})}{r_{\text{anc}}\varepsilon_L} \right\rceil \qquad (\texttt{adaNL});$$

and total cascade sample size $N_{\text{cas}} := n_1 + \sum_{r=2}^{L-1} n_r + n_L$ (`adaSampleSize`).

Why does $\sum_{r=2}^{L} n_r$ avoid geometric blow-up as ε_r decays from $1/20$ to $\Theta(\alpha/L)$? Each n_r scales as $\sqrt{M_{r,\max}}/\varepsilon_r$, while $M_{r,\max}$ is dominated by $N_{r,\max} \propto \varepsilon_{r-1}^2/\gamma^2$. For interior validators $2 \leq r < L$, the ratio $\varepsilon_{r-1}/\varepsilon_r = \alpha^{-1/(L-1)} = \rho = \Theta(1)$ cancels; the anchor $r = L$ has $\varepsilon_{L-1}/\varepsilon_L = (128/5)L\rho = \Theta(L)$ and $M_{L,\max} = O(L^3)$, whose extra factor of L is included explicitly in the anchor calculation below.

Lemma A.2 (Evaluation of the Cascade Sample Bound)**.** *For all $k \geq 1$, $\alpha \in (0, 1/16)$, and $\beta \in (0, 1/2)$, N_{cas} in Definition A.1 satisfies*

$$N_{\text{cas}} \leq \frac{32000\sqrt{k}\,\chi_1}{\alpha} + \frac{16000L_1^3}{\alpha^2} + 1 + \frac{10^8}{\alpha^2}L^{7/2}\ln\left(\frac{64LkM_{2,\max}}{\delta_{\text{val}}}\right)\sqrt{L_{\text{val}}}$$

(`adaExplicitBound`), and with $\tilde{\chi}_1 := \sqrt{\ln(1/(\alpha\beta))}$ if $\beta \geq \alpha^5$ and $\ln(1/(\alpha\beta))$ otherwise, we have $N_{\text{cas}} \leq 10^{12}\big[\frac{\sqrt{k}\,\tilde{\chi}_1}{\alpha} + \frac{1}{\alpha^2}\big(\ln^3(\frac{1}{\alpha\beta}) + \ln^{7/2}(\frac{1}{\alpha})\ln(\frac{k\ln(1/\alpha)}{\alpha\beta})\sqrt{\ln(\frac{\ln(1/\alpha)}{\alpha\beta})}\big)\big]$ $(10^{12} \cdot$ `adaAsymp`$)$, proving the sample bound of Theorem 1.1.

Proof. Since $\alpha \in (0, 1/16)$, $\ln(1/\alpha) > 4\ln 2 > 2.7725$, so $\lceil \ln(1/\alpha) \rceil \geq 3$, $L = \lceil \ln(1/\alpha) \rceil + 2 \geq 5$, and $\ln(1/\alpha) \leq L - 2 < L - 1$. If $\lceil \ln(1/\alpha) \rceil = 3$ then $\frac{\ln(1/\alpha)}{L-1} > \ln 2 > \frac{3}{5}$, while for $\lceil \ln(1/\alpha) \rceil = m \geq 4$, $\frac{\ln(1/\alpha)}{L-1} > \frac{m-1}{m+1} \geq \frac{3}{5}$. Thus $\rho := \alpha^{-1/(L-1)} = \exp(\frac{\ln(1/\alpha)}{L-1}) \in (e^{3/5}, e) \subset (1.822, 2.7183)$, giving $\rho^{-2} < e^{-6/5} < 0.3012$, $\rho^2 > e^{6/5} > 3$, and $\sum_{s=r}^{L} \rho^{-2(s-r)} < (1-\rho^{-2})^{-1} < (1-e^{-6/5})^{-1} < 1.4311$.

Since $128L/\gamma^2 = 8192L^3/\alpha^2$, we have $N_{s,\max} < 8192L^3\varepsilon_{s-1}^2/\alpha^2 + 1$ $(3 \leq s \leq L)$. Set $z_r := L^3\varepsilon_{r-1}^2/\alpha^2$. At $r = L$, $\varepsilon_{L-1} = \frac{1}{20}\alpha\rho$, so $z_L = L^3\rho^2/400 > 125e^{6/5}/400 > 1$. For $3 \leq r \leq L$, since

$\varepsilon_{s-1}/\varepsilon_{r-1} = \rho^{-(s-r)}$ and $z_r = z_L\rho^{2(L-r)} > 3^{L-r}$, we have $L - r + 2 \le 2 \cdot 3^{L-r} < 2z_r$, giving

$$M_{r,\max} < (L - r + 2) + 8192 z_r \sum_{s=r}^{L} \rho^{-2(s-r)} < (2 + 8192 \cdot 1.4311)\,\frac{L^3\varepsilon_{r-1}^2}{\alpha^2} < 12000\,\frac{L^3\varepsilon_{r-1}^2}{\alpha^2}.$$

In particular, $M_{L,\max} = 1 + N_{L,\max} < 2 + 20.48\rho^2 L^3 < 152L^3$, while at $r = 2$, $T_{1,\max} < 600/\alpha^2 + 1$ and $\sum_{s=3}^{L} \varepsilon_{s-1}^2 < \frac{0.4311}{400}$ yield $M_{2,\max} < 2 + 600/\alpha^2 + (L-2) + 8.83L^3/\alpha^2 < 30L^3/\alpha^2 = 12000L^3\varepsilon_1^2/\alpha^2$. Since $\sqrt{12000} < 110$, we get $\sqrt{M_{r,\max}} < 110L^{3/2}\varepsilon_{r-1}/\alpha$ $(2 \le r \le L-1)$ and $\sqrt{M_{L,\max}} < 12.33L^{3/2}$.

Set $\Lambda_{\mathrm{SVT}} := \ln(64LkM_{2,\max}/\delta_{\mathrm{val}})$, so $1/b_{\mathrm{SVT}} = 256L\Lambda_{\mathrm{SVT}}/\alpha$. For $2 \le r \le L-1$, $\varepsilon_{r-1}/\varepsilon_r = \rho < 2.7183$ gives $n_r < 1 + 110 \cdot 110 \cdot 256\,\rho\, L^{5/2}\Lambda_{\mathrm{SVT}}\sqrt{L_{\mathrm{val}}}/\alpha^2 < 1 + 8.421 \times 10^6 L^{5/2}\Lambda_{\mathrm{SVT}}\sqrt{L_{\mathrm{val}}}/\alpha^2$, summing over $L - 2 < L$ stages to at most $(L-2) + 8.421 \times 10^6 L^{7/2}\Lambda_{\mathrm{SVT}}\sqrt{L_{\mathrm{val}}}/\alpha^2$. For the anchor, $1/\varepsilon_L = 512L/\alpha$ and $r_{\mathrm{anc}}\varepsilon_L = 4\varepsilon_L^2$ give

$$n_L < 1 + 55 \cdot 12.33 \cdot 256 \cdot 512\,\frac{L^{7/2}\Lambda_{\mathrm{SVT}}\sqrt{L_{\mathrm{val}}}}{\alpha^2} + 4 \cdot 512^2\,\frac{L^2\ln(6k/\delta_{\mathrm{val}})}{\alpha^2}.$$

Here $55 \cdot 12.33 \cdot 256 \cdot 512 < 8.891 \times 10^7$, $\ln(6k/\delta_{\mathrm{val}}) < \Lambda_{\mathrm{SVT}}$, and $L^{3/2}\sqrt{L_{\mathrm{val}}} > 49$, so the last term is $< 2.2 \times 10^4 L^{7/2}\Lambda_{\mathrm{SVT}}\sqrt{L_{\mathrm{val}}}/\alpha^2$. Absorbing $(L-2)+1$ into $(10^8 - 9.74 \times 10^7)L^{7/2}\Lambda_{\mathrm{SVT}}\sqrt{L_{\mathrm{val}}}/\alpha^2$ and adding $n_1 < 32000\sqrt{k}\,\chi_1/\alpha + 16000L_1^3/\alpha^2 + 1$ proves $N_{\mathrm{cas}} \le \texttt{adaExplicitBound}(k, \alpha, \beta)$. Substituting $L \le 2.1\ln(1/\alpha)$, $L_1 \le 4\ln(1/(\alpha\beta))$, $L_{\mathrm{val}} \le 16\ln(\ln(1/\alpha)/(\alpha\beta))$, and $\Lambda_{\mathrm{SVT}} \le 20\ln(k\ln(1/\alpha)/(\alpha\beta))$ yields $N_{\mathrm{cas}} \le 10^{12} \cdot \texttt{adaAsymp}(k, \alpha, \beta)$ (and log-k absorption gives Corollary 4.7). □

A.2 Interval-Validation ADA Parameters, Sample Size, and Bit Budget

All logarithms in Definition A.3 are base-2 logarithms ($\log_2$).

Definition A.3 (Locked Validation ADA Parameters)**.** For caps $M, J, R \in \mathbb{N}$ and $a_{\mathrm{req}} \in (0, 1/10)$, $\xi_{\mathrm{req}} \in (0, 1/2)$, Algorithm 2 (`alg2Params` in `F2e2e/Spec.lean` §2.1–§2.4) sets:

(V1) *Dyadic inputs and depth:* $a := 2^{-\lceil \log_2(1/a_{\mathrm{req}})\rceil} \in (a_{\mathrm{req}}/2, a_{\mathrm{req}}]$ $(a \le 1/16)$ and $\xi_A := 2^{-\lceil \log_2(1/\xi_{\mathrm{req}})\rceil} \in (\xi_{\mathrm{req}}/2, \xi_{\mathrm{req}}]$ (`dyadicDown`); depth $p := \min\{p' \ge 2 : 4^{p'-1}(a/(8p'))^2 \ge 1\}$ (`valP`); margin $\tau := a/(8p)$ (`valTau`); precision unit $h := \tau/64 = a/(512p)$ (`valH`); and learner step $\kappa := \tau/4$ (`valKappa`).

(V2) *Caps, phase budgets, and log factor:* $T_j^{\max} := \lceil 6/(4^j\tau^2)\rceil$ for $0 \le j < p$ (`valTmax`) and 0 otherwise (`valTcap`); phase budgets (`valC`) $C_0 := T_0^{\max} + T_1^{\max} + J + R + 2$, $C_p := T_{p-2}^{\max} + T_{p-1}^{\max} + 1$, and

$$C_j := T_{j-2}^{\max} + 2T_{j-1}^{\max} + 2T_j^{\max} + T_{j+1}^{\max} + 2 \qquad (1 \le j < p, \text{ with } T_{-1}^{\max} := 0);$$

draw bound $Q := 100(p+1)(M + R + J + 2 + \sum_{j=0}^{p-1} T_j^{\max})$ (`valQ`); $\xi_j := \xi_A/(8(p+1))$ (`valXiJ`); $\delta := h\xi_j/24$ (`valDelta`); and $\chi := \lceil \log_2(2^{40}(p+1)Q/(h\xi_A))\rceil$ (`valChi`).

(V3) *Sensitivities and block sizes:* $d_j := \lceil \sqrt{p\,4^j}\rceil$ $(0 \le j < p)$ and $d_p := \lceil 4/h\rceil$ (`valD`, $\vartheta_j := 1/d_j$); block sizes $n_j := \lceil 2^{20}\chi\, d_j \lceil \sqrt{C_j\chi}\rceil/h + 2^{12}\chi/h^2\rceil$ $(0 \le j \le p$, `valNBlock`); and total $N := \sum_{j=0}^{p} n_j$.

(V4) *Lattice step, noise scales, and bounded sampler:* $h_A := 2^{-q_A}$ is the largest power of two $\le \min\{h/(2^{20}(p+1)\chi),\, 1/(2^{20}\max_{0\le j\le p} n_j)\}$ (`valHAExp`); rational scales $b_{\mathrm{int}} := h/(128\chi)$ (thresholds $2b_{\mathrm{int}}$, tests $4b_{\mathrm{int}}$), $b_{\mathrm{out}} := a/(512\chi)$ (threshold $2b_{\mathrm{out}}$, tests $4b_{\mathrm{out}}$), $b_{\mathrm{num}} := a/(64\chi)$; and sampler budget $\zeta := \delta/(1000Q)$ (`valZeta`), $B_0 := \lceil \log_2(16/\zeta)\rceil$. A draw of scale $b \in \mathbb{Q}_{>0}$ reduces $\varrho := b/(b + h_A) = A/D_\varrho$ to lowest terms and outputs $h_A(G_1 - G_2)$ from two geometric counts capped at $K_s := \lceil ((b + h_A)/h_A)B_0\rceil$, with $r_s := \lceil \log_2(32(K_s+1)/\zeta)\rceil$ rejection trials of $\lceil \log_2 D_\varrho\rceil$ fair bits per coin (`latticeSample`).

In middle blocks ($1 \le j < p$), $d_j \asymp 2^j\sqrt{p}$ and $\sqrt{C_j} \asymp 2^{-j}\tau^{-1}$ cancel 2^j, keeping all n_j balanced.

Lemma A.4 (Validation ADA Sample Size and Bit-Space Accounting)**.** *Under Definition A.3 with $K := J + R + 1$, we have $12 \le p < 6\ln(1/a_{\mathrm{req}})$, the sample size satisfies*

$$N = \sum_{j=0}^{p} n_j \le 2^{64}\left[\left(\frac{p^{7/2}}{a^2} + \frac{p^{3/2}\sqrt{K}}{a}\right)\chi^{3/2} + \frac{p^3\chi}{a^2}\right]$$
$$\le 2^{84}\left(\frac{\sqrt{K}}{a_{\mathrm{req}}} + \frac{1}{a_{\mathrm{req}}^2}\right)\ln^3\left(\frac{1}{a_{\mathrm{req}}}\right)\ln^2\left(\frac{M+J+R+2}{a_{\mathrm{req}}\xi_{\mathrm{req}}}\right),$$

and the uniform oracle-machine implementation of Algorithm 2 runs within $O(N(b_{\mathcal{U}} + \chi^2))$ bits at every step while retaining no past queries, with explicit certified bound $2^{30}N(b_{\mathcal{U}} + \chi^2)$ bits in `F2e2e/BSpace/` (`Theorem_B_space`, `Theorem_B_joint`).

Proof. Write $a = 2^{-e}$ with $e = \lceil \log_2(1/a_{\mathrm{req}}) \rceil \ge 4$. Testing $p' = 2e + 4$ gives $4^{p'-1}(a/(8p'))^2 = 2^{2e}/(2e+4)^2 \ge 1$; since $e < \log_2(1/a_{\mathrm{req}}) + 1$, $\ln 2 > 2/3$, and $\ln(1/a_{\mathrm{req}}) > 2$, we obtain $12 \le p \le 2e + 4 < 6\ln(1/a_{\mathrm{req}})$. By minimality of p, $1 \le 4^{p-1}\tau^2 < 4$, whence $T_{p-1}^{\max} \le 6$, $T_{p-2}^{\max} \le 24$, and $C_p \le 31$. At $j = 0$, $C_0 \le 7.5/\tau^2 + K + 3 \le 10/\tau^2 + K$ (as $\tau \le 1/16$). For $1 \le j < p$, setting $v_j := 4^{-j}\tau^{-2} > 1/4$ gives $C_j \le (96 + 48 + 12 + 1.5)v_j + 8 = 157.5v_j + 8 < 190v_j$.

Using $1/h = 512p/a$, $1/\tau^2 = 64p^2/a^2$, $d_j \le 2\cdot 2^j\sqrt{p}$ ($j < p$), $d_p \le 5/h$, and $\lceil\sqrt{C_j\chi}\rceil \le 2\sqrt{C_j\chi}$, we bound $n_0 \le 2^{36}p^{5/2}\chi^{3/2}/a^2 + 2^{31}p^{3/2}\sqrt{K}\,\chi^{3/2}/a + 2^{30}p^2\chi/a^2 + 1$, $n_j \le 2^{38}p^{5/2}\chi^{3/2}/a^2 + 2^{30}p^2\chi/a^2 + 1$ ($1 \le j < p$, as 2^j in d_j cancels 2^{-j} in $\sqrt{v_j}$), and $n_p \le 2^{44}p^2\chi^{3/2}/a^2 + 2^{30}p^2\chi/a^2 + 1$. Summing over $0 \le j \le p$ yields $N \le 2^{45}[(p^{7/2}/a^2 + p^{3/2}\sqrt{K}/a)\chi^{3/2} + p^3\chi/a^2] \le$ `valNExplicit`. With $L_{\mathrm{B}} := (M + J + R + 2)/(a_{\mathrm{req}}\xi_{\mathrm{req}}) > 40$, $2^{40}(p+1)Q/(h\xi_A) \le 2^{62}L_{\mathrm{B}}^8$ implies $\chi \le 63 + 8\log_2 L_{\mathrm{B}} \le 31\ln L_{\mathrm{B}}$, giving $N \le 2^{84} \cdot$ `valNAsymp`.

For bit space, the N sample codes in $[0, 2^{b_{\mathcal{U}}})$ occupy $O(Nb_{\mathcal{U}})$ bits. Between requests, `alg2RetainedOf` stores only counters, h_A-grid thresholds ($O(p\chi)$ bits), and in-block scores $G_{j,i} \in (h_A/4)\mathbb{Z}$ ($j < p$, $i < n_j$) bounded by $|G_{j,i}| \le T_j^{\max} + T_{j-1}^{\max}/4$, taking $O(\log_2(1/h_A) + \log_2(1/\tau)) = O(\chi)$ bits per point ($O(N\chi)$ bits total). Within a request, rounded evaluations $\hat{f}(X_m) \in h_A\mathbb{Z} \cap [0,1]$ take $O(N\chi)$ bits; exact rational KKT water-filling (Lemma 5.3) over common denominator $k_{\mathrm{act}}n_j\,\mathrm{denom}(\kappa h_A/4)$ with $k_{\mathrm{act}} \le n_j$ takes $O(\chi)$ bits per coordinate and $O(n_j\chi^2)$ bits across sorting and prefix-sum workspace; and each bounded sampler draw (Lemma 5.2) uses $O(\chi)$ live bits without storing unread coins. Because `bitEncodeVal` charges $2\,\mathtt{bitLen}(z) + 3$ bits per integer and $2 + m$ overhead bits per m-element list, every primitive step satisfies `opWork` $\le 8(|v_a| + |v_b|) + 64$, and all loops use constant-depth iterative continuations (Section 7.1), the small-step sum $\mathtt{confBits}(c) + \mathtt{confBits}(c') + \mathtt{stepWork}(c) + \mathtt{ctrlWork}(c)$ is $O(N(b_{\mathcal{U}} + \chi^2))$ at every transition; `F2e2e/BSpace/` mechanically certifies the concrete constant 2^{30}. □

A.3 Turnstile F_2 Global Hierarchy, Dyadic Grids, and Bit Accounting

Definition A.5 (Global Turnstile F_2 Parameters and Total-Behavior Ranges)**.** For integers $n, m, U \ge 1$ and reals $\varepsilon, \beta \in (0, 1/10)$ with dyadic exponents $e_\varepsilon := \lceil\log_2(1/\varepsilon)\rceil$ and $e_\beta := \lceil\log_2(1/\beta)\rceil$ (`dyadicExp`), `mkStreamParams` (`Spec.lean`) builds the record `StreamParams` with fields $(n, m, U, \mathtt{epsExp} = e_\varepsilon, \mathtt{betaExp} = e_\beta)$. In `F2e2e/C/ParamsDefs.lean`, set $\varepsilon_0 := 2^{-e_\varepsilon} \in (\varepsilon/2, \varepsilon]$ (`eps0`), $\beta_0 := 2^{-e_\beta} \in (\beta/2, \beta]$ (`beta0`), $\Lambda := \log_2[2^{100}(n+2)(m+2)(U+2)/(\varepsilon\beta)]$ (`masterLambda`), and integer bit scale $\mathcal{L} := \lceil\log_2[2^{100}(n+2)(m+2)(U+2)/(\varepsilon_0\beta_0)]\rceil = 100 + e_\varepsilon + e_\beta + \lceil\log_2((n+2)(m+2)(U+2))\rceil \in [100, \Lambda + 3]$ (`StreamParams.calL`), and define sequentially:

(F1) *Tree and dimension:* $H := 1 + \lceil\log_2 \max\{1, m\}\rceil$, $T := 2^{12}(m+1)(H+1)^3$, $D_* := n + T + 1$ (guard coordinate $g(d,t) := n + H(t-1) + d + 1$ for $0 \le d < H$, $1 \le t \le m$), and $W := nU + 1$.

(F2) *Accuracy and envelopes:* $\eta := 2^{-\lceil \log_2(2^{20}(H+1)^2/\varepsilon_0)\rceil}$, $\gamma_g := 2^{-\lceil \log_2(8\sqrt{T})\rceil}$, $B := 16/\gamma_g$, $V_0 := W$ and $V_{d+1} := (1+2B^d)V_d$ $(0 \le d \le H)$, $\mathcal{U}_{d,0} := 2^{\lceil \log_2(2V_d^2)\rceil}$, $\mathcal{U}_* := \mathcal{U}_{H-1,0}$, $V := 8B^{H+1}V_H + 2$, $G := B^{H+2}$, $\nu := \gamma_g\eta 2^{-H-20}$, and $E_{\min} := 2^{-8}\eta^2\nu^2$.

(F3) *Epochs, banks, and fingerprint:* $K_{\text{epoch}} := \lceil \log_2(2^{20}\mathcal{U}_* B^{2H}/(\eta^2\nu^2))\rceil$, $K_0 := \lceil 128/\eta^2\rceil$, $J_* := K_0(K_{\text{epoch}}+1)$, $J_{\max} := 2J_*$, $R_{\max} := 4J_* + 2$, $s := \log_2(1/\eta) + 22$, $L_{\text{clip}} := 16s$, $a_{\text{num}} := \eta/(2^{14}L_{\text{clip}})$, $a_{\text{val}} := \eta/(16L_{\text{clip}})$, and $\beta_B := \beta_0/(64m(H+1))$. Each bank (Algorithm 4) runs Algorithm 2 with $(m, J_{\max}, R_{\max}, a_{\text{num}}, \beta_B/2)$, row field $\mathbb{F}_{2^w}$ $(w := \lceil \log_2(D_*+1)\rceil)$, and $R_{\max}$ coarse groups of $32g$ rows $(g := 32\lceil \log_2(4R_{\max}/\beta_B)\rceil)$. Algorithm 3 uses the least prime $q \ge \lceil 10nm(U+1)/\beta_0\rceil^3$ and $e_\beta + 10 = \lceil \log_2(1000/\beta_0)\rceil$ seed trials.

(F4) *Total-behavior ranges and failure transitions:* $Z_{\max}$, $C_{\max}$, $A_{\max}$ are the least powers of two with $Z_{\max} \ge 8D_*(V+1)$, $C_{\max} \ge 8(B^H+1)$, $A_{\max} \ge 16(V+1)^2$. Row counters silently saturate to $[-Z_{\max}, Z_{\max}]$ on $h_b\mathbb{Z}$; stored tags are range-checked in $[0, C_{\max}]$ on $h_C\mathbb{Z}$ and stored caches/candidates/estimates/leaf accumulators in $[0, A_{\max}]$; step increments with $P_d > 0, \widehat{Q}_d > 0$ clamp to $[-1/2, 1/2]$ on $h_{\text{step}}\mathbb{Z}$; and any failed range, cap $(J_{\text{used},d} < J_*$, $k_d \le K_{\text{epoch}})$, positivity $(P_d > 0, \widehat{Q}_d > 0)$, or bank check enters the global absorbing zero-output failure state (Algorithms 4 and 5).

Definition A.6 (Committed Dyadic Precision Grids)**.** With $\text{round}_h(z) := h\lfloor z/h + 1/2\rfloor$ (ties toward $+\infty$), Algorithms 4 and 5 commit to the dyadic grids:

$$h_g := \gamma_g\eta\nu B^{-2H}/2^{20}, \qquad e_C := h_g/(2^{10}\gamma_g V), \qquad h_C := 2^{-\lceil \log_2(TG/e_C)\rceil} \le e_C/(TG),$$
$$\sigma := 2^{-\lceil \log_2(2^{40}TG^2(V+1)^2D_*/(\eta^4\nu^2))\rceil}, \qquad \omega := \sigma^2, \qquad h_P := 2^{-\lceil \log_2(2^{40}TG^2D_*/(\eta^3\nu^2))\rceil},$$
$$h_b := 2^{-\lceil \log_2(2^{40}(T+1)GD_*(V+1)/(\eta\omega))\rceil}, \quad h_{\text{step}} := 2^{-\lceil \log_2(2^{20}/\eta^2)\rceil} = \eta^2/2^{20}, \qquad h_{\text{leaf}} := h_g^2 h_{\text{step}}^{2H}.$$

Here $h_g B^{2H} = \gamma_g\eta\nu/2^{20}$ absorbs cumulative child rescaling $\le B^H$, and $h_g, \sigma \in h_b\mathbb{Z}$ are exact on $h_b\mathbb{Z}$.

Table 3: Binary precision ceilings $\log_2(1/\cdot)$ in terms of $\mathcal{L} \ge 100$ (grids defined in `F2e2e/C/ParamsDefs.lean`, ceilings proved in `F2e2e/C/SpaceParams.lean`).

Quantity	Lean Identifier	Ceiling on $\log_2(1/\cdot)$	Operational Role in Algorithms 4 and 5
h_g	`StreamParams.hg`	$5\mathcal{L}^2$	Refined guard coordinate grid ($h_g B^{2H} = \gamma_g\eta\nu/2^{20}$).
e_C, h_C	`StreamParams.eC, hC`	$4\mathcal{L}^3$, $5\mathcal{L}^3$	Scale-tag tolerance ($e_C\gamma_g V = 2^{-10}h_g$) and tag grid $\widehat{C} \in h_C\mathbb{Z}$.
σ, $\omega = \sigma^2$	`StreamParams.sigma, omega`	$7\mathcal{L}^3$, $14\mathcal{L}^3$	Coordinate-D_* regularizer ($h_b \mid \sigma$) and quadratic floor in ReleaseNumerical.
h_P	`StreamParams.hP`	$7\mathcal{L}^2$	Shadow-cache norm estimate grid (Lemma 6.7).
h_b	`StreamParams.hb`	$18\mathcal{L}^3$	Finest linear sketch counter grid ($h_b \mid h_g$ and $h_b \mid \sigma$).
h_{step}, h_{leaf}	`StreamParams.hstep, hleaf`	$5\mathcal{L}$, $20\mathcal{L}^2$	Geometric step grid ($a_{\text{step}} \in h_{\text{step}}\mathbb{Z}$) and exact leaf grid ($\Delta^2 \in h_{\text{leaf}}\mathbb{Z}$).

Lemma A.7 (Register Bit Lengths and Total Bit-Space Bound)**.** *Under Definitions A.5 and A.6, all ceilings in Table 3 hold,* $\log_2 V_H \le 2\mathcal{L}^3$, $K_{\text{epoch}} \le 6\mathcal{L}^3$, $J_* \le 2^{10}\eta^{-2}\mathcal{L}^3$, *each bank has* $N_{\text{bank}} \le 2^{200}\eta^{-2}\mathcal{L}^7$ *rows with registers of* $\le 2^{30}\mathcal{L}^3$ *bits, and the small-step implementation (`RunsWithinBits`) uses* $O(\varepsilon^{-2}\Lambda^{15})$ *bits at every step across the* $\le H \le \mathcal{L}$ *live banks, with explicit certified bound* $2^{512}\varepsilon^{-2}\Lambda^{15}$ *bits in `F2e2e/C/Impl/` (`Theorem_C`).*

Proof. Let $c := \lceil \log_2((n+2)(m+2)(U+2)) \rceil$, so $c+100 \le \mathcal{L}$. Direct evaluation gives $H \le c+1 \le \mathcal{L}$, $\log_2 W \le c \le \mathcal{L}$, $\log_2 T, \log_2(1/\eta) \le 2\mathcal{L}$, $\log_2 B \le c+23 \le 2\mathcal{L}$, $\log_2 D_* \le 3\mathcal{L}$, $\log_2 G \le 3\mathcal{L}^2$, and $\log_2(1/\nu) \le 6\mathcal{L}$. Since $V_{d+1} = (1+2B^d)V_d \le 4B^d V_d$, induction gives $V_d \le W \cdot 4^d B^{d^2}$ and $\log_2 V_H \le c+2(c+1)+(c+1)^2(c+23) \le 2\mathcal{L}^3$, whence $\log_2 \mathcal{U}_* \le 5\mathcal{L}^3$, $\log_2 V \le 3\mathcal{L}^3$, $\log_2 Z_{\max} \le 4\mathcal{L}^3$, $\log_2 C_{\max} \le 2\mathcal{L}^2$, $\log_2 A_{\max} \le 7\mathcal{L}^3$, and every row of Table 3 follows by substitution. Likewise $K_{\text{epoch}} \le 6\mathcal{L}^3$ and $J_* = 128\eta^{-2}(K_{\text{epoch}}+1) \le 2^{10}\eta^{-2}\mathcal{L}^3$.

For each bank, $s \le 3\mathcal{L}$, $a^{-1} \le 2^{21}\eta^{-1}\mathcal{L}$, $K = J_{\max} + R_{\max} + 1 \le 2^{13}\eta^{-2}\mathcal{L}^3$, and $p, \chi \le 2^{12}\mathcal{L}$; plugging into Lemma A.4 gives $\sum_{j=0}^{p} n_j \le 2^{199}\eta^{-2}\mathcal{L}^7$, while $32gR_{\max} \le 2^{26}\eta^{-2}\mathcal{L}^4$, so $N_{\text{bank}} \le 2^{200}\eta^{-2}\mathcal{L}^7$. Row counters use $\le 22\mathcal{L}^3 + 2$ bits, row seeds $(4s+2)w = O(\mathcal{L}^2)$ bits, and transient rationals/mantissas $\le 512\mathcal{L}^3 \le 2^{30}\mathcal{L}^3$ bits (`SpaceRegs.lean`). Across $\le H$ active-path banks (retired before replacement), the live mathematical state uses at most $(H+1)\cdot 2^{80}N_{\text{bank}}\mathcal{L}^3 + 2^{80}\mathcal{L}^5 \le 2^{328}\varepsilon_0^{-2}\mathcal{L}^{15}$ bits (since $\eta^{-2} \le 2^{46}\varepsilon_0^{-2}\mathcal{L}^4$ and $\log_2 q \le 3\mathcal{L}$). Under `bitEncodeVal` ($2\,\texttt{bitLen}(z)+3$ bits per integer, $2+m$ overhead bits per list), `opWork` $\le 8(|v_a|+|v_b|)+64$, and constant-depth iterative DSL loops (Section 7.1), peak step space $\texttt{confBits}(c) + \texttt{confBits}(c') + \texttt{stepWork}(c) + \texttt{ctrlWork}(c)$ is within $O(1)$ of live state bits, proving $O(\varepsilon^{-2}\Lambda^{15})$ bits; `SpaceBound.lean` certifies the concrete 2^{167} small-step simulation multiplier, and $\varepsilon_0^{-2} < 4\varepsilon^{-2}$, $\mathcal{L} < \Lambda + 3 < 2\Lambda$ ($167 + 328 + 2 + 15 = 512$) yield

$$\underbrace{O(\mathcal{L})}_{\text{live banks}} \cdot \underbrace{O(\varepsilon^{-2}\mathcal{L}^4)}_{\eta^{-2}} \cdot \underbrace{O(\mathcal{L}^7)}_{\text{rows beyond } \eta^{-2}} \cdot \underbrace{O(\mathcal{L}^3)}_{\text{bits/row + workspace}} \le 2^{512}\varepsilon^{-2}\Lambda^{15} \text{ bits.} \qquad \square$$

A.4 Symbol Correspondence with the Lean Specification

Table 4 maps our unified clash-free notation across Sections 4 to 6 to `F2e2e/Spec.lean` and `F2e2e/C/ParamsDefs.lean`.

Table 4: Symbol dictionary between this paper's unified notation and `F2e2e/Spec.lean` / `F2e2e/C/ParamsDefs.lean`.

Scope	Paper	Lean Identifier	Meaning & Disambiguation
Theorem 1.1	N_{cas}, n_1, n_r, n_L	`adaSampleSize, adaN1, adaNr, adaNL`	Cascade sample/block sizes (n is stream dim.).
Theorem 1.1	$L, \gamma, u_0, b_{\mathrm{SVT}}$	`adaL, adaGamma, adaU0, adaBsvt`	Depth, margin, tail unit, and SVT scale.
Theorem 1.1	L_1, χ_1, p_1, q_B	`adaL1, adaChi1, adaP1`	Stage-1 factors and Beta(p_1, p_1) ($q_B = (\texttt{adaP1}\,\alpha\,\beta) - 1$).
Theorem 1.1	$\ell, \delta_{\ell,i}, \Pi$	`m, ADAField.delta m, adaPublicOutputs`	Validator index (m is stream length) and transcript.
Theorem 1.2	$a_{\mathrm{req}}, \xi_{\mathrm{req}}$; a, ξ_A	`aReq, ξReq; dyadicDown`	Validation accuracy/failure and dyadic roundings.
Theorem 1.2	p, τ, h, κ, χ	`valP, valTau, valH, valKappa, valChi`	Depth, margin $a/(8p)$, unit $\tau/64$, step, and log factor.
Theorem 1.2	$T_j^{\max}, C_j, Q, d_j$	`valTcap, valC, valQ, valD`	Edge caps, phase budgets, draw bound, sensitivities.
Theorem 1.2	N, n_j, h_A	`Alg2Params.N, valNBlock, Alg2Params.hA`	Sample size (`valNExplicit`, `valNAsymp`) and grid.
Theorem 1.2	$2^{30} N(b_{\mathcal{U}} + \chi^2)$	`BSpace.bSpaceBound 30`	Bit budget and retained state `alg2RetainedOf`.
Theorem 1.3	$n, m, U, \varepsilon, \beta$	`mkStreamParams; StreamParams`	Stores $(n, m, U,$ `epsExp`, `betaExp`$)$; ε_0, β_0 via `eps0, beta0`.
Theorem 1.3	Λ, $\mathcal{L}$	`masterLambda, StreamParams.calL`	Master log factor and integer bit scale ($\mathcal{L} \geq 100$).
Theorem 1.3	$\gamma_g, \eta, B, V_d, \mathcal{U}_*$	`StreamParams.gamma, eta, B, Vd, Ustar`	Guard scale (clashes with γ), accuracy, envelopes.
Theorem 1.3	s, $L_{\mathrm{clip}} = 16s$	`StreamParams.s, StreamParams.L`	Moment order and cutoff (clashes with cascade L).
Theorem 1.3	$2^{512} \varepsilon^{-2} \Lambda^{15}$	`spaceBoundC`	Certified worst-case bit bound (`RunsWithinBits`).

References

[1] Noga Alon, László Babai, and Alon Itai. A fast and simple randomized parallel algorithm for the maximal independent set problem. *Journal of Algorithms*, 7(4):567–583, 1986.

[2] Noga Alon, Yossi Matias, and Mario Szegedy. The space complexity of approximating the frequency moments. *Journal of Computer and System Sciences*, 58(1):137–147, 1999.

[3] Kazuoki Azuma. Weighted sums of certain dependent random variables. *Tôhoku Mathematical Journal*, 19(3):357–367, 1967.

[4] Raef Bassily, Kobbi Nissim, Adam Smith, Thomas Steinke, Uri Stemmer, and Jonathan Ullman. Algorithmic stability for adaptive data analysis. *SIAM Journal on Computing*, 50(3):STOC16–377–STOC16–405, 2021. Preliminary version in STOC 2016.

[5] Mihir Bellare and John Rompel. Randomness-efficient oblivious sampling. In *Proceedings of the 35th Annual IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 276–287, 1994.

[6] Omri Ben-Eliezer, Talya Eden, and Krzysztof Onak. Adversarially robust streaming via dense-sparse trade-offs. In *Proceedings of the 5th Symposium on Simplicity in Algorithms (SOSA)*, pages 214–227, 2022.

[7] Omri Ben-Eliezer, Rajesh Jayaram, David P. Woodruff, and Eylon Yogev. A framework for adversarially robust streaming algorithms. *Journal of the ACM*, 69(2):17:1–17:33, 2022.

[8] Rajendra Bhatia and Chandler Davis. A better bound on the variance. *The American Mathematical Monthly*, 107(4):353–357, 2000.

[9] Mark Bun and Thomas Steinke. Concentrated differential privacy: Simplifications, extensions, and lower bounds. In *Proceedings of the 14th Theory of Cryptography Conference (TCC), Part I*, pages 635–658, 2016.

[10] J. Lawrence Carter and Mark N. Wegman. Universal classes of hash functions. *Journal of Computer and System Sciences*, 18(2):143–154, 1979.

[11] Yeshwanth Cherapanamjeri and Jelani Nelson. On adaptive distance estimation. In *Advances in Neural Information Processing Systems 33 (NeurIPS)*, 2020.

[12] Edith Cohen, Jelani Nelson, Tamás Sarlós, and Uri Stemmer. Tricking the hashing trick: A tight lower bound on the robustness of CountSketch to adaptive inputs. In *Proceedings of the 37th AAAI Conference on Artificial Intelligence (AAAI)*, pages 7235–7243, 2023.

[13] Yuval Dagan and Gil Kur. A bounded-noise mechanism for differential privacy. In *Proceedings of the 35th Conference on Learning Theory (COLT)*, pages 625–661, 2022.

[14] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28*, volume 12699 of *Lecture Notes in Computer Science*, pages 625–635. Springer, 2021.

[15] Itai Dinur, Uri Stemmer, David P. Woodruff, and Samson Zhou. On differential privacy and adaptive data analysis with bounded space. In *Advances in Cryptology – EUROCRYPT 2023, Part III*, pages 35–65, 2023.

[16] Cynthia Dwork, Vitaly Feldman, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Aaron Roth. Generalization in adaptive data analysis and holdout reuse. In *Advances in Neural Information Processing Systems 28 (NIPS)*, pages 2350–2358, 2015.

[17] Cynthia Dwork, Vitaly Feldman, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Aaron Roth. Preserving statistical validity in adaptive data analysis. In *Proceedings of the 47th Annual ACM Symposium on Theory of Computing (STOC)*, pages 117–126, 2015.

[18] Cynthia Dwork and Aaron Roth. The algorithmic foundations of differential privacy. *Foundations and Trends in Theoretical Computer Science*, 9(3–4):211–407, 2014.

[19] Cynthia Dwork, Guy N. Rothblum, and Salil Vadhan. Boosting and differential privacy. In *Proceedings of the 51st IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 51–60, 2010.

[20] David A. Freedman. On tail probabilities for martingales. *The Annals of Probability*, 3(1):100–118, 1975.

[21] Elena Gribelyuk, Honghao Lin, David P. Woodruff, Huacheng Yu, and Samson Zhou. A strong separation for adversarially robust ℓ_0 estimation for linear sketches. In *Proceedings of the 65th IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 2318–2343, 2024.

[22] Elena Gribelyuk, Honghao Lin, David P. Woodruff, Huacheng Yu, and Samson Zhou. Lifting linear sketches: Optimal bounds and adversarial robustness. In *Proceedings of the 57th Annual ACM Symposium on Theory of Computing (STOC)*, pages 395–406, 2025.

[23] Elena Gribelyuk, Honghao Lin, David P. Woodruff, Huacheng Yu, and Samson Zhou. Adversarial robustness on insertion-deletion streams. In *Proceedings of the 58th Annual ACM Symposium on Theory of Computing (STOC)*, 2026. `arXiv:2602.20854`.

[24] Moritz Hardt and Jonathan Ullman. Preventing false discovery in interactive data analysis is hard. In *Proceedings of the 55th IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 454–463, 2014.

[25] Moritz Hardt and David P. Woodruff. How robust are linear sketches to adaptive inputs? In *Proceedings of the 45th Annual ACM Symposium on Theory of Computing (STOC)*, pages 121–130, 2013.

[26] Avinatan Hassidim, Haim Kaplan, Yishay Mansour, Yossi Matias, and Uri Stemmer. Adversarially robust streaming algorithms via differential privacy. In *Advances in Neural Information Processing Systems 33 (NeurIPS)*, 2020.

[27] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.

[28] Christopher Jung, Katrina Ligett, Seth Neel, Aaron Roth, Saeed Sharifi-Malvajerdi, and Moshe Shenfeld. A new analysis of differential privacy's generalization guarantees. In *Proceedings of the 11th Innovations in Theoretical Computer Science Conference (ITCS)*, pages 31:1–31:17, 2020.

[29] Haim Kaplan, Yishay Mansour, Kobbi Nissim, and Uri Stemmer. Separating adaptive streaming from oblivious streaming using the bounded storage model. In *Advances in Cryptology – CRYPTO 2021, Part III*, pages 94–121, 2021.

[30] Daniel M. Kane, Jelani Nelson, and David P. Woodruff. On the exact space complexity of sketching and streaming small norms. In *Proceedings of the 21st Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 1161–1178, 2010.

[31] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP)*, pages 367–381, 2020.

[32] Jacob T. Schwartz. Fast probabilistic algorithms for verification of polynomial identities. *Journal of the ACM*, 27(4):701–717, 1980.

[33] Shai Shalev-Shwartz. Online learning and online convex optimization. *Foundations and Trends in Machine Learning*, 4(2):107–194, 2012.

[34] Thomas Steinke and Jonathan Ullman. Interactive fingerprinting codes and the hardness of preventing false discovery. In *Proceedings of the 28th Conference on Learning Theory (COLT)*, pages 1588–1628, 2015.

[35] Salil P. Vadhan and Tianhao Wang. Concurrent composition of differential privacy. In *Proceedings of the 19th Theory of Cryptography Conference (TCC), Part II*, pages 582–604, 2021.

[36] David P. Woodruff and Samson Zhou. Tight bounds for adversarially robust streams and sliding windows via difference estimators. In *Proceedings of the 62nd IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 1183–1196, 2021.

[37] Richard Zippel. Probabilistic algorithms for sparse polynomials. In *Symbolic and Algebraic Computation (EUROSAM '79)*, volume 72 of *Lecture Notes in Computer Science*, pages 216–226. Springer, 1979.